

Perfusion Cell Culture Processes for Biopharmaceuticals

Process Development, Design, and Scale-up

-
- Abu-Absi, N. R., Kenty, B. M., Cuellar, M. E., Borys, M. C., Sakhamuri, S., Strachan, D. J., Hausladen, M. C. & Li, Z. J. (2011), 'Real time monitoring of multiple parameters in mammalian cell culture bioreactors using an in-line Raman spectroscopy probe', *Biotechnology and Bioengineering* 108(5), 1215–1221. CrossRef (<http://dx.doi.org/10.1002/bit.23023>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Real+time+monitoring+of+multiple+parameters+in+mammalian+cell+culture+bioreactors+using+an+in-line+Raman+spec+Absi+N.+R.+&author=Kenty+B.+M.+&author=Cuellar+M.+E.+&author=Borys+M.+C.+&author=Sakhamuri+S.+&author=Strachan+D.+J.+&author=Hausladen+M.+C.+&author=Li+Z.+J.+&journal=Biotechn+1221) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21449033>)
-
- Ahmed, S. U., Ranganathan, P., Pandey, A. & Sivaraman, S. (2010), 'Computational fluid dynamics modeling of gas dispersion in multi impeller bioreactor', *Journal of Bioscience and Bioengineering* 109(6), 588–597. CrossRef (<http://dx.doi.org/10.1016/j.jbiosc.2009.11.014>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Computational+fluid+dynamics+modeling+of+gas+dispersion+in+multi+impeller+bioreactor&publication+year=2010&author=Ahmed+S.+U.+&author=Ranganathan+P.+&author=Pandey+597) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20471599>)
-
- Al-Rubeai, M. (2015), *Animal Cell Culture*, Vol. 9, 9th edn, Springer. CrossRef (<http://dx.doi.org/10.1007/978-3-319-10320-4>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Animal+Cell+Culture&publication+year=2015&author=Al-Rubeai+M.)
-
- Allison, G., Cain, Y. T., Cooney, C., Garcia, T., Bizjak, T. G., Holte, O., Jagota, N., Komars, B., Korakianiti, E., Kourti, D., Madurawe, R., Morefield, E., Montgomery, F., Nasr, M., Randolph, W., Robert, J. L., Rudd, D. & Zezza, D. (2015), 'Regulatory and quality considerations for continuous manufacturing May 20–21, 2014 continuous manufacturing symposium', *Journal of Pharmaceutical Sciences* 104(3), 803–812. CrossRef (<http://dx.doi.org/10.1002/jps.24324>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Regulatory+and+quality+considerations+for+continuous+manufacturing+May+20%2E%80%9321+2014+continuous+manufacturing+symposium&publication+year=2015&author=Alliso+812) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25830179>)
-
- Amanullah, A., McFarlane, C. M., Emery, A. N. & Nienow, A. W. (2001), 'Scale-down model to simulate spatial pH variations in large-scale bioreactors', *Biotechnology and Bioengineering* 73(5), 390–399. CrossRef (<http://dx.doi.org/10.1002/bit.1072>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Scale-down+model+to+simulate+spatial+pH+variations+in+large-scale+bioreactors&publication+year=2001&author=Amanullah+A.+&author=McFarlane+C.+M.+&author=Emery+A.+N.+&author=Nienow+A.+W.+&journal=Biotechnology+and+Bioengineering&v+399) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/11320509>)
-
- Anson, S., Esteban, G. & Schmid, G. (2010), 'On-line monitoring of responses to nutrient feed additions by multi-frequency permittivity measurements in fed-batch cultivations of CHO cells', *Cytotechnology* 62(2), 121–132. CrossRef (<http://dx.doi.org/10.1007/s10616-010-9267-z>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=On-line+monitoring+of+responses+to+nutrient+feed+additions+by+multi-frequency+permittivity+measurements+in+fed-batch+cultivations+of+CHO+cells&publication+year=2010&author=Anson+S.+&author=Esteban+G.+&author=Schmid+G.+&journal=Cytotechnology&volume=62&doi=10.1007/s10616-010-9267-z&pages=121-132) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20407823>)
-
- Antonia, S. J., Larkin, J. & Ascierto, P. A. (2014), 'Immuno-oncology combinations: A review of clinical experience and future prospects', *Clinical Cancer Research* 20(24), 6258–6268. CrossRef (<http://dx.doi.org/10.1158/1078-0432.CCR-14-1457>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Immuno-oncology+combinations:+A+review+of+clinical+experience+and+future+prospects&publication+year=2014&author=Antonia+S.+J.+&author=Larkin+J.+&author=Ascierto+P.+A.+&journal=Clinical+0432.CCR-14-1457&pages=6258-6268) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25341541>)
-
- Arnold, L., Lee, K., Rucker-Pezzini, J. & Lee, J. H. (2018), 'Implementation of fully integrated continuous antibody processing: Effects on productivity and CO₂m', *Biotechnology Journal* 14(2), 1–10. Google Scholar (https://scholar.google.com/scholar_lookup?title=Implementation+of+fully+integrated+continuous+antibody+processing:+Effects+on+productivity+and+CO2m&publication+year=2018&author=Arnold+L.+&author=Lee+K.+&author=Ruc+Pezzini+J.+&author=Lee+J.+H.+&journal=Biotechnology+Journal&volume=14&pages=1-10) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/29729129>)
-
- Arnold, S. A., Crowley, J., Woods, N., Harvey, L. M. & McNeil, B. (2003), 'In-situ near infrared spectroscopy to monitor key analytes in mammalian cell cultivation', *Biotechnology and Bioengineering* 84(1), 13–19. CrossRef (<http://dx.doi.org/10.1002/bit.10738>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=In-situ+near+infrared+spectroscopy+to+monitor+key+analytes+in+mammalian+cell+cultivation&publication+year=2003&author=Arnold+S.+A.+&author=Crowley+J.+&author=Woods+N.+&author=19) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/12910538>)
-
- Assirelli, M., Bujalski, W., Eaglesham, A. & Nienow, A. (2005), 'Intensifying micromixing in a semi-batch reactor using a Rushton turbine', *Chemical Engineering Science* 60(8–9), 2333–2339. CrossRef (<http://dx.doi.org/10.1016/j.ces.2004.10.041>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Intensifying+micromixing+in+a+semi-batch+reactor+using+a+Rushton+turbine&publication+year=2005&author=Assirelli+M.+&author=Bujalski+W.+&author=Eaglesham+A.+&author=Nienow+A.+&journal=Chemical+Engineering+Sc+2339)

Atkins, P., De Paula, J. & Friedmand, R. (2010), *Physical Chemistry, 9th edn.*

Google Scholar ([https://scholar.google.com/scholar?q=Atkins+P.+De+Paula+J.+&+Friedmand+R.+\(2010\)+Physical+Chemistry+9th+edn.](https://scholar.google.com/scholar?q=Atkins+P.+De+Paula+J.+&+Friedmand+R.+(2010)+Physical+Chemistry+9th+edn.))

Bacchin, P., Aimar, P. & Field, R. W. (2006), 'Critical and sustainable fluxes: Theory, experiments and applications', *Journal of Membrane Science* 281(1–2), 42–69.

CrossRef (<http://dx.doi.org/10.1016/j.memsci.2006.04.014>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Critical+and+sustainable+fluxes:+Theory+experiments+and+applications&publication+year=2006&author=Bacchin+P.&author=Aimar+P.&author=Field+R.+W.&journal=Journal+of+Mem
69)

Bailey, J. E. & Ollis, D. F. (1986), '*Fundamentals of Biochemical engineering*'.

Google Scholar ([https://scholar.google.com/scholar?q=Bailey+J.+E.+&+Ollis+D.+F.+\(1986\)+%E2%80%98Fundamentals+of+Biochemical+engineering%E2%80%99.](https://scholar.google.com/scholar?q=Bailey+J.+E.+&+Ollis+D.+F.+(1986)+%E2%80%98Fundamentals+of+Biochemical+engineering%E2%80%99.))

Bandyopadhyay, B., Humphrey, A. E. & Taguchi, H. (1967), 'Dynamic measurement of the volumetric oxygen transfer coefficient in fermentation systems', *Biotechnology and Bioengineering* 9(4), 533–544. CrossRef (<http://dx.doi.org/10.1002/bit.260090408>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Dynamic+measurement+of+the+volumetric+oxygen+transfer+coefficient+in+fermentation+systems&publication+year=1967&author=Bandyopadhyay+B.&author=Humphrey+A.+E.+&ai

544)

Baptista, R. P., Fluri, D. A. & Zandstra, P. W. (2013), 'High density continuous production of murine pluripotent cells in an acoustic perfused bioreactor at different oxygen concentrations.',

Biotechnology and Bioengineering 110(2), 648–655. CrossRef (<http://dx.doi.org/10.1002/bit.24717>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=High+density+continuous+production+of+murine+pluripotent+cells+in+an+acoustic+perfused+bioreactor+at+different+oxygen+concentrations&publication+year=2013&author=Bapti

655)

Barbaroux, M., Gerighausen, S. & Hackel, H. (2014), 'An approach to quality and security of supply for single-use bioreactors', *Advances in Biochemical Engineering/Biotechnology* 138, 239–272.

CrossRef (http://dx.doi.org/10.1007/10_2013_198) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=An+approach+to+quality+and+security+of+supply+for+single-use+bioreactors&publication+year=2014&author=Barbaroux+M.&author=Gerighausen+S.&author=Hackel+H.&journal=Advances+in+Biochemical+Engineering/Biotechnology&volume=138
272)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23793913>)

Barberis, M., Klipp, E., Vanoni, M. & Alberghina, L. (2007), 'Cell size at S phase initiation: An emergent property of the G1/S network', *PLoS Computational Biology* 3(4), e64.

CrossRef (<http://dx.doi.org/10.1371/journal.pcbi.0030064>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Cell+size+at+S+phase+initiation:+An+emergent+property+of+the+G1/S+network&publication+year=2007&author=Barberis+M.&author=Klipp+E.&author=Vanoni+M.&author=Alberghir

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/17432928>)

Barrett, S., Franklin, J., Stangl, M., Cvetkovic, A. & He, W. (2018), 'Intensification of a multi-product perfusion platform managing growth characteristics at high cell density for maximized volumetric productivity', *Cell Culture Engineering XVI*.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Intensification+of+a+multi-product+perfusion+platform+managing+growth+characteristics+at+high+cell+density+for+maximized+volumetric+productivity&publication+year=2018&author=Barrett+S.&author=Frankl

18)

Barrett, T. A., Wu, A., Zhang, H., Levy, M. S. & Lye, G. J. (2010), 'Microwell engineering characterization for mammalian cell culture process development', *Biotechnology and Bioengineering* 105(2), 260–275. CrossRef (<http://dx.doi.org/10.1002/bit.22531>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Microwell+engineering+characterization+for+mammalian+cell+culture+process+development&publication+year=2010&author=Barrett+T.+A.&author=Wu+A.&author=Zhang+H.&authi

275)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19739083>)

Bartholomew, D. J. (2010), 'Principal components analysis', in *International Encyclopedia of Education*, Vol. 2, Elsevier, pp. 374–377.

Google Scholar (https://scholar.google.com/scholar_lookup?title=International+Encyclopedia+of+Education&publication+year=2010&author=Bartholomew+D.+J.)

Bates, R. L., Fondy, P. L. & Fenic, J. G. (1966), 'Impeller characteristics and power', in J. Uh L, VW; Gray, ed., *Mixing: Theory and Practice 1*, Vol. 1, pp. 111–178.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Impeller+characteristics+and+power&publication+year=1966&author=Bates+R.+L.&author=Fondy+P.+L.&author=Fenic+J.+G.&journal=Mixing:+Theory+and+Practice+1&volume=1&pa

178)

Baur, D., Angarita, M., Müller-Späth, T. & Morbidelli, M. (2015), 'Optimal model-based design of the twin-column CaptureSMB process improves capacity utilization and productivity in protein A affinity capture', *Biotechnology Journal* 11(1), 135–145. CrossRef (<http://dx.doi.org/10.1002/biot.201500223>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Optimal+model-based+design+of+the+twin-column+CaptureSMB+process+improves+capacity+utilization+and+productivity+in+protein+A+affinity+capture&publication+year=2015&author=Baur+D.&author=Angarita+M.&author=M%Sp%C3%A4th+T.&author=Morbidelli+M.&journal=Biotechnology+Journal&volume=11&doi=10.1002/biot.201500223&pages=135-145)

145)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26308369>)

Baur, D., Angarita, M., Müller-Späth, T., Steinebach, F. & Morbidelli, M. (2016), 'Comparison of batch and continuous multi-column protein A capture processes by optimal design', *Biotechnology Journal* 11(7), 920–931. CrossRef (<http://dx.doi.org/10.1002/biot.201500481>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Comparison+of+batch+and+continuous+multi-column+protein+A+capture+processes+by+optimal+design&publication+year=2016&author=Baur+D.&author=Angarita+M.&author=M%C3%BCler-Sp%C3%A4th+T.&author=Steinebach+F.&author=Morbidelli+M.&journal=Biotechnology+Journal&volume=11&doi=10.1002/biot.201500481&pages=920-931)

931)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26992151>)

Bausch, M., Schultheiss, C. & Sieck, J. B. (2018), 'Recommendations for comparison of productivity between fed-batch and perfusion processes', *Biotechnology Journal* 14(2), 1–4.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Recommendations+for+comparison+of+productivity+between+fed-batch+and+perfusion+processes&publication+year=2018&author=Bausch+M.&author=Schultheiss+C.&author=Sieck+J.+B.&journal=Biotechnology+Journal&volume=14&pages=1-4

4)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30024096>)

Beier, S. P. & Jonsson, G. (2009), 'Critical flux determination by flux-stepping', *Wiley Inter-Science* 56(7), 1739–1747.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Critical+flux+determination+by+flux-stepping&publication+year=2009&author=Beier+S.+P.&author=Jonsson+G.&journal=Wiley+Inter-Science&volume=56&pages=1739-1747)

1747)

- Bendiak, B. & Schachter, H. (1987), 'Control of glycoprotein synthesis: Kinetic mechanism, substrate specificity, and inhibition characteristics of UDP-N-acetylglucosamine:alpha-D-mannoside beta 1-2 N-acetylglucosaminyltransferase II from rat liver', *Journal of Biological Chemistry* 262(12), 5784–5790.
Google Scholar (https://scholar.google.com/scholar_lookup?title=Control+of+glycoprotein+synthesis:+Kinetic+mechanism+substrate+specificity+and+inhibition+characteristics+of+UDP-N-acetylglucosamine:alpha-D-mannoside+beta+1-2+N-acetylglucosaminyltransferase+II+from+rat+liver&publication+year=1987&author=Bendiak+B.&author=Schachter+H.&journal=Journal+of+Biological+Chemistry&volume=262&pages=5784-5790)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/2952645>)
- Benz, G. T. (2011), 'Bioreactor design for chemical engineers', *American Institute of Chemical Engineers* 107, 21–26.
Google Scholar (https://scholar.google.com/scholar_lookup?title=Bioreactor+design+for+chemical+engineers&publication+year=2011&author=Benz+G.+T.&journal=American+Institute+of+Chemical+Engineers&volume=107&pages=21-26)
- Beresford, T. P., Fitzsimons, N. A., Brennan, N. L. & Cogan, T. M. (2001), 'Recent advances in cheese microbiology', *International Dairy Journal* 11, 254–274.
CrossRef ([http://dx.doi.org/10.1016/S0958-6946\(01\)00056-5](http://dx.doi.org/10.1016/S0958-6946(01)00056-5)) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Recent+advances+in+cheese+microbiology&publication+year=2001&author=Beresford+T.+P.&author=Fitzsimons+N.+A.&author=Brennan+N.+L.&author=Cogan+T.+M.&journal=Internat+Dairy+Journal&volume=11&pages=254-274)
- Berg, J. M., Tymoczko, J. L. & Stryer, L. (2007), *Biochemistry*, 6th edn, Sara Tenney.
Google Scholar (https://scholar.google.com/scholar_lookup?title=Biochemistry&publication+year=2007&author=Berg+J.+M.&author=Tymoczko+J.+L.&author=Stryer+L.)
- Berg, P. (1974), 'Potential biohazards of recombinant DNA molecules', *Science* 1114(1973), 1973–1974.
Google Scholar (https://scholar.google.com/scholar_lookup?title=Potential+biohazards+of+recombinant+DNA+molecules&publication+year=1974&author=Berg+P.&journal=Science&volume=1114&pages=1973-1974)
- Berg, P., Baltimore, D., Brenner, S., Roblin, R. & Singer, M. (1975), 'Summary statement of the Asilomar conference on recombinant DNA molecules', *Proceedings of the National Academy of Sciences of the United States of America* 72, 1981–1984. CrossRef (<http://dx.doi.org/10.1073/pnas.72.6.1981>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Summary+statement+of+the+Asilomar+conference+on+recombinant+DNA+molecules&publication+year=1975&author=Berg+P.&author=Baltimore+D.&author=Brenner+S.&author=Roblin+R.&author=Singer+M.&journal=Proceedings+of+the+National+Academy+of+Sciences+of+the+United+States+of+America&volume=72&pages=1981-1984)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/806076>)
- Berg, P. & Mertz, J. E. (2010), 'Personal reflections on the origins and emergence of recombinant DNA technology', *Genetics* 184(1), 9–17.
CrossRef (<http://dx.doi.org/10.1534/genetics.109.112144>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Personal+reflections+on+the+origins+and+emergence+of+recombinant+DNA+technology&publication+year=2010&author=Berg+P.&author=Mertz+J.+E.&journal=Genetics&volume=184&pages=9-17)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20061565>)
- Berry, B. N., Dobrowsky, T. M., Timson, R. C., Kshirsagar, R., Ryll, T. & Wiltberger, K. (2016), 'Quick generation of Raman spectroscopy based in-process glucose control to influence biopharmaceutical protein product quality during mammalian cell culture', *Biotechnology Progress* 32(1), 224–234. CrossRef (<http://dx.doi.org/10.1002/btpr.2205>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Quick+generation+of+Raman+spectroscopy+based+in-process+glucose+control+to+influence+biopharmaceutical+protein+product+quality+during+mammalian+cell+culture&publication+year=2016&author=Berry+B.+N.&author=Dobrowsky+T.+M.&author=Timson+R.+C.&author=Kshirsagar+R.&author=Ryll+T.&author=Wiltberger+K.&journal=Biotechnology+Progress&volume=32&pages=224-234)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26587969>)
- Bertrand, V., Karst, D. J. & Morbidelli, M. (2019), 'Transcriptome and proteome analysis of steady state in a perfusion CHO cell culture process', *Biotechnology and Bioengineering* pp. 1–14.
Google Scholar (<https://scholar.google.com/scholar?q=Bertrand+V.+Karst+D.+J.+&Morbidelli+M.+&publication+year=2019&title=Transcriptome+and+proteome+analysis+of+steady+state+in+a+perfusion+CHO+cell+culture+process&biotechnology+and+bioengineering+pp.+1-14>)
- Bertrand, V., Vogg, S., Villiger, T. K., Stettler, M., Broly, H., Soos, M. & Morbidelli, M. (2018), 'Proteomic analysis of micro-scale bioreactors as scale-down model for a mAb producing CHO industrial fed-batch platform', *Journal of Biotechnology* 279, 27–36. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2018.04.015>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Proteomic+analysis+of+micro-scale+bioreactors+as+scale-down+model+for+a+mAb+producing+CHO+industrial+fed-batch+platform&publication+year=2018&author=Bertrand+V.&author=Vogg+S.&author=Villiger+T.+K.&author=Stettler+M.&author=Broly+H.&author=Soos+M.&author=Morbidelli+M.&journal=Journal+of+Biotechnology&volume=279&pages=27-36)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/29719200>)
- Beyer, B., Schuster, M., Jungbauer, A. & Lingg, N. (2018), 'Microheterogeneity of recombinant antibodies: Analytics and functional impact', *Biotechnology Journal* 13(1), 1–11.
CrossRef (<http://dx.doi.org/10.1002/biot.201700476>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Microheterogeneity+of+recombinant+antibodies:+Analytics+and+functional+impact&publication+year=2018&author=Beyer+B.&author=Schuster+M.&author=Jungbauer+A.&author=Lingg+N.&journal=Biotechnology+Journal&volume=13&pages=1-11)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28862393>)
- Bibila, T. A. & Robinson, D. K. (1995), 'In pursuit of the optimal fed-batch process for monoclonal antibody production', *Biotechnology Progress* 11(1), 1–13.
CrossRef (<http://dx.doi.org/10.1021/bp00031a001>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=In+pursuit+of+the+optimal+fed-batch+process+for+monoclonal+antibody+production&publication+year=1995&author=Bibila+T.+A.&author=Robinson+D.+K.&journal=Biotechnology+Progress&volume=11&doi=10.1021/bp00031a001)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/7765983>)
- Bielser, J.-M., Chappuis, L., Xiao, Y., Souquet, J., Broly, H. & Morbidelli, M. (2019a), 'Perfusion cell culture for the production of conjugated recombinant fusion proteins reduces clipping and quality heterogeneity compared to batch-mode processes', *Journal of Biotechnology* 302, 26–31. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2019.06.006>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Perfusion+cell+culture+for+the+production+of+conjugated+recombinant+fusion+proteins+reduces+clipping+and+quality+heterogeneity+compared+to+batch-mode+processes&publication+year=2019&author=Bielser+J.-M.&author=Chappuis+L.&author=Xiao+Y.&author=Souquet+J.&author=Broly+H.&author=Morbidelli+M.&journal=Journal+of+Biotechnology&volume=302&doi=10.1016/j.jbiotec.2019.06.006)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/31207262>)

-
- Brunner, M., Doppler, P., Klein, T., Herwig, C. & Fricke, J. (2018), 'Elevated pCO₂ affects the lactate metabolic shift in CHO cell culture processes', *Engineering in Life Sciences* 18(3), 204–214.
CrossRef (<http://dx.doi.org/10.1002/elsc.201700131>) |
Google Scholar ([https://scholar.google.com/scholar_lookup?title=Elevated+pCO₂+affects+the+lactate+metabolic+shift+in+CHO+cell+culture+processes&publication+year=2018&author=Brunner+M.&author=Doppler+P.&author=Klein+T.&author=Hen+214](https://scholar.google.com/scholar_lookup?title=Elevated+pCO2+affects+the+lactate+metabolic+shift+in+CHO+cell+culture+processes&publication+year=2018&author=Brunner+M.&author=Doppler+P.&author=Klein+T.&author=Hen+214))
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/32624899>)
-
- Buckley, K. & Ryder, A. G. (2017), 'Applications of Raman spectroscopy in biopharmaceutical manufacturing: A short review', *Applied Spectroscopy* 71(6), 1085–1116.
CrossRef (<http://dx.doi.org/10.1177/0003702817703270>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Applications+of+Raman+spectroscopy+in+biopharmaceutical+manufacturing:+A+short+review&publication+year=2017&author=Buckley+K.&author=Ryder+A.+G.&journal=Applied+Sp+1116)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28534676>)
-
- Bujalski, W., Nienow, A., Chatwin, S. & Cooke, M. (1987), 'The dependency on scale of power numbers of Rushton disc turbines', *Chemical Engineering Science* 42(2), 317–326.
CrossRef ([http://dx.doi.org/10.1016/0009-2509\(87\)85061-3](http://dx.doi.org/10.1016/0009-2509(87)85061-3)) |
Google Scholar ([https://scholar.google.com/scholar_lookup?title=The+dependency+on+scale+of+power+numbers+of+Rushton+disc+turbines&publication+year=1987&author=Bujalski+W.&author=Nienow+A.&author=Chatwin+S.&author=Cooke+M.+8+2509\(87\)85061-3&pages=317-326](https://scholar.google.com/scholar_lookup?title=The+dependency+on+scale+of+power+numbers+of+Rushton+disc+turbines&publication+year=1987&author=Bujalski+W.&author=Nienow+A.&author=Chatwin+S.&author=Cooke+M.+8+2509(87)85061-3&pages=317-326))
-
- Bunnak, P., Allmendinger, R., Ramasamy, S. V., Lettieri, P. & Titchener-Hooker, N. J. (2016), 'Life-cycle and cost of goods assessment of fed-batch and perfusion-based manufacturing processes for mAbs', *Biotechnology Progress* 32(5), 1324–1335. CrossRef (<http://dx.doi.org/10.1002/btpr.2323>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Life-cycle+and+cost+of+goods+assessment+of+fed-batch+and+perfusion-based+manufacturing+processes+for+mAbs&publication+year=2016&author=Bunnak+P.&author=Allmendinger+R.&author=Ramasamy+S.+V.&author=Lettieri+P.&author=Titchener-Hooker+N.+J.&journal=Biotechnology+Progress&volume=32&doi=10.1002/btpr.2323&pages=1324-1335)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27390260>)
-
- Cao, D.-S., Xu, Q.-S., Liang, Y.-Z., Chen, X. & Li, H.-D. (2010), 'Prediction of aqueous solubility of druglike organic compounds using partial least squares, back-propagation network and support vector machine', *Journal of Chemometrics* 24, 584–595.
Google Scholar (https://scholar.google.com/scholar_lookup?title=Prediction+of+aqueous+solubility+of+druglike+organic+compounds+using+partial+least+squares+back-propagation+network+and+support+vector+machine&publication+year=2010&author=Cao+D.-S.&author=Xu+Q.-S.&author=Liang+Y.-Z.&author=Chen+X.&author=Li+H.-D.&journal=Journal+of+Chemometrics&volume=24&pages=584-595)
-
- Caplice, E. & Fitzgerald, G. F. (1999), 'Food fermentations: Role of microorganisms in food production and preservation', *International Journal of Food Microbiology* 50(1–2), 131–149.
CrossRef ([http://dx.doi.org/10.1016/S0168-1605\(99\)00082-3](http://dx.doi.org/10.1016/S0168-1605(99)00082-3)) |
Google Scholar ([https://scholar.google.com/scholar_lookup?title=Food+fermentations:+Role+of+microorganisms+in+food+production+and+preservation&publication+year=1999&author=Caplice+E.&author=Fitzgerald+G.+F.&journal=International+Jc+1605\(99\)00082-3&pages=131-149](https://scholar.google.com/scholar_lookup?title=Food+fermentations:+Role+of+microorganisms+in+food+production+and+preservation&publication+year=1999&author=Caplice+E.&author=Fitzgerald+G.+F.&journal=International+Jc+1605(99)00082-3&pages=131-149))
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/10488849>)
-
- Carrondo, M. J., Alves, P. M., Carinhas, N., Glassey, J., Hesse, F., Merten, O. W., Micheletti, M., Noll, T., Oliveira, R., Reichl, U., Staby, A., Teixeira, A. P., Weichert, H. & Mandenius, C. F. (2012), 'How can measurement, monitoring, modeling and control advance cell culture in industrial biotechnology?', *Biotechnology Journal* 7(12), 1522–1529. CrossRef (<http://dx.doi.org/10.1002/biot.201200226>)
|
Google Scholar (https://scholar.google.com/scholar_lookup?title=How+can+measurement+monitoring+modeling+and+control+advance+cell+culture+in+industrial+biotechnology?&publication+year=2012&author=Carrondo+M.+J.&author=Alves+P.+M.&author=Carinhas+N.&author=Glassey+J.&author=Hesse+F.&author=Merten+O.+W.&author=Micheletti+M.&author=1529)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22949408>)
-
- Carvell, J. P. & Dowd, J. E. (2006), 'On-line measurements and control of viable cell density in cell culture manufacturing processes using radio-frequency impedance', *Cytotechnology* 50(1–3), 35–48.
CrossRef (<http://dx.doi.org/10.1007/s10616-005-3974-x>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=On-line+measurements+and+control+of+viable+cell+density+in+cell+culture+manufacturing+processes+using+radio-frequency+impedance&publication+year=2006&author=Carvell+J.+P.&author=Dowd+J.+E.&journal=Cytotechnology&volume=50&doi=10.1007/s10616-005-3974-x&pages=35-48)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19003069>)
-
- Cervera, A. E., Petersen, N., Lantz, A. E., Larsen, A. & Gernaey, K. V. (2009), 'Application of near-infrared spectroscopy for monitoring and control of cell culture and fermentation', *Biotechnology Progress* 25(6), 1561–1581.
Google Scholar (https://scholar.google.com/scholar_lookup?title=Application+of+near-infrared+spectroscopy+for+monitoring+and+control+of+cell+culture+and+fermentation&publication+year=2009&author=Cervera+A.+E.&author=Petersen+N.&author=Lantz+A.+E.&author+1581)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19787698>)
-
- Chakrabarty, A., Buzzard, G. T. & Rundell, A. E. (2013), 'Model-based design of experiments for cellular processes', *Wiley Interdisciplinary Reviews: Systems Biology and Medicine* 5(2), 181–203.
Google Scholar (https://scholar.google.com/scholar_lookup?title=Model-based+design+of+experiments+for+cellular+processes&publication+year=2013&author=Chakrabarty+A.&author=Buzzard+G.+T.&author=Rundell+A.+E.&journal=Wiley+Interdisciplinary+Ri+203)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23293047>)
-
- Chandrasekharan, K. & Calderbank, P. (1981), 'Further observations on the scale-up of aerated mixing vessels', *Chemical Engineering Science* 36(5), 818–823.
CrossRef ([http://dx.doi.org/10.1016/0009-2509\(81\)85033-6](http://dx.doi.org/10.1016/0009-2509(81)85033-6)) |
Google Scholar ([https://scholar.google.com/scholar_lookup?title=Further+observations+on+the+scale-up+of+aerated+mixing+vessels&publication+year=1981&author=Chandrasekharan+K.&author=Calderbank+P.&journal=Chemical+Engineering+Science&volume=36&doi=10.1016/0009-2509\(81\)85033-6&pages=818-823](https://scholar.google.com/scholar_lookup?title=Further+observations+on+the+scale-up+of+aerated+mixing+vessels&publication+year=1981&author=Chandrasekharan+K.&author=Calderbank+P.&journal=Chemical+Engineering+Science&volume=36&doi=10.1016/0009-2509(81)85033-6&pages=818-823))
-
- Chen, C., Wong, H. E. & Goudar, C. T. (2018), 'Upstream process intensification and continuous manufacturing', *Current Opinion in Chemical Engineering* 22, 191–198.
CrossRef (<http://dx.doi.org/10.1016/j.coche.2018.10.006>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Upstream+process+intensification+and+continuous+manufacturing&publication+year=2018&author=Chen+C.&author=Wong+H.+E.&author=Goudar+C.+T.&journal=Current+Opinion+198)

- Chen, L., Nguang, S. K., Chen, X. D. & Li, X. M. (2004), 'Modelling and optimization of fed-batch fermentation processes using dynamic neural networks and genetic algorithms', *Biochemical Engineering Journal* 22(1), 51–61. CrossRef (<http://dx.doi.org/10.1016/j.bej.2004.07.012>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Modelling+and+optimization+of+fed-batch+fermentation+processes+using+dynamic+neural+networks+and+genetic+algorithms&publication+year=2004&author=Chen+L.&author=Nguang+S.+K.&author=Chen+X.+D.&author=61)
- Chen, T.-T. (2013), 'Immuno-oncology', *Journal for ImmunoTherapy of Cancer* 1(18), 1–9. CrossRef (<http://dx.doi.org/10.1186/2051-1426-1-18>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Immuno-oncology&publication+year=2013&author=Chen+T.-T.&journal=Journal+for+ImmunoTherapy+of+Cancer&volume=1&doi=10.1186/2051-1426-1-18&pages=1-9) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24829754>)
- Chisti, Y. (2000), 'Animal-cell damage in sparged bioreactors', *Trends in Biotechnology* 18(10), 420–432. CrossRef ([http://dx.doi.org/10.1016/S0167-7799\(00\)01474-8](http://dx.doi.org/10.1016/S0167-7799(00)01474-8)) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Animal-cell+damage+in+sparged+bioreactors&publication+year=2000&author=Chisti+Y.&journal=Trends+in+Biotechnology&volume=18&doi=10.1016/S0167-7799\(00\)01474-8&pages=420-432](https://scholar.google.com/scholar_lookup?title=Animal-cell+damage+in+sparged+bioreactors&publication+year=2000&author=Chisti+Y.&journal=Trends+in+Biotechnology&volume=18&doi=10.1016/S0167-7799(00)01474-8&pages=420-432)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/10998508>)
- Chisti, Y. (2001), 'Hydrodynamic damage to animal cells', *Critical Reviews in Biotechnology* 21(2), 67–110. CrossRef (<http://dx.doi.org/10.1080/20013891081692>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Hydrodynamic+damage+to+animal+cells&publication+year=2001&author=Chisti+Y.&journal=Critical+Reviews+in+Biotechnology&volume=21&doi=10.1080/20013891081692&pages=67-110) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/11451047>)
- Chotteau, V. (2017), 'Process development in screening scale bioreactors and perspectives for very high cell density perfusion', *Integrated Continuous Biomanufacturing III*, Suzanne Farid, University College London, United Kingdom Chetan Goudar, Amgen, USA Paula Alves, IBET, Portugal Veena Warikoo, Axcella Health, Inc., USA Eds, ECI Symposium Series. p. 10691. Google Scholar (https://scholar.google.com/scholar_lookup?title=Integrated+Continuous+Biomanufacturing+III&publication+year=2017&author=Chotteau+V.)
- Chu, L. & Robinson, D. K. (2001), 'Industrial choices for protein production by large-scale cell culture', *Current Opinion in Biotechnology* 12(2), 180–187. CrossRef ([http://dx.doi.org/10.1016/S0958-1669\(00\)00197-X](http://dx.doi.org/10.1016/S0958-1669(00)00197-X)) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Industrial+choices+for+protein+production+by+large-scale+cell+culture&publication+year=2001&author=Chu+L.&author=Robinson+D.+K.&journal=Current+Opinion+in+Biotechnology&volume=12&doi=10.1016/S0958-1669\(00\)00197-X&pages=180-187](https://scholar.google.com/scholar_lookup?title=Industrial+choices+for+protein+production+by+large-scale+cell+culture&publication+year=2001&author=Chu+L.&author=Robinson+D.+K.&journal=Current+Opinion+in+Biotechnology&volume=12&doi=10.1016/S0958-1669(00)00197-X&pages=180-187)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/11287235>)
- Chugh, P. & Roy, V. (2014), 'Biosimilars: Current scientific and regulatory considerations', *Current Clinical Pharmacology* 9(1), 53–63. CrossRef (<http://dx.doi.org/10.2174/15748847113089990066>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Biosimilars:+Current+scientific+and+regulatory+considerations&publication+year=2014&author=Chugh+P.&author=Roy+V.&journal=Current+Clinical+Pharmacology&volume=9&doi=163) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23952143>)
- Chuppa, S., Tsai, Y. S., Yoon, S., Shackelford, S., Rozales, C., Bhat, R., Tsay, G., Matanguihan, C., Konstantinov, K. & Naveh, D. (1997), 'Fermentor temperature as a tool for control of high-density perfusion cultures of mammalian cells', *Biotechnology and Bioengineering* 55(2), 328–338. CrossRef ([http://dx.doi.org/10.1002/\(SICI\)1097-0290\(19970720\)55:2<328::AID-BIT10>3.0.CO;2-D](http://dx.doi.org/10.1002/(SICI)1097-0290(19970720)55:2<328::AID-BIT10>3.0.CO;2-D)) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Fermentor+temperature+as+a+tool+for+control+of+high-density+perfusion+cultures+of+mammalian+cells&publication+year=1997&author=Chuppa+S.&author=Tsai+Y.+S.&author=Yoon+S.&author=Shackelford+S.&author=Rozales+C.&author=Bh0290\(19970720\)55:2%3C328::AID-BIT10%3E3.0.CO;2-D&pages=328-338](https://scholar.google.com/scholar_lookup?title=Fermentor+temperature+as+a+tool+for+control+of+high-density+perfusion+cultures+of+mammalian+cells&publication+year=1997&author=Chuppa+S.&author=Tsai+Y.+S.&author=Yoon+S.&author=Shackelford+S.&author=Rozales+C.&author=Bh0290(19970720)55:2%3C328::AID-BIT10%3E3.0.CO;2-D&pages=328-338)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18636491>)
- Clincke, M. F., Mölleryd, C., Samani, P. K., Lindskog, E., Fäldt, E., Walsh, K. & Chotteau, V. (2013b), 'Very high density of Chinese hamster ovary cells in perfusion by alternating tangential flow or tangential flow filtration in WAVE bioreactor, Part II: Applications for antibody production and cryopreservation', *Biotechnology Progress* 29(3), 768–777. CrossRef (<http://dx.doi.org/10.1002/btpr.1703>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Very+high+density+of+Chinese+hamster+ovary+cells+in+perfusion+by+alternating+tangential+flow+or+tangential+flow+filtration+in+WAVE+bioreactor+Part+II:+Applications+for+antib777) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23436783>)
- Clincke, M. F., Mölleryd, C., Zhang, Y., Lindskog, E., Walsh, K. & Chotteau, V. (2011), 'Study of a recombinant CHO cell line producing a monoclonal antibody by ATF or TFF external filter perfusion in a WAVE Bioreactor', *BMC Proceedings* 5(8), 105. CrossRef (<http://dx.doi.org/10.1186/1753-6561-5-S8-P105>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Study+of+a+recombinant+CHO+cell+line+producing+a+monoclonal+antibody+by+ATF+or+TFF+external+filter+perfusion+in+a+WAVE+Bioreactor&publication+year=2011&author=Clinck6561-5-S8-P105) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22373105>)
- Clincke, M. F., Mölleryd, C., Zhang, Y., Lindskog, E., Walsh, K. & Chotteau, V. (2013a), 'Very high density of CHO cells in perfusion by ATF or TFF in WAVE bioreactor, Part I: Effect of the cell density on the process', *Biotechnology Progress* 29(3), 754–767. CrossRef (<http://dx.doi.org/10.1002/btpr.1704>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Very+high+density+of+CHO+cells+in+perfusion+by+ATF+or+TFF+in+WAVE+bioreactor+Part+I:+Effect+of+the+cell+density+on+the+process&publication+year=2013&author=Clincke+M.+767) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23436789>)
- Coffman, J., Lin, H., Wang, S., Godfrey, S., Orozco, R., Yildirim, S., Salm, J., Hiller, G., Gagnon, M., Farnier, R., Kottmeier, B. & Sullivan, D. (2017), 'Balancing continuous, integrated, and batch processing', in *Integrated Continuous Biomanufacturing III*. Google Scholar (https://scholar.google.com/scholar_lookup?title=Balancing+continuous+integrated+and+batch+processing&publication+year=2017&author=Coffman+J.&author=Lin+H.&author=Wang+S.&author=Godfrey+S.&author=Orozco+R.+a9)
- Cohen, S. N., Chang, A. C. Y. & Hsu, L. (1972), 'Non-chromosomal antibiotic resistance in bacteria: genetic transformation of *Escherichia coli* by R-factor DNA', *Proceedings of the National Academy of Sciences of the United States of America* 69, 2110–2114. CrossRef (<http://dx.doi.org/10.1073/pnas.69.8.2110>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Non-chromosomal+antibiotic+resistance+in+bacteria:+genetic+transformation+of+Escherichia+coli+by+R-factor+DNA&publication+year=1972&author=Cohen+S.+N.&author=Chang+A.+C.+Y.&author=Hsu+L.&journal=Proceedings+of+the+National+Academy+of+Sciences+of+the+United+States+o2114)

-
- Cohen, S. N., Chang, A. C. Y., Boyer, H. W. & Helling, R. B. (1973), 'Construction of biologically functional bacterial plasmids in vitro', *Proceedings of the National Academy of Sciences of the United States of America* 70(11), 3240–3244. [CrossRef \(http://dx.doi.org/10.1073/pnas.70.11.3240\)](http://dx.doi.org/10.1073/pnas.70.11.3240) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Construction+of+biologically+functional+bacterial+plasmids+in+vivo&publication+year=1973&author=Cohen+S.+N.&author=Chang+A.+C.+Y.&author=Boyer+H.+W.&author=Helling+R.3244\)](https://scholar.google.com/scholar_lookup?title=Construction+of+biologically+functional+bacterial+plasmids+in+vivo&publication+year=1973&author=Cohen+S.+N.&author=Chang+A.+C.+Y.&author=Boyer+H.+W.&author=Helling+R.3244) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/4594039\)](https://www.ncbi.nlm.nih.gov/pubmed/4594039)
-
- Colosimo, A., Goncz, K., Holmes, A., Kunzelmann, K., Bennet, M. & Gruenert, D. (2000), 'Transfer and expression of foreign genes in mammalian cells', *BioTechniques* 29(2), 314–331. [CrossRef \(http://dx.doi.org/10.2144/00292rv01\)](http://dx.doi.org/10.2144/00292rv01) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Transfer+and+expression+of+foreign+genes+in+mammalian+cells&publication+year=2000&author=Colosimo+A.&author=Goncz+K.&author=Holmes+A.&author=Kunzelmann+K.&author=Bennet+M.&author=Gruenert+D.331\)](https://scholar.google.com/scholar_lookup?title=Transfer+and+expression+of+foreign+genes+in+mammalian+cells&publication+year=2000&author=Colosimo+A.&author=Goncz+K.&author=Holmes+A.&author=Kunzelmann+K.&author=Bennet+M.&author=Gruenert+D.331) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/10948433\)](https://www.ncbi.nlm.nih.gov/pubmed/10948433)
-
- Coronel, J., Klausning, S., Heinrich, C., Noll, T., Figueredo-Cardero, A. & Castilho, L. R. (2016), 'Valeric acid supplementation combined to mild hypothermia increases productivity in CHO cell cultivations', *Biochemical Engineering Journal* 114, 101–109. [CrossRef \(http://dx.doi.org/10.1016/j.bej.2016.06.031\)](http://dx.doi.org/10.1016/j.bej.2016.06.031) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Valeric+acid+supplementation+combined+to+mild+hypothermia+increases+productivity+in+CHO+cell+cultivations&publication+year=2016&author=Coronel+J.&author=Klausning+S.&author=Heinrich+C.&author=Noll+T.&author=Figueredo+Cardero+A.&author=Castilho+L.+R.&journal=Biochemical+Engineering+Journal&volume=114&doi=10.1016/j.bej.2016.06.031&pages=101-109\)](https://scholar.google.com/scholar_lookup?title=Valeric+acid+supplementation+combined+to+mild+hypothermia+increases+productivity+in+CHO+cell+cultivations&publication+year=2016&author=Coronel+J.&author=Klausning+S.&author=Heinrich+C.&author=Noll+T.&author=Figueredo+Cardero+A.&author=Castilho+L.+R.&journal=Biochemical+Engineering+Journal&volume=114&doi=10.1016/j.bej.2016.06.031&pages=101-109)
-
- Croughan, M. S., Konstantinov, K. B. & Cooney, C. (2015), 'The future of industrial bioprocessing: Batch or continuous?', *Biotechnology and Bioengineering* 112(4), 648–651. [CrossRef \(http://dx.doi.org/10.1002/bit.25529\)](http://dx.doi.org/10.1002/bit.25529) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=The+future+of+industrial+bioprocessing:+Batch+or+continuous&publication+year=2015&author=Croughan+M.+S.&author=Konstantinov+K.+B.&author=Cooney+C.&journal=Biotechnology+and+Bioengineering&volume=112&doi=10.1002/bit.25529&pages=648-651\)](https://scholar.google.com/scholar_lookup?title=The+future+of+industrial+bioprocessing:+Batch+or+continuous&publication+year=2015&author=Croughan+M.+S.&author=Konstantinov+K.+B.&author=Cooney+C.&journal=Biotechnology+and+Bioengineering&volume=112&doi=10.1002/bit.25529&pages=648-651) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/25694022\)](https://www.ncbi.nlm.nih.gov/pubmed/25694022)
-
- Davey, C. L., Davey, H. M., Kell, D. B. & Todd, R. W. (1993), 'Introduction to the dielectric estimation of cellular biomass in real time, with special emphasis on measurements at high volume fractions', *Analytica Chimica Acta* 279(1), 155–161. [CrossRef \(http://dx.doi.org/10.1016/0003-2670\(93\)85078-X\)](http://dx.doi.org/10.1016/0003-2670(93)85078-X) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Introduction+to+the+dielectric+estimation+of+cellular+biomass+in+real+time+with+special+emphasis+on+measurements+at+high+volume+fractions&publication+year=1993&author=Davey+C.L.&author=Davey+H.M.&author=Kell+D.B.&author=Todd+R.W.&journal=Analytica+Chimica+Acta&volume=279&doi=10.1016/0003-2670\(93\)85078-X&pages=155-161\)](https://scholar.google.com/scholar_lookup?title=Introduction+to+the+dielectric+estimation+of+cellular+biomass+in+real+time+with+special+emphasis+on+measurements+at+high+volume+fractions&publication+year=1993&author=Davey+C.L.&author=Davey+H.M.&author=Kell+D.B.&author=Todd+R.W.&journal=Analytica+Chimica+Acta&volume=279&doi=10.1016/0003-2670(93)85078-X&pages=155-161)
-
- Davis, D., Delia, S., Safc, L., Ross, S., Lyons, D. & Hodzic, I. (2015), 'Modeling perfusion at small scale using ambr[®] 15', in *ECI Digital Archives*. [Google Scholar \(https://scholar.google.com/scholar?q=Davis+D.+Delia+S.+Safc+L.+Ross+S.+Lyons+D.+&+Hodzic+I.+&+2015+&+E2%80%98Modeling+perfusion+at+small+scale+using+ambr%C2%A9+15+E2%80%99+in+ECI+Digital+Archives.\)](https://scholar.google.com/scholar?q=Davis+D.+Delia+S.+Safc+L.+Ross+S.+Lyons+D.+&+Hodzic+I.+&+2015+&+E2%80%98Modeling+perfusion+at+small+scale+using+ambr%C2%A9+15+E2%80%99+in+ECI+Digital+Archives.)
-
- De Jesus, M. J., Girard, P., Bourgeois, M., Baumgartner, G., Jacko, B., Amstutz, H. & Wurm, F. M. (2004), 'TubeSpin satellites: A fast track approach for process development with animal cells using shaking technology', *Biochemical Engineering Journal* 17(3), 217–223. [CrossRef \(http://dx.doi.org/10.1016/S1369-703X\(03\)00180-3\)](http://dx.doi.org/10.1016/S1369-703X(03)00180-3) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=TubeSpin+satellites:+A+fast+track+approach+for+process+development+with+animal+cells+using+shaking+technology&publication+year=2004&author=De+Jesus+M.+J.&author=Girard+P.&author=Bourgeois+M.&author=Baumgartner+G.&author=Jacko+B.&author=Amstutz+H.&author=Wurm+F.M.&journal=Biochemical+Engineering+Journal&volume=17&doi=10.1016/S1369-703X\(03\)00180-3&pages=217-223\)](https://scholar.google.com/scholar_lookup?title=TubeSpin+satellites:+A+fast+track+approach+for+process+development+with+animal+cells+using+shaking+technology&publication+year=2004&author=De+Jesus+M.+J.&author=Girard+P.&author=Bourgeois+M.&author=Baumgartner+G.&author=Jacko+B.&author=Amstutz+H.&author=Wurm+F.M.&journal=Biochemical+Engineering+Journal&volume=17&doi=10.1016/S1369-703X(03)00180-3&pages=217-223)
-
- Demain, A. L. (2007), 'The business of biotechnology', *Industrial Biotechnology* 3(3), 269–283. [CrossRef \(http://dx.doi.org/10.1089/ind.2007.3.269\)](http://dx.doi.org/10.1089/ind.2007.3.269) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=The+business+of+biotechnology&publication+year=2007&author=Demain+A.+L.&journal=Industrial+Biotechnology&volume=3&doi=10.1089/ind.2007.3.269&pages=269-283\)](https://scholar.google.com/scholar_lookup?title=The+business+of+biotechnology&publication+year=2007&author=Demain+A.+L.&journal=Industrial+Biotechnology&volume=3&doi=10.1089/ind.2007.3.269&pages=269-283)
-
- Deschênes, J.-S., Desbiens, A., Perrier, M. & Kamen, A. (2006), 'Use of cell bleed in a high cell density perfusion culture and multivariable control of biomass and metabolite concentrations', *Asia-Pacific Journal of Chemical Engineering* 1(1–2), 82–91. [CrossRef \(http://dx.doi.org/10.1002/apj.10\)](http://dx.doi.org/10.1002/apj.10) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Use+of+cell+bleed+in+a+high+cell+density+perfusion+culture+and+multivariable+control+of+biomass+and+metabolite+concentrations&publication+year=2006&author=Desch%C3%A9nes+J.-S.&author=Desbiens+A.&author=Perrier+M.&author=Kamen+A.&journal=Asia-Pacific+Journal+of+Chemical+Engineering&volume=1&doi=10.1002/apj.10&pages=82-91\)](https://scholar.google.com/scholar_lookup?title=Use+of+cell+bleed+in+a+high+cell+density+perfusion+culture+and+multivariable+control+of+biomass+and+metabolite+concentrations&publication+year=2006&author=Desch%C3%A9nes+J.-S.&author=Desbiens+A.&author=Perrier+M.&author=Kamen+A.&journal=Asia-Pacific+Journal+of+Chemical+Engineering&volume=1&doi=10.1002/apj.10&pages=82-91)
-
- Deshpande, N. S. & Barigou, M. (1999), 'Performance characteristics of novel mechanical foam breakers in a stirred tank reactor', *Journal of Chemical Technology and Biotechnology* 987(May), 979–987. [CrossRef \(http://dx.doi.org/10.1002/\(SICI\)1097-4660\(199910\)74:10<979::AID-JCTB139>3.0.CO;2-8\)](http://dx.doi.org/10.1002/(SICI)1097-4660(199910)74:10<979::AID-JCTB139>3.0.CO;2-8) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Performance+characteristics+of+novel+mechanical+foam+breakers+in+a+stirred+tank+reactor&publication+year=1999&author=Deshpande+N.S.&author=Barigou+M.&journal=Journal+of+Chemical+Technology+and+Biotechnology&volume=987&doi=10.1002/\(SICI\)1097-4660\(199910\)74:10%3C979::AID-JCTB139%3E3.0.CO;2-8&pages=979-987\)](https://scholar.google.com/scholar_lookup?title=Performance+characteristics+of+novel+mechanical+foam+breakers+in+a+stirred+tank+reactor&publication+year=1999&author=Deshpande+N.S.&author=Barigou+M.&journal=Journal+of+Chemical+Technology+and+Biotechnology&volume=987&doi=10.1002/(SICI)1097-4660(199910)74:10%3C979::AID-JCTB139%3E3.0.CO;2-8&pages=979-987)
-
- Deshpande, R. R. & Heinzle, E. (2004), 'On-line oxygen uptake rate and culture viability measurement of animal cell culture using microplates with integrated oxygen sensors', *Biotechnology Letters* 26(9), 763–767. [CrossRef \(http://dx.doi.org/10.1023/B:BILE.0000024101.57683.6d\)](http://dx.doi.org/10.1023/B:BILE.0000024101.57683.6d) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=On-line+oxygen+uptake+rate+and+culture+viability+measurement+of+animal+cell+culture+using+microplates+with+integrated+oxygen+sensors&publication+year=2004&author=Deshpande+R.R.&author=Heinzle+E.&journal=Biotechnology+Letters&volume=26&doi=10.1023/B:BILE.0000024101.57683.6d\)](https://scholar.google.com/scholar_lookup?title=On-line+oxygen+uptake+rate+and+culture+viability+measurement+of+animal+cell+culture+using+microplates+with+integrated+oxygen+sensors&publication+year=2004&author=Deshpande+R.R.&author=Heinzle+E.&journal=Biotechnology+Letters&volume=26&doi=10.1023/B:BILE.0000024101.57683.6d) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/15195979\)](https://www.ncbi.nlm.nih.gov/pubmed/15195979)
-
- D'Este, M., Alvarado-Morales, M. & Angelidaki, I. (2017), 'Amino acids production focusing on fermentation technologies: A review', *Biotechnology Advances* 36(1), 14–25. [CrossRef \(http://dx.doi.org/10.1016/j.biotechadv.2017.09.001\)](http://dx.doi.org/10.1016/j.biotechadv.2017.09.001) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Amino+acids+production+focusing+on+fermentation+technologies:+A+review&publication+year=2017&author=D%E2%80%99Este+M.&author=Alvarado-Morales+M.&author=Angelidaki+I.&journal=Biotechnology+Advances&volume=36&doi=10.1016/j.biotechadv.2017.09.001&pages=14-25\)](https://scholar.google.com/scholar_lookup?title=Amino+acids+production+focusing+on+fermentation+technologies:+A+review&publication+year=2017&author=D%E2%80%99Este+M.&author=Alvarado-Morales+M.&author=Angelidaki+I.&journal=Biotechnology+Advances&volume=36&doi=10.1016/j.biotechadv.2017.09.001&pages=14-25) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/28888551\)](https://www.ncbi.nlm.nih.gov/pubmed/28888551)
-
- Devi, T. T. & Kumar, B. (2017), 'Mass transfer and power characteristics of stirred tank with Rushton and curved blade impeller', *Engineering Science and Technology, an International Journal* 20(2), 730–737. [CrossRef \(http://dx.doi.org/10.1016/j.jestch.2016.11.005\)](http://dx.doi.org/10.1016/j.jestch.2016.11.005) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Mass-transfer+and+power+characteristics+of+stirred+tank+with+Rushton+and+curved+blade+impeller&publication+year=2017&author=Devi+T.+T.&author=Kumar+B.&journal=Engineering+Science+and+Technology+an+International+Journal&volume=20&doi=10.1016/j.jestch.2016.11.005\)](https://scholar.google.com/scholar_lookup?title=Mass-transfer+and+power+characteristics+of+stirred+tank+with+Rushton+and+curved+blade+impeller&publication+year=2017&author=Devi+T.+T.&author=Kumar+B.&journal=Engineering+Science+and+Technology+an+International+Journal&volume=20&doi=10.1016/j.jestch.2016.11.005)

- Dhir, S., Morrow, K. J., Rhinehart, R. R. & Wiesner, T. (2000), 'Dynamic optimization of hybridoma growth in a fed-batch bioreactor', *Biotechnology and Bioengineering* 67(2), 197–205. [CrossRef \(http://dx.doi.org/10.1002/\(SICI\)1097-0290\(20000120\)67:2<197::AID-BIT9>3.0.CO;2-W\)](http://dx.doi.org/10.1002/(SICI)1097-0290(20000120)67:2<197::AID-BIT9>3.0.CO;2-W) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Dynamic+optimization+of+hybridoma+growth+in+a+fed-batch+bioreactor&publication+year=2000&author=Dhir+S.&author=Morrow+K.+J.&author=Rhinehart+R.+R.&author=Wiesner+T.&journal=Biotechnology+and+Bioengineering&volume=67&0290\(20000120\)67:2%3C197::AID-BIT9%3E3.0.CO;2-W&pages=197-205\)](https://scholar.google.com/scholar_lookup?title=Dynamic+optimization+of+hybridoma+growth+in+a+fed-batch+bioreactor&publication+year=2000&author=Dhir+S.&author=Morrow+K.+J.&author=Rhinehart+R.+R.&author=Wiesner+T.&journal=Biotechnology+and+Bioengineering&volume=67&0290(20000120)67:2%3C197::AID-BIT9%3E3.0.CO;2-W&pages=197-205)
- Dorival-García, N. & Bones, J. (2017), 'Monitoring leachables from single-use bioreactor bags for mammalian cell culture by dispersive liquid-liquid microextraction followed by ultra high performance liquid chromatography quadrupole time of flight mass spectrometry', *Journal of Chromatography A* 1512, 51–60. [CrossRef \(http://dx.doi.org/10.1016/j.chroma.2017.06.077\)](http://dx.doi.org/10.1016/j.chroma.2017.06.077) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Monitoring+leachables+from+single-use+bioreactor+bags+for+mammalian+cell+culture+by+dispersive+liquid-liquid+microextraction+followed+by+ultra+high+performance+liquid+chromatography+quadrupole+time+of+flight+mass+spectrometry&publication+year=2017&author=Dorival-Garc%C3%ADa+N.&author=Bones+J.&journal=Journal+of+Chromatography+A&volume=1512&doi=10.1016/j.chroma.2017.06.077&pages=51-60\)](https://scholar.google.com/scholar_lookup?title=Monitoring+leachables+from+single-use+bioreactor+bags+for+mammalian+cell+culture+by+dispersive+liquid-liquid+microextraction+followed+by+ultra+high+performance+liquid+chromatography+quadrupole+time+of+flight+mass+spectrometry&publication+year=2017&author=Dorival-Garc%C3%ADa+N.&author=Bones+J.&journal=Journal+of+Chromatography+A&volume=1512&doi=10.1016/j.chroma.2017.06.077&pages=51-60) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/28720219\)](https://www.ncbi.nlm.nih.gov/pubmed/28720219)
- Dowd, J. E., Jubb, A., Kwok, K. E. & Piret, J. M. (2003), 'Optimization and control of perfusion cultures using a viable cell probe and cell specific perfusion rates', *Cytotechnology* 42(1), 35–45. [CrossRef \(http://dx.doi.org/10.1023/A:1026192228471\)](http://dx.doi.org/10.1023/A:1026192228471) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Optimization+and+control+of+perfusion+cultures+using+a+viable+cell+probe+and+cell+specific+perfusion+rates&publication+year=2003&author=Dowd+J.+E.&author=Jubb+A.&author=Kwok+K.+E.&author=Piret+J.+M.&journal=Cytotechnology&volume=42&doi=10.1023/A:1026192228471\)](https://scholar.google.com/scholar_lookup?title=Optimization+and+control+of+perfusion+cultures+using+a+viable+cell+probe+and+cell+specific+perfusion+rates&publication+year=2003&author=Dowd+J.+E.&author=Jubb+A.&author=Kwok+K.+E.&author=Piret+J.+M.&journal=Cytotechnology&volume=42&doi=10.1023/A:1026192228471) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/19002926\)](https://www.ncbi.nlm.nih.gov/pubmed/19002926)
- Dowd, J. E., Weber, I., Rodriguez, B., Piret, J. M. & Kwok, K. E. (1999), 'Predictive control of hollow-fiber bioreactors for the production of monoclonal antibodies', *Biotechnology and Bioengineering* 63(4), 484–492. [CrossRef \(http://dx.doi.org/10.1002/\(SICI\)1097-0290\(19990520\)63:4<484::AID-BIT12>3.0.CO;2-5\)](http://dx.doi.org/10.1002/(SICI)1097-0290(19990520)63:4<484::AID-BIT12>3.0.CO;2-5) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Predictive+control+of+hollow-fiber+bioreactors+for+the+production+of+monoclonal+antibodies&publication+year=1999&author=Dowd+J.+E.&author=Weber+I.&author=Rodriguez+B.&author=Piret+J.+M.&author=Kwok+K.+E.&journal=Biotechnology+and+Bioengineering&volume=63&doi=10.1002/\(SICI\)1097-0290\(19990520\)63:4%3C484::AID-BIT12%3E3.0.CO;2-5&pages=484-492\)](https://scholar.google.com/scholar_lookup?title=Predictive+control+of+hollow-fiber+bioreactors+for+the+production+of+monoclonal+antibodies&publication+year=1999&author=Dowd+J.+E.&author=Weber+I.&author=Rodriguez+B.&author=Piret+J.+M.&author=Kwok+K.+E.&journal=Biotechnology+and+Bioengineering&volume=63&doi=10.1002/(SICI)1097-0290(19990520)63:4%3C484::AID-BIT12%3E3.0.CO;2-5&pages=484-492) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/10099629\)](https://www.ncbi.nlm.nih.gov/pubmed/10099629)
- Du, Z., Treiber, D., Mccarter, J. D., Fomina-Yadlin, D., Saleem, R. A., Mccoy, R. E., Zhang, Y., Tharmalingam, T., Leith, M., Follstad, B. D., Dell, B., Grism, B., Zupke, C., Heath, C., Morris, A. E. & Reddy, P. (2015), 'Use of a small molecule cell cycle inhibitor to control cell growth and improve specific productivity and product quality of recombinant proteins in CHO cell cultures', *Biotechnology and Bioengineering* 112(1), 141–155. [CrossRef \(http://dx.doi.org/10.1002/bit.25332\)](http://dx.doi.org/10.1002/bit.25332) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Use+of+a+small+molecule+cell+cycle+inhibitor+to+control+cell+growth+and+improve+specific+productivity+and+product+quality+of+recombinant+proteins+in+CHO+cell+cultures&publication+year=2015&author=Du+Z.&author=Saleem+R.+A.&author=Mccoy+R.+E.&author=Zhang+Y.&author=Tharmalingam+T.&author=Leith+M.&author=Follstad+B.+D.&author=Dell+B.&author=Grism+B.&author=Zupke+C.&author=Heath+C.&author=Morris+A.+E.&author=Reddy+P.&journal=Biotechnology+and+Bioengineering&volume=112&doi=10.1002/bit.25332\)](https://scholar.google.com/scholar_lookup?title=Use+of+a+small+molecule+cell+cycle+inhibitor+to+control+cell+growth+and+improve+specific+productivity+and+product+quality+of+recombinant+proteins+in+CHO+cell+cultures&publication+year=2015&author=Du+Z.&author=Saleem+R.+A.&author=Mccoy+R.+E.&author=Zhang+Y.&author=Tharmalingam+T.&author=Leith+M.&author=Follstad+B.+D.&author=Dell+B.&author=Grism+B.&author=Zupke+C.&author=Heath+C.&author=Morris+A.+E.&author=Reddy+P.&journal=Biotechnology+and+Bioengineering&volume=112&doi=10.1002/bit.25332) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/25042542\)](https://www.ncbi.nlm.nih.gov/pubmed/25042542)
- Ducommun, P., Bolzonella, I., Marison, I., von Stockar, U. & Rhiel, M. (2002), 'Real-time in situ monitoring of freely suspended and immobilized cell cultures based on mid-infrared spectroscopic measurements', *Biotechnology and Bioengineering* 77(2), 174–185. [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Real-time+in+situ+monitoring+of+freely+suspended+and+immobilized+cell+cultures+based+on+mid-infrared+spectroscopic+measurements&publication+year=2002&author=Ducommun+P.&author=Bolzonella+I.&author=Marison+I.&author=von+Stockar+U.&author=Rhiel+M.&journal=Biotechnology+and+Bioengineering&volume=77&doi=10.1002/\(SICI\)1097-0290\(20020120\)77:2<174::AID-BIT77>3.0.CO;2-5\)](https://scholar.google.com/scholar_lookup?title=Real-time+in+situ+monitoring+of+freely+suspended+and+immobilized+cell+cultures+based+on+mid-infrared+spectroscopic+measurements&publication+year=2002&author=Ducommun+P.&author=Bolzonella+I.&author=Marison+I.&author=von+Stockar+U.&author=Rhiel+M.&journal=Biotechnology+and+Bioengineering&volume=77&doi=10.1002/(SICI)1097-0290(20020120)77:2<174::AID-BIT77>3.0.CO;2-5)
- Ducommun, P., Bolzonella, I., Rhiel, M., Pugeaud, P., Von Stockar, U. & Marison, I. W. (2001a), 'On-line determination of animal cell concentration', *Biotechnology and Bioengineering* 72(5), 515–522. [CrossRef \(http://dx.doi.org/10.1002/1097-0290\(20010305\)72:5<515::AID-BIT1015>3.0.CO;2-Q\)](http://dx.doi.org/10.1002/1097-0290(20010305)72:5<515::AID-BIT1015>3.0.CO;2-Q) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=On-line+determination+of+animal+cell+concentration&publication+year=2001&author=Ducommun+P.&author=Bolzonella+I.&author=Rhiel+M.&author=Pugeaud+P.&author=Stockar+U.&author=Marison+I.+W.&journal=Biotechnology+and+Bioengineering&volume=72&doi=10.1002/1097-0290\(20010305\)72:5%3C515::AID-BIT1015%3E3.0.CO;2-Q&pages=515-522\)](https://scholar.google.com/scholar_lookup?title=On-line+determination+of+animal+cell+concentration&publication+year=2001&author=Ducommun+P.&author=Bolzonella+I.&author=Rhiel+M.&author=Pugeaud+P.&author=Stockar+U.&author=Marison+I.+W.&journal=Biotechnology+and+Bioengineering&volume=72&doi=10.1002/1097-0290(20010305)72:5%3C515::AID-BIT1015%3E3.0.CO;2-Q&pages=515-522) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/11460241\)](https://www.ncbi.nlm.nih.gov/pubmed/11460241)
- Duetz, W. A. (2007), 'Microtiter plates as mini-bioreactors: Miniaturization of fermentation methods', *Trends in Microbiology* 15(10), 469–475. [CrossRef \(http://dx.doi.org/10.1016/j.tim.2007.09.004\)](http://dx.doi.org/10.1016/j.tim.2007.09.004) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Microtiter+plates+as+mini-bioreactors+Miniaturization+of+fermentation+methods&publication+year=2007&author=Duetz+W.+A.&journal=Trends+in+Microbiology&volume=15&doi=10.1016/j.tim.2007.09.004&page=475\)](https://scholar.google.com/scholar_lookup?title=Microtiter+plates+as+mini-bioreactors+Miniaturization+of+fermentation+methods&publication+year=2007&author=Duetz+W.+A.&journal=Trends+in+Microbiology&volume=15&doi=10.1016/j.tim.2007.09.004&page=475) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/17920276\)](https://www.ncbi.nlm.nih.gov/pubmed/17920276)
- Ecker, D. M., Jones, S. D. & Levine, H. L. (2015), 'The therapeutic monoclonal antibody market', *mAbs* 7(1), 9–14. [CrossRef \(http://dx.doi.org/10.4161/19420862.2015.989042\)](http://dx.doi.org/10.4161/19420862.2015.989042) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=The+therapeutic+monoclonal+antibody+market&publication+year=2015&author=Ecker+D.+M.&author=Jones+S.+D.&author=Levine+H.+L.&journal=mAbs&volume=7&doi=10.4161/19420862.2015.989042\)](https://scholar.google.com/scholar_lookup?title=The+therapeutic+monoclonal+antibody+market&publication+year=2015&author=Ecker+D.+M.&author=Jones+S.+D.&author=Levine+H.+L.&journal=mAbs&volume=7&doi=10.4161/19420862.2015.989042) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/25529996\)](https://www.ncbi.nlm.nih.gov/pubmed/25529996)
- Eibl, R., Kaiser, S., Lombriser, R. & Eibl, D. (2010), 'Disposable bioreactors: The current state-of-the-art and recommended applications in biotechnology', *Applied Microbiology and Biotechnology* 86(1), 41–49. [CrossRef \(http://dx.doi.org/10.1007/s00253-009-2422-9\)](http://dx.doi.org/10.1007/s00253-009-2422-9) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Disposable+bioreactors+The+current+state-of-the-art+and+recommended+applications+in+biotechnology&publication+year=2010&author=Eibl+R.&author=Kaiser+S.&author=Lombriser+R.&author=Eibl+D.&journal=Applied+Microbiology+and+Biotechnology&volume=86&doi=10.1007/s00253-009-2422-9&pages=41-49\)](https://scholar.google.com/scholar_lookup?title=Disposable+bioreactors+The+current+state-of-the-art+and+recommended+applications+in+biotechnology&publication+year=2010&author=Eibl+R.&author=Kaiser+S.&author=Lombriser+R.&author=Eibl+D.&journal=Applied+Microbiology+and+Biotechnology&volume=86&doi=10.1007/s00253-009-2422-9&pages=41-49) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/20094714\)](https://www.ncbi.nlm.nih.gov/pubmed/20094714)
- Eleftherios, P. (1991), 'Media additives for protecting freely suspended animal cells against agitation and aeration damage', *Tibtech* 9, 316–324. [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Media+additives+for+protecting+freely+suspended+animal+cells+against+agitation+and+aeration+damage&publication+year=1991&author=Eleftherios+P.&journal=Tibtech&volume=9&pages=316-324\)](https://scholar.google.com/scholar_lookup?title=Media+additives+for+protecting+freely+suspended+animal+cells+against+agitation+and+aeration+damage&publication+year=1991&author=Eleftherios+P.&journal=Tibtech&volume=9&pages=316-324)
- Eon-Duval, A., Gleixner, R., Valax, P., Soos, M., Neunstoecklin, B., Morbidelli, M. & Broly, H. (2013), 'Quality by design applied to a Fc-fusion protein: A case study', in *Therapeutic Fc-Fusion Proteins*, Wiley-VCH Verlag GmbH and Co. KGaA, Weinheim, pp. 155–189. [CrossRef \(http://dx.doi.org/10.1002/9783527675272.ch06\)](http://dx.doi.org/10.1002/9783527675272.ch06) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Therapeutic+Fc-Fusion+Proteins&publication+year=2013&author=Eon-Duval+A.&author=Gleixner+R.&author=Valax+P.&author=Soos+M.&author=Neunstoecklin+B.&author=Morbidelli+M.&author=Broly+H.&pages=155-189\)](https://scholar.google.com/scholar_lookup?title=Therapeutic+Fc-Fusion+Proteins&publication+year=2013&author=Eon-Duval+A.&author=Gleixner+R.&author=Valax+P.&author=Soos+M.&author=Neunstoecklin+B.&author=Morbidelli+M.&author=Broly+H.&pages=155-189)
- Eon-Duval, A., Valax, P., Solacroup, T., Broly, H., Gleixner, R., Strat, C. L. & Sutter, J. (2012), 'Application of the quality by design approach to the drug substance manufacturing process of an Fc fusion protein: Towards a global multi-step design space', *Journal of Pharmaceutical Sciences* 101(10), 3604–3618. [CrossRef \(http://dx.doi.org/10.1002/jps.23273\)](http://dx.doi.org/10.1002/jps.23273) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Application+of+the+quality+by+design+approach+to+the+drug+substance+manufacturing+process+of+an+Fc+fusion+protein+Towards+a+global+multi-step+design+space&publication+year=2012&author=Eon-Duval+A.&author=Valax+P.&author=Solacroup+T.&author=Broly+H.&author=Gleixner+R.&author=Strat+C.+L.&author=Sutter+J.&journal=Journal+of+Pharmaceutical+Sciences&volume=101&doi=10.1002/jps.23273\)](https://scholar.google.com/scholar_lookup?title=Application+of+the+quality+by+design+approach+to+the+drug+substance+manufacturing+process+of+an+Fc+fusion+protein+Towards+a+global+multi-step+design+space&publication+year=2012&author=Eon-Duval+A.&author=Valax+P.&author=Solacroup+T.&author=Broly+H.&author=Gleixner+R.&author=Strat+C.+L.&author=Sutter+J.&journal=Journal+of+Pharmaceutical+Sciences&volume=101&doi=10.1002/jps.23273) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/22327323\)](https://www.ncbi.nlm.nih.gov/pubmed/22327323)

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Application+of+the+quality+by+design+approach+to+the+drug+substance+manufacturing+process+of+an+Fc+fusion+protein:+Towards+a+global+multi%C3%A2%C5%81%C5%82step+Duval+A.&author=Valax+P.&author=Solacroup+T.&author=Broly+H.&author=Gleixner+R.&author=Strat+C.+L.&author=Sutter+J.&journal=Journal+of+Pharmaceutical+Sciences&volume=101

Escandar, G. M., Damiani, P. C., Goicoechea, H. C. & Olivieri, A. C. (2006), 'A review of multivariate calibration methods applied to biomedical analysis', *Microchemical Journal* 82(1), 29–42.

CrossRef (<http://dx.doi.org/10.1016/j.microc.2005.07.001>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=A+review+of+multivariate+calibration+methods+applied+to+biomedical+analysis&publication+year=2006&author=Escandar+G.+M.&author=Damiani+P.+C.&author=Goicoechea+H.+C.42)

Esmonde-White, K. A., Cuellar, M., Uerpmann, C., Lenain, B. & Lewis, I. R. (2017), 'Raman spectroscopy as a process analytical technology for pharmaceutical manufacturing and bioprocessing',

Analytical and Bioanalytical Chemistry 409(3), 637–649. CrossRef (<http://dx.doi.org/10.1007/s00216-016-9824-1>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Raman+spectroscopy+as+a+process+analytical+technology+for+pharmaceutical+manufacturing+and+bioprocessing&publication+year=2017&author=Esmonde-White+K.+A.&author=Cuellar+M.&author=Uerpmann+C.&author=Lenain+B.&author=Lewis+I.+R.&journal=Analytical+and+Bioanalytical+Chemistry&volume=409&doi=10.1007/s00216-016-9824-1&pages=637-649)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27491299>)

Feidl, F., Vogt, S., Wolf, M., Podobnik, M., Ruggeri, C., Ulmer, N., ... & Morbidelli, M. (2020). Process-wide control and automation of an integrated continuous manufacturing platform for antibodies.

Biotechnology and Bioengineering. CrossRef (<http://dx.doi.org/10.1002/bit.27296>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Process-

wide+control+and+automation+of+an+integrated+continuous+manufacturing+platform+for+antibodies&publication+year=2020&author=Feidl+F.&author=Vogt+S.&author=Wolf+M.&author=

Feidl, F. (2019), Digitalization Platform and Supervisory Control of a Continuous Integrated, PhD thesis, ETH Zürich.

Google Scholar ([https://scholar.google.com/scholar?q=Feidl+F.+\(2019\)+Digitalization+Platform+and+Supervisory+Control+of+a+Continuous+Integrated+PhD+thesis+ETH+Z%C3%BCrich.](https://scholar.google.com/scholar?q=Feidl+F.+(2019)+Digitalization+Platform+and+Supervisory+Control+of+a+Continuous+Integrated+PhD+thesis+ETH+Z%C3%BCrich.))

Feidl, F., Garbellini, S., Vogt, S., Sokolov, M., Souquet, J., Broly, H., Butt, A. & Morbidelli, M. (2019), 'A new flow cell and chemometric protocol for implementing inâ€line Raman spectroscopy in chromatography', *Biotechnology Progress* (March), e2847. CrossRef (<http://dx.doi.org/10.1002/btpr.2847>) |

Google Scholar (<https://scholar.google.com/scholar?q=Feidl+F.+Garbellini+S.+Vogt+S.+Sokolov+M.+Souquet+J.+Broly+H.+Butt%C3%A9+A.+&+Morbidelli+M.+>

(2019)+%E2%80%98A+new+flow+cell+and+chemometric+protocol+for+implementing+in+C3%A2%C5%81%C5%82line+Raman+spectroscopy+in+chromatography%E2%80%99+Biotechnolog(March)+e2847.)

Finn, B., Harvey, L. M., McNeil, B. & McNeil, B. (2006), 'Near-infrared spectroscopic monitoring of biomass, glucose, ethanol and protein content in a high cell density baker's yeast fed-batch

bioprocess', *Yeast* 23(7), 507–517. CrossRef (<http://dx.doi.org/10.1002/yea.1371>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Near-

infrared+spectroscopic+monitoring+of+biomass+glucose+ethanol+and+protein+content+in+a+high+cell+density+baker%E2%80%99s+yeast+fed-batch+bioprocess&publication+year=2006&author=Finn+B.&author=Harvey+L.+M.&author=McNeil+B.&author=McNeil+B.&journal=Yeast&volume=23&doi=10.1002/yea.1371&pages=507-517)

Fisher, A. C. C., Kamga, M.-H. H., Agarabi, C., Brorson, K., Lee, S. L. L. & Yoon, S. (2018), 'The current scientific and regulatory landscape in advancing integrated continuous biopharmaceutical

manufacturing', *Trends in Biotechnology* 37(3), 253–267. CrossRef (<http://dx.doi.org/10.1016/j.tibtech.2018.08.008>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=The+current+scientific+and+regulatory+landscape+in+advancing+integrated+continuous+biopharmaceutical+manufacturing&publication+year=2018&author=Fisher+A.+C.+C.&author=H.+H.&author=Agarabi+C.&author=Brorson+K.&author=Lee+S.+L.+L.&author=Yoon+S.&journal=Trends+in+Biotechnology&volume=37&doi=10.1016/j.tibtech.2018.08.008&pages=253-267)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30241924>)

Fleischaker, R. J. & Sinskey, A. J. (1981), 'Oxygen demand and supply in cell culture', *European Journal of Applied Microbiology and Biotechnology* 12(4), 193–197.

CrossRef (<http://dx.doi.org/10.1007/BF00499486>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Oxygen+demand+and+supply+in+cell+culture&publication+year=1981&author=Fleischaker+R.+J.&author=Sinskey+A.+J.&journal=European+Journal+of+Applied+Microbiology+and+Bio197)

Fleming, A. (1929), 'On the antibacterial action of cultures of a penicillium, with special reference to their use in the isolation of B. influenzae. 1929', *British Journal of Experimental Pathology* 10(3), 226–236.

Google Scholar (https://scholar.google.com/scholar_lookup?

title=On+the+antibacterial+action+of+cultures+of+a+penicillium+with+special+reference+to+their+use+in+the+isolation+of+B.+influenzae.+1929&publication+year=1929&author=Fleming+236)

Fogler, H. S. (2008), *Elements of Chemical Reaction Engineering*, 5th edn, Pearson Education.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Elements+of+Chemical+Reaction+Engineering&publication+year=2008&author=Fogler+H.+S.)

Frenzel, A., Hust, M. & Schirrmann, T. (2013), 'Expression of recombinant antibodies', *Frontiers in Immunology* 4(July), 1–20. CrossRef (<http://dx.doi.org/10.3389/fimmu.2013.00217>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Expression+of+recombinant+antibodies&publication+year=2013&author=Frenzel+A.&author=Hust+M.&author=Schirrmann+T.&journal=Frontiers+in+Immunology&volume=4&doi=10.20)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23908655>)

Froment, G. F., Bischoff, K. B. & De Wilde, J. (1990), *Chemical Reactor Analysis and Design*, Vol. 2, Wiley New York.

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Chemical+Reactor+Analysis+and+Design&publication+year=1990&author=Froment+G.+F.&author=Bischoff+K.+B.&author=De+Wilde+J.)

Gagnon, M., Hiller, G., Luan, Y. T., Kittredge, A., Defelice, J. & Drapeau, D. (2011), 'High-end pH-controlled delivery of glucose effectively suppresses lactate accumulation in CHO fed-batch cultures',

Biotechnology and Bioengineering 108(6), 1328–1337. CrossRef (<http://dx.doi.org/10.1002/bit.23072>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=High-end+pH-controlled+delivery+of+glucose+effectively+suppresses+lactate+accumulation+in+CHO+fed-

batch+cultures&publication+year=2011&author=Gagnon+M.&author=Hiller+G.&author=Luan+Y.+T.&author=Kittredge+A.&author=Defelice+J.&author=Drapeau+D.&journal=Biotechnology+1337)

García-Muñoz, S. & Polizzi, M. (2012), 'WSPLS: A new approach towards mixture modeling and accelerated product development', *Chemometrics and Intelligent Laboratory Systems* 114, 116–121.

CrossRef (<http://dx.doi.org/10.1016/j.chemolab.2012.03.009>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?](https://scholar.google.com/scholar_lookup?title=WSPLS:+A+new+approach+towards+mixture+modeling+and+accelerated+product+development&publication+year=2012&author=Garc%C3%ADa-Mu%C3%B1oz+S.&author=Polizzi+M.&journal=Chemometrics+and+Intelligent+Laboratory+Systems&volume=114&doi=10.1016/j.chemolab.2012.03.009&pages=116-121)

[title=WSPLS:+A+new+approach+towards+mixture+modeling+and+accelerated+product+development&publication+year=2012&author=Garc%C3%ADa-Mu%C3%B1oz+S.&author=Polizzi+M.&journal=Chemometrics+and+Intelligent+Laboratory+Systems&volume=114&doi=10.1016/j.chemolab.2012.03.009&pages=116-121](https://scholar.google.com/scholar_lookup?title=WSPLS:+A+new+approach+towards+mixture+modeling+and+accelerated+product+development&publication+year=2012&author=Garc%C3%ADa-Mu%C3%B1oz+S.&author=Polizzi+M.&journal=Chemometrics+and+Intelligent+Laboratory+Systems&volume=114&doi=10.1016/j.chemolab.2012.03.009&pages=116-121))

Garnier, A., Voyer, R., Tom, R., Perret, S., Jardin, B. & Kamen, A. (1996), 'Dissolved carbon dioxide accumulation in a large scale and high density production of TGF β receptor with baculovirus infected Sf-9 cells', *Cytotechnology* 22(1), 53–63. [CrossRef \(http://dx.doi.org/10.1007/BF00353924\)](http://dx.doi.org/10.1007/BF00353924) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Dissolved+carbon+dioxide+accumulation+in+a+large+scale+and+high+density+production+of+TGF%CE%B2+receptor+with+9+cells&publication+year=1996&author=Garnier+A.&author=Voyer+R.&author=Tom+R.&author=Perret+S.&author=Jardin+B.&author=Kamen+A.&journal=Cytotechnology&volume=22&doi=63)

Gautam, R., Vanga, S., Ariese, F. & Umapathy, S. (2015), 'Review of multidimensional data processing approaches for Raman and infrared spectroscopy', *EPJ Techniques and Instrumentation* 2(1), 8. [CrossRef \(http://dx.doi.org/10.1140/epjti/s40485-015-0018-6\)](http://dx.doi.org/10.1140/epjti/s40485-015-0018-6) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Review+of+multidimensional+data+processing+approaches+for+Raman+and+infrared+spectroscopy&publication+year=2015&author=Gautam+R.&author=Vanga+S.&author=Ariese+F+015-0018-6)

Glassey, J., Gernaey, K. V., Clemens, C., Schulz, T. W., Oliveira, R., Striedner, G. & Mandenius, C.-F. (2011), 'Process analytical technology (PAT) for biopharmaceuticals', *Biotechnology Journal* 6(4), 369–377. [CrossRef \(http://dx.doi.org/10.1002/biot.201000356\)](http://dx.doi.org/10.1002/biot.201000356) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Process+analytical+technology+\(PAT\)+for+biopharmaceuticals&publication+year=2011&author=Glassey+J.&author=Gernaey+K.+V.&author=Clemens+C.&author=Schulz+T.+W.&author=Oliveira+R.&author=Striedner+G.+a+F.&journal=Biotechnology+Journal&volume=6&doi=10.1002/biot.201000356&pages=369-377](https://scholar.google.com/scholar_lookup?title=Process+analytical+technology+(PAT)+for+biopharmaceuticals&publication+year=2011&author=Glassey+J.&author=Gernaey+K.+V.&author=Clemens+C.&author=Schulz+T.+W.&author=Oliveira+R.&author=Striedner+G.+a+F.&journal=Biotechnology+Journal&volume=6&doi=10.1002/biot.201000356&pages=369-377))
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21416609>)

Glassey, J. & von Stosch, M., eds (2018), *Hybrid Modeling in Process Industries*, CRC Press. [CrossRef \(http://dx.doi.org/10.1201/9781351184373\)](http://dx.doi.org/10.1201/9781351184373) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Hybrid+Modeling+in+Process+Industries&publication+year=2018&author=Glassey+J.&author=von+Stosch+M.)

Godawat, R., Konstantinov, K., Rohani, M. & Warikoo, V. (2015), 'End-to-end integrated fully continuous production of recombinant monoclonal antibodies', *Journal of Biotechnology* 213, 13–19. [CrossRef \(http://dx.doi.org/10.1016/j.jbiotec.2015.06.393\)](http://dx.doi.org/10.1016/j.jbiotec.2015.06.393) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=End-to-end+integrated+fully+continuous+production+of+recombinant+monoclonal+antibodies&publication+year=2015&author=Godawat+R.&author=Konstantinov+K.&author=Rohani+M.&author=19)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26073998>)

Godoy Silva, R., Berdugo, C. & Chalmers, J. J. (2010), 'Aeration, mixing, and hydrodynamics, animal cell bioreactors', in *Encyclopedia of Industrial Biotechnology*, American Cancer Society, pp. 1–27. [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Encyclopedia+of+Industrial+Biotechnology&publication+year=2010&author=Godoy+Silva+R.&author=Berdugo+C.&author=Chalmers+J.+J.&pages=1-27\)](https://scholar.google.com/scholar_lookup?title=Encyclopedia+of+Industrial+Biotechnology&publication+year=2010&author=Godoy+Silva+R.&author=Berdugo+C.&author=Chalmers+J.+J.&pages=1-27)

Goh, P. S., Ismail, A. F. & Ng, B. C. (2017), 'Raman spectroscopy', *Membrane Characterization* 72(12), 31–46. [CrossRef \(http://dx.doi.org/10.1016/B978-0-444-63776-5.00002-4\)](http://dx.doi.org/10.1016/B978-0-444-63776-5.00002-4) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Raman+spectroscopy&publication+year=2017&author=Goh+P.+S.&author=Ismail+A.+F.&author=Ng+B.+C.&journal=Membrane+Characterization&volume=72&doi=10.1016/B978-0-444-63776-5.00002-4&pages=31-46)

Goletz, S., Stahn, R. & Kreye, S. (2016), Patent WO 2016/193083 A1.

Google Scholar ([https://scholar.google.com/scholar?q=Goletz+S.+Stahn+R.+&+Kreye+S.+\(2016\)+Patent+WO+2016/193083+A1.](https://scholar.google.com/scholar?q=Goletz+S.+Stahn+R.+&+Kreye+S.+(2016)+Patent+WO+2016/193083+A1.))

Gomez, N., Ambhaikar, M., Zhang, L., Huang, C.-J. J., Barkhordarian, H., Lull, J. & Gutierrez, C. (2017), 'Analysis of tubespins as a suitable scale-down model of bioreactors for high cell density CHO cell culture', *Biotechnology Progress* 33(2), 490–499. [CrossRef \(http://dx.doi.org/10.1002/btpr.2418\)](http://dx.doi.org/10.1002/btpr.2418) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Analysis+of+tubespins+as+a+suitable+scale-down+model+of+bioreactors+for+high+cell+density+CHO+cell+culture&publication+year=2017&author=Gomez+N.&author=Ambhaikar+M.&author=Zhang+L.&author=Huang+C.-J.+J.&author=Barkhordarian+H.&author=Lull+J.&author=Gutierrez+C.&journal=Biotechnology+Progress&volume=33&doi=10.1002/btpr.2418&pages=490-499)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27977914>)

Gorenflo, V. M., Angepat, S., Bowen, B. D. & Piret, J. M. (2003), 'Optimization of an acoustic cell filter with a novel air-backflush system', *Biotechnology Progress* 19(1), 30–36.

[CrossRef \(http://dx.doi.org/10.1021/bp025625a\)](http://dx.doi.org/10.1021/bp025625a) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Optimization+of+an+acoustic+cell+filter+with+a+novel+air-backflush+system&publication+year=2003&author=Gorenflo+V.+M.&author=Angepat+S.&author=Bowen+B.+D.&author=Piret+J.+M.&journal=Biotechnology+Progress&volume=19&doi=10.1021/bp025625a)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/12573003>)

Goudar, C., Stevens, J., Le, K., Gupta, S., Tan, C. & Munro, T. (2017), 'Enabling next-generation cell line development using continuous perfusion and nanofluidic technologies', in *Integrated Continuous Biomanufacturing III*.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Enabling+next-generation+cell+line+development+using+continuous+perfusion+and+nanofluidic+technologies&publication+year=2017&author=Goudar+C.&author=Stevens+J.&author=Le+K.&author=C)

Goudar, C. T., Matanguihan, R., Long, E., Cruz, C., Zhang, C., Piret, J. M. & Konstantinov, K. B. (2007), 'Decreased pCO₂ accumulation by eliminating bicarbonate addition to high cell-density cultures', *Biotechnology and Bioengineering* 96(6), 1107–1117. [CrossRef \(http://dx.doi.org/10.1002/bit.21116\)](http://dx.doi.org/10.1002/bit.21116) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Decreased+pCO2+accumulation+by+eliminating+bicarbonate+addition+to+high+cell-density+cultures&publication+year=2007&author=Goudar+C.+T.&author=Matanguihan+R.&author=Long+E.&author=Cruz+C.&author=Zhang+C.&author=Piret+J.+M.&author=Konstantinov+1117)

Goudar, C. T., Piret, J. M. & Konstantinov, K. B. (2011), 'Estimating cell specific oxygen uptake and carbon dioxide production rates for mammalian cells in perfusion culture', *Biotechnology Progress* 27(5), 1347–1357. [CrossRef \(http://dx.doi.org/10.1002/btpr.646\)](http://dx.doi.org/10.1002/btpr.646) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Estimating+cell+specific+oxygen+uptake+and+carbon+dioxide+production+rates+for+mammalian+cells+in+perfusion+culture&publication+year=2011&author=Goudar+C.+T.&author=1357)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21626724>)

Gramer, M. J., Eckblad, J. J., Donahue, R., Brown, J., Shultz, C., Vickerman, K., Priem, P., van den Bremer, E. T., Gerritsen, J. & van Berkel, P. H. (2011), 'Modulation of antibody galactosylation through feeding of uridine, manganese chloride, and galactose', *Biotechnology and Bioengineering* 108(7), 1591–1602. [CrossRef \(http://dx.doi.org/10.1002/bit.23075\)](http://dx.doi.org/10.1002/bit.23075) |

Google Scholar ([https://scholar.google.com/scholar_lookup?](https://scholar.google.com/scholar_lookup?title=Modulation+of+antibody+galactosylation+through+feeding+of+uridine+manganese+chloride+and+galactose&publication+year=2011&author=Gramer+M.+J.+&author=Eckblad+J.+J.+&au1602)

[title=Modulation+of+antibody+galactosylation+through+feeding+of+uridine+manganese+chloride+and+galactose&publication+year=2011&author=Gramer+M.+J.+&author=Eckblad+J.+J.+&au1602](https://scholar.google.com/scholar_lookup?title=Modulation+of+antibody+galactosylation+through+feeding+of+uridine+manganese+chloride+and+galactose&publication+year=2011&author=Gramer+M.+J.+&author=Eckblad+J.+J.+&au1602))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21328321>)

Gray, D. R., Chen, S., Howarth, W., Inlow, D. & Maiorella, B. L. (1996), 'CO₂ in large-scale and high-density CHO cell perfusion culture', *Cytotechnology* 22(1–3), 65–78.

CrossRef (<http://dx.doi.org/10.1007/BF00353925>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=CO₂+in+large-scale+and+high-](https://scholar.google.com/scholar_lookup?title=CO2+in+large-scale+and+high-density+CHO+cell+perfusion+culture&publication+year=1996&author=Gray+D.+R.+&author=Chen+S.+&author=Howarth+W.+&author=Inlow+D.+&author=Maiorella+B.+L.+&journal=Cytotechnolo78)

[density+CHO+cell+perfusion+culture&publication+year=1996&author=Gray+D.+R.+&author=Chen+S.+&author=Howarth+W.+&author=Inlow+D.+&author=Maiorella+B.+L.+&journal=Cytotechnolo78](https://scholar.google.com/scholar_lookup?title=CO2+in+large-scale+and+high-density+CHO+cell+perfusion+culture&publication+year=1996&author=Gray+D.+R.+&author=Chen+S.+&author=Howarth+W.+&author=Inlow+D.+&author=Maiorella+B.+L.+&journal=Cytotechnolo78))

Grillberger, L., Kreil, T. R., Nasr, S. & Reiter, M. (2009), 'Emerging trends in plasma-free manufacturing of recombinant protein therapeutics expressed in mammalian cells', *Biotechnology Journal* 4(2), 186–201. CrossRef (<http://dx.doi.org/10.1002/biot.200800241>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Emerging+trends+in+plasma-](https://scholar.google.com/scholar_lookup?title=Emerging+trends+in+plasma-free+manufacturing+of+recombinant+protein+therapeutics+expressed+in+mammalian+cells&publication+year=2009&author=Grillberger+L.+&author=Kreil+T.+R.+&author=Nasr+S.+&author=201)

[free+manufacturing+of+recombinant+protein+therapeutics+expressed+in+mammalian+cells&publication+year=2009&author=Grillberger+L.+&author=Kreil+T.+R.+&author=Nasr+S.+&author=201](https://scholar.google.com/scholar_lookup?title=Emerging+trends+in+plasma-free+manufacturing+of+recombinant+protein+therapeutics+expressed+in+mammalian+cells&publication+year=2009&author=Grillberger+L.+&author=Kreil+T.+R.+&author=Nasr+S.+&author=201))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19226552>)

Gunther, J., Conner, J. & Seborg, D. (2007), 'Fault detection and diagnosis in an industrial fed-batch cell culture process', *Biotechnology Progress* 23(4), 851–857.

CrossRef (<http://dx.doi.org/10.1002/bp070063m>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Fault+detection+and+diagnosis+in+an+industrial+fed-](https://scholar.google.com/scholar_lookup?title=Fault+detection+and+diagnosis+in+an+industrial+fed-batch+cell+culture+process&publication+year=2007&author=Gunther+J.+&author=Conner+J.+&author=Seborg+D.+&journal=Biotechnology+Progress&volume=23&doi=10.1002/bp070063m&p;857)

[batch+cell+culture+process&publication+year=2007&author=Gunther+J.+&author=Conner+J.+&author=Seborg+D.+&journal=Biotechnology+Progress&volume=23&doi=10.1002/bp070063m&p;857](https://scholar.google.com/scholar_lookup?title=Fault+detection+and+diagnosis+in+an+industrial+fed-batch+cell+culture+process&publication+year=2007&author=Gunther+J.+&author=Conner+J.+&author=Seborg+D.+&journal=Biotechnology+Progress&volume=23&doi=10.1002/bp070063m&p;857))

Hammond, M., Marghitoiu, L., Lee, H., Perez, L., Rogers, G., Nashed-Samuel, Y., Nunn, H. & Kline, S. (2014), 'A cytotoxic leachable compound from single-use bioprocess equipment that causes poor cell growth performance', *Biotechnology Progress* 30(2), 332–337. CrossRef (<http://dx.doi.org/10.1002/btpr.1869>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=A+cytotoxic+leachable+compound+from+single-](https://scholar.google.com/scholar_lookup?title=A+cytotoxic+leachable+compound+from+single-use+bioprocess+equipment+that+causes+poor+cell+growth+performance&publication+year=2014&author=Hammond+M.+&author=Marghitoiu+L.+&author=Lee+H.+&author=Perez+L.+&authcSamuel+Y.+&author=Nunn+H.+&author=Kline+S.+&journal=Biotechnology+Progress&volume=30&doi=10.1002/btpr.1869&pages=332-337)

[use+bioprocess+equipment+that+causes+poor+cell+growth+performance&publication+year=2014&author=Hammond+M.+&author=Marghitoiu+L.+&author=Lee+H.+&author=Perez+L.+&authcSamuel+Y.+&author=Nunn+H.+&author=Kline+S.+&journal=Biotechnology+Progress&volume=30&doi=10.1002/btpr.1869&pages=332-337](https://scholar.google.com/scholar_lookup?title=A+cytotoxic+leachable+compound+from+single-use+bioprocess+equipment+that+causes+poor+cell+growth+performance&publication+year=2014&author=Hammond+M.+&author=Marghitoiu+L.+&author=Lee+H.+&author=Perez+L.+&authcSamuel+Y.+&author=Nunn+H.+&author=Kline+S.+&journal=Biotechnology+Progress&volume=30&doi=10.1002/btpr.1869&pages=332-337))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24497314>)

Heidemann, R., Lünse, S., Tran, D. & Zhang, C. (2010), 'Characterization of cell-banking parameters for the cryopreservation of mammalian cell lines in 100-mL cryobags', *Biotechnology Progress* 26(4), 1154–1163.

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Characterization+of+cell-banking+parameters+for+the+cryopreservation+of+mammalian+cell+lines+in+100-](https://scholar.google.com/scholar_lookup?title=Characterization+of+cell-banking+parameters+for+the+cryopreservation+of+mammalian+cell+lines+in+100-mL+cryobags&publication+year=2010&author=Heidemann+R.+&author=L%C3%BCnse+S.+&author=Tran+D.+&author=Zhang+C.+&journal=Biotechnology+Progress&volume=26&pages=1154-1163)

[mL+cryobags&publication+year=2010&author=Heidemann+R.+&author=L%C3%BCnse+S.+&author=Tran+D.+&author=Zhang+C.+&journal=Biotechnology+Progress&volume=26&pages=1154-1163](https://scholar.google.com/scholar_lookup?title=Characterization+of+cell-banking+parameters+for+the+cryopreservation+of+mammalian+cell+lines+in+100-mL+cryobags&publication+year=2010&author=Heidemann+R.+&author=L%C3%BCnse+S.+&author=Tran+D.+&author=Zhang+C.+&journal=Biotechnology+Progress&volume=26&pages=1154-1163))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20730770>)

Heidemann, R., Mered, M., Wang, D. Q., Gardner, B., Zhang, C., Michaels, J., Henzler, H. J., Abbas, N. & Konstantinov, K. (2002), 'A new seed-train expansion method for recombinant mammalian cell lines', *Cytotechnology* 38(1–3), 99–108. CrossRef (<http://dx.doi.org/10.1023/A:1021114300958>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=A+new+seed-](https://scholar.google.com/scholar_lookup?title=A+new+seed-train+expansion+method+for+recombinant+mammalian+cell+lines&publication+year=2002&author=Heidemann+R.+&author=Mered+M.+&author=Wang+D.+Q.+&author=Gardner+B.+&author=108)

[train+expansion+method+for+recombinant+mammalian+cell+lines&publication+year=2002&author=Heidemann+R.+&author=Mered+M.+&author=Wang+D.+Q.+&author=Gardner+B.+&author=108](https://scholar.google.com/scholar_lookup?title=A+new+seed-train+expansion+method+for+recombinant+mammalian+cell+lines&publication+year=2002&author=Heidemann+R.+&author=Mered+M.+&author=Wang+D.+Q.+&author=Gardner+B.+&author=108))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19003091>)

Helenius, A. & Aebi, M. (2001), 'Intracellular functions of N-linked glycans', *Science* 291(5512), 2364–2369. CrossRef (<http://dx.doi.org/10.1126/science.291.5512.2364>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Intracellular+functions+of+N-](https://scholar.google.com/scholar_lookup?title=Intracellular+functions+of+N-linked+glycans&publication+year=2001&author=Helenius+A.+&author=Aebi+M.+&journal=Science&volume=291&doi=10.1126/science.291.5512.2364&pages=2364-2369)

[linked+glycans&publication+year=2001&author=Helenius+A.+&author=Aebi+M.+&journal=Science&volume=291&doi=10.1126/science.291.5512.2364&pages=2364-2369](https://scholar.google.com/scholar_lookup?title=Intracellular+functions+of+N-linked+glycans&publication+year=2001&author=Helenius+A.+&author=Aebi+M.+&journal=Science&volume=291&doi=10.1126/science.291.5512.2364&pages=2364-2369))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/11269317>)

Henry, O., Kwok, E. & Piret, J. M. (2008), 'Simpler non-instrumented batch and semi-continuous cultures provide mammalian cell kinetic data comparable to continuous and perfusion cultures', *Biotechnology Progress* 24(4), 921–931. CrossRef (<http://dx.doi.org/10.1002/btpr.17>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Simpler+non-instrumented+batch+and+semi-](https://scholar.google.com/scholar_lookup?title=Simpler+non-instrumented+batch+and+semi-continuous+cultures+provide+mammalian+cell+kinetic+data+comparable+to+continuous+and+perfusion+cultures&publication+year=2008&author=Henry+O.+&author=Kwok+E.+&author=Pir931)

[continuous+cultures+provide+mammalian+cell+kinetic+data+comparable+to+continuous+and+perfusion+cultures&publication+year=2008&author=Henry+O.+&author=Kwok+E.+&author=Pir931](https://scholar.google.com/scholar_lookup?title=Simpler+non-instrumented+batch+and+semi-continuous+cultures+provide+mammalian+cell+kinetic+data+comparable+to+continuous+and+perfusion+cultures&publication+year=2008&author=Henry+O.+&author=Kwok+E.+&author=Pir931))

Higel, F., Seidl, A., Sörgel, F. & Friess, W. (2016), 'N-glycosylation heterogeneity and the influence on structure, function and pharmacokinetics of monoclonal antibodies and Fc fusion proteins', *European Journal of Pharmaceutics and Biopharmaceutics* 100, 94–100. CrossRef (<http://dx.doi.org/10.1016/j.ejpb.2016.01.005>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=N-glycosylation+heterogeneity+and+the+influence+on+structure+function+and+pharmacokinetics+of+monoclonal+antibodies+and+Fc+fusion+proteins&publication+year=2016&author=Hige100)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26775146>)

Hiller, G. W., Ovalle, A. M., Gagnon, M. P., Curran, M. L. & Wang, W. (2017), 'Cell-controlled hybrid perfusion fed-batch CHO cell process provides significant productivity improvement over conventional fed-batch cultures', *Biotechnology and Bioengineering* 114(7), 1438–1447. CrossRef (<http://dx.doi.org/10.1002/bit.26259>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Cell-controlled+hybrid+perfusion+fed-batch+CHO+cell+process+provides+significant+productivity+improvement+over+co](https://scholar.google.com/scholar_lookup?title=Cell-controlled+hybrid+perfusion+fed-batch+CHO+cell+process+provides+significant+productivity+improvement+over+co+batch+cultures&publication+year=2017&author=Hiller+G.+W.+&author=Ovalle+A.+M.+&author=Gagnon+M.+P.+&author=Curran+M.+L.+&author=Wang+W.+&journal=Biotechnology+and+Bioeng1447)

[batch+cultures&publication+year=2017&author=Hiller+G.+W.+&author=Ovalle+A.+M.+&author=Gagnon+M.+P.+&author=Curran+M.+L.+&author=Wang+W.+&journal=Biotechnology+and+Bioeng1447](https://scholar.google.com/scholar_lookup?title=Cell-controlled+hybrid+perfusion+fed-batch+CHO+cell+process+provides+significant+productivity+improvement+over+co+batch+cultures&publication+year=2017&author=Hiller+G.+W.+&author=Ovalle+A.+M.+&author=Gagnon+M.+P.+&author=Curran+M.+L.+&author=Wang+W.+&journal=Biotechnology+and+Bioeng1447))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28128436>)

Hoos, A. (2016), 'Development of immuno-oncology drugs: From CTLA4 to PD1 to the next generations', *Nature Reviews Drug Discovery* 15(4), 235–247.

CrossRef (<http://dx.doi.org/10.1038/nrd.2015.35>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Development+of+immuno-](https://scholar.google.com/scholar_lookup?title=Development+of+immuno-oncology+drugs:+From+CTLA4+to+PD1+to+the+next+generations&publication+year=2016&author=Hoos+A.+&journal=Nature+Reviews+Drug+Discovery&volume=15&doi=10.1038/nrd.2015.247)

[oncology+drugs:+From+CTLA4+to+PD1+to+the+next+generations&publication+year=2016&author=Hoos+A.+&journal=Nature+Reviews+Drug+Discovery&volume=15&doi=10.1038/nrd.2015.247](https://scholar.google.com/scholar_lookup?title=Development+of+immuno-oncology+drugs:+From+CTLA4+to+PD1+to+the+next+generations&publication+year=2016&author=Hoos+A.+&journal=Nature+Reviews+Drug+Discovery&volume=15&doi=10.1038/nrd.2015.247))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26965203>)

Hossler, P., Khattak, S. F. & Li, Z. J. (2009), 'Optimal and consistent protein glycosylation in mammalian cell culture'. *Glycobiology* 19(9), 936–949.

CrossRef (<http://dx.doi.org/10.1093/glycob/cwp079>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Optimal+and+consistent+protein+glycosylation+in+mammalian+cell+culture&publication+year=2009&author=Hossler+P.&author=Khattak+S.+F.&author=Li+Z.+J.&journal=Glycobiology 949)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19494347>)

Howard, D. H., Bach, P. B., Berndt, E. R. & Rena, M. C. (2015), 'Pricing in the market for anticancer drugs', *Journal of Economic Perspectives* 29(1), 1689–1699.

CrossRef (<http://dx.doi.org/10.1257/jep.29.1.139>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Pricing+in+the+market+for+anticancer+drugs&publication+year=2015&author=Howard+D.+H.&author=Bach+P.+B.&author=Berndt+E.+R.&author=Rena+M.+C.&journal=Journal+of+Eco 1699)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28441702>)

Hsie, A. W., Recio, L., Katz, D. S., Lee, C. Q., Wagner, M. & Schenley, R. L. (1986), 'Evidence for reactive oxygen species inducing mutations in mammalian cells.', *Proceedings of the National Academy of Sciences* 83(24), 9616–9620. CrossRef (<http://dx.doi.org/10.1073/pnas.83.24.9616>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Evidence+for+reactive+oxygen+species+inducing+mutations+in+mammalian+cells&publication+year=1986&author=Hsie+A.+W.&author=Recio+L.&author=Katz+D.+S.&author=Lee+C.+ 9620)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/2432598>)

Hu, W.-S. (2012), *Cell Culture Bioprocess Engineering*. Springer.

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Cell+Culture+Bioprocess+Engineering&publication+year=2012&author=Hu+W.-S.\)](https://scholar.google.com/scholar_lookup?title=Cell+Culture+Bioprocess+Engineering&publication+year=2012&author=Hu+W.-S.))

Hubert, M. & Engelen, S. (2004), 'Robust PCA and classification in biosciences', *Bioinformatics* 20(11), 1728–1736. CrossRef (<http://dx.doi.org/10.1093/bioinformatics/bth158>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Robust+PCA+and+classification+in+biosciences&publication+year=2004&author=Hubert+M.&author=Engelen+S.&journal=Bioinformatics&volume=20&doi=10.1093/bioinformatics/bth 1736)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/14988110>)

Hughes, S. S. (2001), 'Making dollars out of DNA: The first major patent in biotechnology and the commercialization of molecular biology, 1974–1980', *Isis* 92(3), 541–575.

CrossRef (<http://dx.doi.org/10.1086/385281>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Making+dollars+out+of+DNA+The+first+major+patent+in+biotechnology+and+the+commercialization+of+molecular+biology+1974%E2%80%931980&publication+year=2001&author=I 575)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/11810894>)

Hughmark, G. A. (1980), 'Power requirements and interfacial area in gas–liquid turbine agitated systems', *Industrial and Engineering Chemistry Process Design and Development* 19(4), 638–641.

CrossRef (<http://dx.doi.org/10.1021/i260076a023>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Power+requirements+and+interfacial+area+in+gas%E2%80%93liquid+turbine+agitated+systems&publication+year=1980&author=Hughmark+G.+A.&journal=Industrial+and+Engineeri 641)

Indahl, U. (2005), 'A twist to partial least squares regression', *Journal of Chemometrics* 19(1), 32–44. CrossRef (<http://dx.doi.org/10.1002/cem.904>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=A+twist+to+partial+least+square+regression&publication+year=2005&author=Indahl+U.&journal=Journal+of+Chemometrics&volume=19&doi=10.1002/cem.904&pages=32-44)

Ishida, M., Haga, R., Nishimura, N., Matuzaki, H. & Nakano, R. (1990), 'High cell density suspension culture of mammalian anchorage independent cells: Oxygen transfer by gas sparging and defoaming with a hydrophobic net', *Cytotechnology* 4(3), 215–225. CrossRef (<http://dx.doi.org/10.1007/BF00563782>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=High+cell+density+suspension+culture+of+mammalian+anchorage+independent+cells:+Oxygen+transfer+by+gas+sparging+and+defoaming+with+a+hydrophobic+net&publication+y 225)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/1366989>)

Ivarsson, M., Noh, H., Morbidelli, M. & Soos, M. (2015), 'Insights into pH-induced metabolic switch by flux balance analysis', *Biotechnology Progress* 31(2), 347–357.

CrossRef (<http://dx.doi.org/10.1002/btpr.2043>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Insights+into+pH-

induced+metabolic+switch+by+flux+balance+analysis&publication+year=2015&author=Ivarsson+M.&author=Noh+H.&author=Morbidelli+M.&author=Soos+M.&journal=Biotechnology+Pro 357)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25906421>)

Jackson, D., Symons, R. H. & Berg, P. (1972), 'Biochemical method for inserting new genetic information into DNA of simian virus 40: Circular DNA molecules containing lambda phage genes and the galactose operon of Escherichia coli', *Proceedings of the National Academy of Sciences of the United States of America* 69, 2904–2909. CrossRef (<http://dx.doi.org/10.1073/pnas.69.10.2904>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Biochemical+method+for+inserting+new+genetic+information+into+DNA+of+simian+virus+40:+Circular+DNA+molecules+containing+lambda+phage+genes+and+the+galactose+oper 2909)

Jacquemart, R., Vandersluis, M., Zhao, M., Sukhija, K., Sidhu, N. & Stout, J. (2016), 'A single-use strategy to enable manufacturing of affordable biologics', *Computational and Structural Biotechnology Journal* 14, 309–318. CrossRef (<http://dx.doi.org/10.1016/j.csbj.2016.06.007>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=A+single-

use+strategy+to+enable+manufacturing+of+affordable+biologics&publication+year=2016&author=Jacquemart+R.&author=Vandersluis+M.&author=Zhao+M.&author=Sukhija+K.&author=S 318)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27570613>)

Jagschies, G., Lindskog, E., Lacki, K. & Galliher, P. M. (2018), *Biopharmaceutical Processing: Development, Design, and Implementation of Manufacturing Processes*, Elsevier Science.

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Biopharmaceutical+Processing:+Development+Design+and+Implementation+of+Manufacturing+Processes&publication+year=2018&author=Jagschies+G.&author=Lindskog+E.&authi

Jang, J. D. & Barford, J. P. (2000), 'An unstructured kinetic model of macromolecular metabolism in batch and fed-batch cultures of hybridoma cells producing monoclonal antibody', *Biochemical Engineering Journal* 4(2), 153–168. CrossRef ([http://dx.doi.org/10.1016/S1369-703X\(99\)00041-8](http://dx.doi.org/10.1016/S1369-703X(99)00041-8)) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=An+unstructured+kinetic+model+of+macromolecular+metabolism+in+batch+and+fed-batch+cultures+of+hybridoma+cells+producing+monoclonal+antibody&publication+year=2000&author=Jang+J.+D.&author=Barford+J.+P.&journal=Biochemical+Engineering+Journal&volun703X\(99\)00041-8&pages=153-168](https://scholar.google.com/scholar_lookup?title=An+unstructured+kinetic+model+of+macromolecular+metabolism+in+batch+and+fed-batch+cultures+of+hybridoma+cells+producing+monoclonal+antibody&publication+year=2000&author=Jang+J.+D.&author=Barford+J.+P.&journal=Biochemical+Engineering+Journal&volun703X(99)00041-8&pages=153-168))

Janoschek, S., Schulze, M., Zijlstra, G., Greller, G. & Matuszczyk, J. (2018), 'A protocol to transfer a fed-batch platform process into semi-perfusion mode: The benefit of automated small scale bioreactors compared to shake flasks as scale-down model', *Biotechnology Progress* 35(2), 8–10.

Google Scholar (https://scholar.google.com/scholar_lookup?title=A+protocol+to+transfer+a+fed-batch+platform+process+into+semi-perfusion+mode:+The+benefit+of+automated+small+scale+bioreactors+compared+to+shake+flasks+as+scale-down+model&publication+year=2018&author=Janoschek+S.&author=Schulze+M.&author=Zijlstra+G.&author=Greller+G.&author=Matuszczyk+J.&journal=Biotechnology+Progress&volume10)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30479066>)

Jefferis, R. (2005), 'Glycosylation of recombinant antibody therapeutics', *Biotechnology Progress* 21(1), 11–16. CrossRef (<http://dx.doi.org/10.1021/bp040016j>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Glycosylation+of+recombinant+antibody+therapeutics&publication+year=2005&author=Jefferis+R.&journal=Biotechnology+Progress&volume=21&doi=10.1021/bp040016j&pages=11-16)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/15903235>)

Jefferis, R. (2009), 'Recombinant antibody therapeutics: The impact of glycosylation on mechanisms of action', *Trends in Pharmacological Sciences* 30(7), 356–362.

CrossRef (<http://dx.doi.org/10.1016/j.tips.2009.04.007>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Recombinant+antibody+therapeutics:+The+impact+of+glycosylation+on+mechanisms+of+action&publication+year=2009&author=Jefferis+R.&journal=Trends+in+Pharmacological+Sci362)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19552968>)

Jesus, M. D. & Wurm, F. M. (2011), 'Manufacturing recombinant proteins in kg-ton quantities using animal cells in bioreactors', *European Journal of Pharmaceutics and Biopharmaceutics* 78(2), 184–188. CrossRef (<http://dx.doi.org/10.1016/j.ejpb.2011.01.005>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Manufacturing+recombinant+proteins+in+kg-ton+quantities+using+animal+cells+in+bioreactors&publication+year=2011&author=Jesus+M.+D.&author=Wurm+F.+M.&journal=European+Journal+of+Pharmaceutics+and+Biopharmaceut188)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21256214>)

Jiang, M., Severson, K. A., Love, J. C., Madden, H., Swann, P., Zang, L. & Braatz, R. D. (2017), 'Opportunities and challenges of real-time release testing in biopharmaceutical manufacturing', *Biotechnology and Bioengineering* 114(11), 2445–2456. CrossRef (<http://dx.doi.org/10.1002/bit.26383>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Opportunities+and+challenges+of+real-time+release+testing+in+biopharmaceutical+manufacturing&publication+year=2017&author=Jiang+M.&author=Severson+K.+A.&author=Love+J.+C.&author=Madden+H.&author=Swann+P.2456)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28710854>)

Jiang, R., Hoesli, N., Mueller, R., Kretz, T., Chen, H., Xu, S. & Bowers, J. (2018), 'Probing lactate metabolism variations in large-scale bioreactors', *Biotechnology Progress* 34(3), 756–766.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Probing+lactate+metabolism+variations+in+large-scale+bioreactors&publication+year=2018&author=Jiang+R.&author=Hoesli+N.&author=Mueller+R.&author=Kretz+T.&author=Chen+H.&author=Xu+S.&author=Bowers+J.&journal=Biotechn766)

Jimenez del Val, I., Nagy, J. M. & Kontoravdi, C. (2011), 'A dynamic mathematical model for monoclonal antibody N-linked glycosylation and nucleotide sugar donor transport within a maturing Golgi apparatus', *Biotechnology Progress* 27(6), 1730–1743. CrossRef (<http://dx.doi.org/10.1002/btpr.688>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=A+dynamic+mathematical+model+for+monoclonal+antibody+N-linked+glycosylation+and+nucleotide+sugar+donor+transport+within+a+maturing+Golgi+apparatus&publication+year=2011&author=Jimenez+del+Val+I.&author=Nagy+J.+M.&author=Kont1743)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21956887>)

Joao De Jesus, M. & Wurm, F. M. (2013), 'Scale-up and predictability in process development with suspension cultures of mammalian cells for recombinant protein manufacture: comments on a trend reversal', *Pharmaceutical Bioprocessing* 1(4), 1–3.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Scale-up+and+predictability+in+process+development+with+suspension+cultures+of+mammalian+cells+for+recombinant+protein+manufacture:+comments+on+a+trend+reversal&publication+3)

Joeris, K., Frerichs, J.-G., Konstantinov, K. & Scheper, T. (2002), 'In-situ microscopy: Online process monitoring of mammalian cell cultures', *Cytotechnology* 38(1–3), 129–134.

CrossRef (<http://dx.doi.org/10.1023/A:1021170502775>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=In-situ+microscopy:+Online+process+monitoring+of+mammalian+cell+cultures&publication+year=2002&author=Joeris+K.&author=Frerichs+J.-G.&author=Konstantinov+K.&author=Scheper+T.&journal=Cytotechnology&volume=38&doi=10.1023/A:1021170502775&pages=129-134)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19003094>)

Jolliffe, I. (2005), 'Principal component analysis', in *Encyclopedia of Statistics in Behavioral Science*, John Wiley and Sons.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Encyclopedia+of+Statistics+in+Behavioral+Science&publication+year=2005&author=Jolliffe+I.)

Jordan, M. & Jenkins, N. (2007), 'Tools for high-throughput medium and process optimization', *Methods in Biotechnology* 24, 193–202. CrossRef (http://dx.doi.org/10.1007/978-1-59745-399-8_7) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Tools+for+high-throughput+medium+and+process+optimization&publication+year=2007&author=Jordan+M.&author=Jenkins+N.&journal=Methods+in+Biotechnology&volume=24&doi=10.1007/978-1-59745-399-8+7&pages=193-202)

Jordan, M., Kinnon, N. M., Monchois, V., Stettler, M. & Broly, H. (2018), 'Intensification of large-scale cell culture processes', *Current Opinion in Chemical Engineering* 22, 253–257.

CrossRef (<http://dx.doi.org/10.1016/j.coche.2018.11.008>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Intensification+of+large-scale+cell+culture+processes&publication+year=2018&author=Jordan+M.&author=Kinnon+N.+M.&author=Monchois+V.&author=Stettler+M.&author=Broly+H.&journal=Current+Opinion+in257)

Jordan, M., Voisard, D., Berthoud, A. & Tercier, L. (2012), 'Cell culture medium improvement by rigorous shuffling of components using media blending', *Cytotechnology* 65(1), 31–40.

CrossRef (<http://dx.doi.org/10.1007/s10616-012-9462-1>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?](https://scholar.google.com/scholar_lookup?title=Cell+culture+medium+improvement+by+rigorous+shuffling+of+components+using+media+blending&publication+year=2012&author=Jordan+M.&author=Voisard+D.&author=Berthou012-9462-1&pages=31-40)

[title=Cell+culture+medium+improvement+by+rigorous+shuffling+of+components+using+media+blending&publication+year=2012&author=Jordan+M.&author=Voisard+D.&author=Berthou012-9462-1&pages=31-40](https://scholar.google.com/scholar_lookup?title=Cell+culture+medium+improvement+by+rigorous+shuffling+of+components+using+media+blending&publication+year=2012&author=Jordan+M.&author=Voisard+D.&author=Berthou012-9462-1&pages=31-40))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22695856>)

Karst, D. J., Scibona, E., Serra, E., Bielser, J.-M. M., Souquet, J., Stettler, M., Broly, H., Soos, M., Morbidelli, M. & Villiger, T. K. (2017b), 'Modulation and modeling of monoclonal antibody N-linked glycosylation in mammalian cell perfusion reactors', *Biotechnology and Bioengineering* 114(9), 1–37. CrossRef (<http://dx.doi.org/10.1002/bit.26315>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Modulation+and+modeling+of+monoclonal+antibody+N-linked+glycosylation+in+mammalian+cell+perfusion+reactors&publication+year=2017&author=Karst+D.+J.&author=Scibona+E.&author=Serra+E.&author=Bielser+J.-M.+M.&author=Souquet+J.&author=Stettler+M.&author=Broly+H.&author=Soos+M.&author=Morbidelli+M.&author=Villiger+T.+K.&journal=Biotechnology+and+Bioengineering&volume=1137)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28409838>)

Karst, D. J., Serra, E., Villiger, T. K., Soos, M. & Morbidelli, M. (2016), 'Characterization and comparison of ATF and TFF in stirred bioreactors for continuous mammalian cell culture processes', *Biochemical Engineering Journal* 110, 17–26. CrossRef (<http://dx.doi.org/10.1016/j.bej.2016.02.003>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Characterization+and+comparison+of+ATF+and+TFF+in+stirred+bioreactors+for+continuous+mammalian+cell+culture+processes&publication+year=2016&author=Karst+D.+J.&author

[title=Characterization+and+comparison+of+ATF+and+TFF+in+stirred+bioreactors+for+continuous+mammalian+cell+culture+processes&publication+year=2016&author=Karst+D.+J.&author](https://scholar.google.com/scholar_lookup?title=Characterization+and+comparison+of+ATF+and+TFF+in+stirred+bioreactors+for+continuous+mammalian+cell+culture+processes&publication+year=2016&author=Karst+D.+J.&author)

[title=Characterization+and+comparison+of+ATF+and+TFF+in+stirred+bioreactors+for+continuous+mammalian+cell+culture+processes&publication+year=2016&author=Karst+D.+J.&author](https://scholar.google.com/scholar_lookup?title=Characterization+and+comparison+of+ATF+and+TFF+in+stirred+bioreactors+for+continuous+mammalian+cell+culture+processes&publication+year=2016&author=Karst+D.+J.&author)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/29289800>)

Karst, D. J., Steinebach, F., Soos, M. & Morbidelli, M. (2017a), 'Process performance and product quality in an integrated continuous antibody production process', *Biotechnology and Bioengineering* 114(2), 298–307. CrossRef (<http://dx.doi.org/10.1002/bit.26069>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Process+performance+and+product+quality+in+an+integrated+continuous+antibody+production+process&publication+year=2017&author=Karst+D.+J.&author=Steinebach+F.&author

[title=Process+performance+and+product+quality+in+an+integrated+continuous+antibody+production+process&publication+year=2017&author=Karst+D.+J.&author=Steinebach+F.&author](https://scholar.google.com/scholar_lookup?title=Process+performance+and+product+quality+in+an+integrated+continuous+antibody+production+process&publication+year=2017&author=Karst+D.+J.&author=Steinebach+F.&author)

[title=Process+performance+and+product+quality+in+an+integrated+continuous+antibody+production+process&publication+year=2017&author=Karst+D.+J.&author=Steinebach+F.&author](https://scholar.google.com/scholar_lookup?title=Process+performance+and+product+quality+in+an+integrated+continuous+antibody+production+process&publication+year=2017&author=Karst+D.+J.&author=Steinebach+F.&author)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27997765>)

Karst, D. J., Steinhoff, R. F., Kopp, M. R., Soos, M., Zenobi, R. & Morbidelli, M. (2017d), 'Isotope labeling to determine the dynamics of metabolic response in CHO cell perfusion bioreactors using MALDI-TOF-MS', *Biotechnology Progress* 33(6), 1630–1639. CrossRef (<http://dx.doi.org/10.1002/btpr.2539>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Isotope+labeling+to+determine+the+dynamics+of+metabolic+response+in+CHO+cell+perfusion+bioreactors+using+MALDI+MS&publication+year=2017&author=Karst+D.+J.&author=Steinhoff+R.+F.&author=Kopp+M.+R.&author=Soos+M.&author=Zenobi+R.&author=Morbidelli+M.&journal=Biotechnology+Progress

[title=Isotope+labeling+to+determine+the+dynamics+of+metabolic+response+in+CHO+cell+perfusion+bioreactors+using+MALDI+MS&publication+year=2017&author=Karst+D.+J.&author=Steinhoff+R.+F.&author=Kopp+M.+R.&author=Soos+M.&author=Zenobi+R.&author=Morbidelli+M.&journal=Biotechnology+Progress](https://scholar.google.com/scholar_lookup?title=Isotope+labeling+to+determine+the+dynamics+of+metabolic+response+in+CHO+cell+perfusion+bioreactors+using+MALDI+MS&publication+year=2017&author=Karst+D.+J.&author=Steinhoff+R.+F.&author=Kopp+M.+R.&author=Soos+M.&author=Zenobi+R.&author=Morbidelli+M.&journal=Biotechnology+Progress)

[title=Isotope+labeling+to+determine+the+dynamics+of+metabolic+response+in+CHO+cell+perfusion+bioreactors+using+MALDI+MS&publication+year=2017&author=Karst+D.+J.&author=Steinhoff+R.+F.&author=Kopp+M.+R.&author=Soos+M.&author=Zenobi+R.&author=Morbidelli+M.&journal=Biotechnology+Progress](https://scholar.google.com/scholar_lookup?title=Isotope+labeling+to+determine+the+dynamics+of+metabolic+response+in+CHO+cell+perfusion+bioreactors+using+MALDI+MS&publication+year=2017&author=Karst+D.+J.&author=Steinhoff+R.+F.&author=Kopp+M.+R.&author=Soos+M.&author=Zenobi+R.&author=Morbidelli+M.&journal=Biotechnology+Progress)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/10397813>)

Kawase, Y., Halard, B. & Moo-Young, M. (1992), 'Liquid-phase mass transfer coefficients in bioreactors', *Biotechnology and Bioengineering* 39(11), 1133–1140. CrossRef (<http://dx.doi.org/10.1002/bit.260391109>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Liquid-phase+mass+transfer+coefficients+in+bioreactors&publication+year=1992&author=Kawase+Y.&author=Halard+B.&author=Moo-Young+M.&journal=Biotechnology+and+Bioengineering&volume=39&doi=10.1002/bit.260391109&pages=1133-1140)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18600915>)

Kelly, P. S., McSweeney, S., Coleman, O., Carillo, S., Henry, M., Chandran, D., Kellett, A., Bones, J., Clynes, M., Meleady, P. & Barron, N. (2016), 'Process-relevant concentrations of the leachable bDtbPP impact negatively on CHO cell production characteristics', *Biotechnology Progress* 32(6), 1547–1558. CrossRef (<http://dx.doi.org/10.1002/btpr.2345>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

[title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author](https://scholar.google.com/scholar_lookup?title=Process-relevant+concentrations+of+the+leachable+bDtbPP+impact+negatively+on+CHO+cell+production+characteristics&publication+year=2016&author=Kelly+P.+S.&author=McSweeney+S.&author)

Google Scholar (https://scholar.google.com/scholar_lookup?

title=PCA+and+PLS+with+very+large+data+sets&publication+year=2005&author=Kettaneh+N.&author=Berglund+A.&author=Wold+S.&journal=Computational+Statistics+and+Data+Analysis 85)

Khawli, L. A., Goswami, S., Hutchinson, R., Kwong, Z. W., Yang, J., Wang, X., Yao, Z., Sreedhara, A., Cano, T., Tesar, D., Nijem, I., Allison, D. E., Wong, P. Y., Kao, Y. H., Quan, C., Joshi, A., Harris, R. J. & Motchnik, P. (2010), 'Charge variants in IgG1: Isolation, characterization, in vitro binding properties and pharmacokinetics in rats', *mAbs* 2(6), 613–624. CrossRef (<http://dx.doi.org/10.4161/mabs.2.6.13333>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Charge+variants+in+IgG1:+Isolation+characterization+in+vitro+binding+properties+and+pharmacokinetics+in+rats&publication+year=2010&author=Khawli+L.+A.&author=Goswami+S 624)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20818176>)

Kimura, R. & Miller, W. M. (1996), 'Effects of elevated pCO₂(2) and/or osmolality on the growth and recombinant tPA production of CHO cells.', *Biotechnology and Bioengineering* 52(1), 152–160.

CrossRef ([http://dx.doi.org/10.1002/\(SICI\)1097-0290\(19961005\)52:1<152::AID-BIT15>3.0.CO;2-Q](http://dx.doi.org/10.1002/(SICI)1097-0290(19961005)52:1<152::AID-BIT15>3.0.CO;2-Q)) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Effects+of+elevated+pCO2)+and/or+osmolality+on+the+growth+and+recombinant+tPA+production+of+CHO+cells&publication+year=1996&author=Kimura+R.&author=Miller+W.+M. 8 0290(19961005)52:1%3C152::AID-BIT15%3E3.0.CO;2-Q&pages=152-160)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18629861>)

Kiparissides, A., Koutinas, M., Kontoravdi, C., Mantalaris, A. & Pistikopoulos, E. N. (2011), 'Closing the loop' in biological systems modeling: From the in silico to the in vitro', *Automatica* 47(6), 1147–1155. CrossRef (<http://dx.doi.org/10.1016/j.automatica.2011.01.013>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Closing+the+loop%E2%80%99+in+biological+systems+modeling:+From+the+in+silico+to+the+in+vitro&publication+year=2011&author=Kiparissides+A.&author=Koutinas+M.&author=K 1155)

Kirdar, A. O., Chen, G., Weidner, J. & Rathore, A. S. (2010), 'Application of near-infrared (NIR) spectroscopy for screening of raw materials used in the cell culture medium for the production of a recombinant therapeutic protein', *Biotechnology Progress* 26(2), 527–531.

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Application+of+near-infrared+\(NIR\)+spectroscopy+for+screening+of+raw+materials+used+in+the+cell+culture+medium+for+the+production+of+a+recombinant+therapeutic+protein&publication+year=2010&author=Ki](https://scholar.google.com/scholar_lookup?title=Application+of+near-infrared+(NIR)+spectroscopy+for+screening+of+raw+materials+used+in+the+cell+culture+medium+for+the+production+of+a+recombinant+therapeutic+protein&publication+year=2010&author=Ki)

531)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19938040>)

Kiviharju, K., Salonen, K., Moilanen, U. & Eerikainen, T. (2008), 'Biomass measurement online: The performance of in situ measurements and software sensors', *Journal of Industrial Microbiology and Biotechnology* 35(7), 657–665. CrossRef (<http://dx.doi.org/10.1007/s10295-008-0346-5>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Biomass+measurement+online:+The+performance+of+in+situ+measurements+and+software+sensors&publication+year=2008&author=Kiviharju+K.&author=Salonen+K.&author=Moil 008-0346-5&pages=657-665)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18392869>)

Klutz, S., Holtmann, L., Lobedann, M. & Schembecker, G. (2016), 'Cost evaluation of antibody production processes in different operation modes', *Chemical Engineering Science* 141, 63–74.

CrossRef (<http://dx.doi.org/10.1016/j.ces.2015.10.029>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Cost+evaluation+of+antibody+production+processes+in+different+operation+modes&publication+year=2016&author=Klutz+S.&author=Holtmann+L.&author=Lobedann+M.&author=S 74)

Klutz, S., Magnus, J., Lobedann, M., Schwan, P., Maier, B., Niklas, J., Temming, M. & Schembecker, G. (2015), 'Developing the biofacility of the future based on continuous processing and single-use technology', *Journal of Biotechnology* 213, 120–130. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2015.06.388>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Developing+the+biofacility+of+the+future+based+on+continuous+processing+and+single-use+technology&publication+year=2015&author=Klutz+S.&author=Magnus+J.&author=Lobedann+M.&author=Schwan+P.&author=Maier+B.&author=Niklas+J.&author=Temming+M.+auth

130)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26091773>)

Kochanowski, N., Blanchard, F., Cacan, R., Chirat, F., Guedon, E., Marc, A. & Goergen, J. L. (2008), 'Influence of intracellular nucleotide and nucleotide sugar contents on recombinant interferon- γ glycosylation during batch and fed-batch cultures of CHO cells', *Biotechnology and Bioengineering* 100(4), 721–733. CrossRef (<http://dx.doi.org/10.1002/bit.21816>) |

Google Scholar (<https://scholar.google.com/scholar?q=Kochanowski+N.+Blanchard+F.+Cacan+R.+Chirat+F.+Guedon+E.+Marc+A.+&Goergen+J.+L.+>

(2008)+%E2%80%98Influence+of+intracellular+nucleotide+and+nucleotide+sugar+contents+on+recombinant+interferon-%CE%B3+glycosylation+during+batch+and+fed-batch+cultures+of+CHO+cells%E2%80%99+Biotechnology+and+Bioengineering+100(4)+721%E2%80%93733.)

Kohrt, H. E., Tumeh, P. C., Benson, D., Bhardwaj, N., Brody, J., Formenti, S., Fox, B. A., Galon, J., June, C. H., Kalos, M., Kirsch, I., Kleen, T., Kroemer, G., Lanier, L., Levy, R., Lyster, H. K., Maecker, H., Marabelle, A., Melenhorst, J., Miller, J., Melero, I., Odunsi, K., Palucka, K., Peoples, G., Ribas, A., Robins, H., Robinson, W., Serafini, T., Sondel, P., Vivier, E., Weber, J., Wolchok, J., Zitvogel, L., Disis, M. L. & Cheever, M. A. (2016), 'Immunodynamics: A cancer immunotherapy trials network review of immune monitoring in immuno-oncology clinical trials', *Journal for ImmunoTherapy of Cancer* 4(1), 1–16. CrossRef (<http://dx.doi.org/10.1186/s40425-016-0118-0>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Immunodynamics:+A+cancer+immunotherapy+trials+network+review+of+immune+monitoring+in+immuno-oncology+clinical+trials&publication+year=2016&author=Kohrt+H.+E.&author=Tumeh+P.+C.&author=Benson+D.&author=Bhardwaj+N.&author=Brody+J.&author=Formenti+S.&author=Fox-](https://scholar.google.com/scholar_lookup?title=Immunodynamics:+A+cancer+immunotherapy+trials+network+review+of+immune+monitoring+in+immuno-oncology+clinical+trials&publication+year=2016&author=Kohrt+H.+E.&author=Tumeh+P.+C.&author=Benson+D.&author=Bhardwaj+N.&author=Brody+J.&author=Formenti+S.&author=Fox-016-0118-0&pages=1-16)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26981245>)

Kolwyck, D., Mcsweeney, M. & Johns, J. (2017), *Biomanufacturing Technology Roadmap: Supply Partnership Management, technical report*, BioPhorum Operations Group Ltd.

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Biomanufacturing+Technology+Roadmap:+Supply+Partnership+Management+technical+report&publication+year=2017&author=Kolwyck+D.&author=Mcsweeney+M.&author=Johns+

Konstantinov, K. B. & Cooney, C. L. (2015), 'White paper on continuous bioprocessing May 20–21, 2014 continuous manufacturing symposium', *Journal of Pharmaceutical Sciences* 104(3), 813–820.

CrossRef (<http://dx.doi.org/10.1002/jps.24268>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=White+paper+on+continuous+bioprocessing+May+20%E2%80%9321+2014+continuous+manufacturing+symposium&publication+year=2015&author=Konstantinov+K.+B.&author=Coo 820)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28756843>)

Konstantinov, K. B., Yeong-shou, Tsai, Moles, D. & Matanguihan, R. (1996), 'Control of long-term perfusion Chinese hamster ovary cell culture by glucose auxostat', *Biotechnology Progress* 12(1), 100–109. CrossRef (<http://dx.doi.org/10.1021/bp950044p>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Control+of+long-term+perfusion+Chinese+hamster+ovary+cell+culture+by+glucose+auxostat&publication+year=1996&author=Konstantinov+K.+B.&author=Yeong-shou+Tsai&author=Moles+D.&author=Matanguihan+R.&journal=Biotechnology+Progress&volume=12&doi=10.1021/bp950044p&pages=100-109)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/8845100>)

Konstantinov, K., Chuppa, S., Sajan, E., Tsai, Y., Yoon, S. & Golini, F. (1994), 'Real-time biomass-concentration monitoring in animal-cell cultures', *Trends in Biotechnology* 12(8), 324–333. CrossRef ([http://dx.doi.org/10.1016/0167-7799\(94\)90049-3](http://dx.doi.org/10.1016/0167-7799(94)90049-3)) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Real-time+biomass-concentration+monitoring+in+animal-cell+cultures&publication+year=1994&author=Konstantinov+K.&author=Chuppa+S.&author=Sajan+E.&author=Tsai+Y.&author=Yoon+S.&author=Golini+F.&journal=Trends+in+Biotechnology&volume=12&doi=10.1016/0167-7799\(94\)90049-3&pages=324-333](https://scholar.google.com/scholar_lookup?title=Real-time+biomass-concentration+monitoring+in+animal-cell+cultures&publication+year=1994&author=Konstantinov+K.&author=Chuppa+S.&author=Sajan+E.&author=Tsai+Y.&author=Yoon+S.&author=Golini+F.&journal=Trends+in+Biotechnology&volume=12&doi=10.1016/0167-7799(94)90049-3&pages=324-333))
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/7765262>)

Konstantinov, K., Goudar, C., Ng, M., Meneses, R., Thrift, J., Chuppa, S., Matanguihan, C., Michaels, J. & Naveh, D. (2006), 'The "push-to-low" approach for optimization of high-density perfusion cultures of animal cells', *Advances in Biochemical Engineering/Biotechnology* 101(July), 75–98. CrossRef (http://dx.doi.org/10.1007/10_016) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=The+%E2%80%9Cpush-to-low%E2%80%9D+approach+for+optimization+of+high-density+perfusion+cultures+of+animal+cells&publication+year=2006&author=Konstantinov+K.&author=Goudar+C.&author=Ng+M.&author=Meneses+R.&author=Thrift+J.&author=Chuppa+S.&journal=Advances+in+Biochemical+Engineering+Biotechnology&volume=101&doi=10.1007/10_016)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/16989258>)

Kontoravdi, C., Pistikopoulos, E. N. & Mantalaris, A. (2010), 'Systematic development of predictive mathematical models for animal cell cultures', *Computers and Chemical Engineering* 34(8), 1192–1198. CrossRef (<http://dx.doi.org/10.1016/j.compchemeng.2010.03.012>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Systematic+development+of+predictive+mathematical+models+for+animal+cell+cultures&publication+year=2010&author=Kontoravdi+C.&author=Pistikopoulos+E.+N.&author=Mantalaris+A.&journal=Computers+and+Chemical+Engineering&volume=34&doi=10.1016/j.compchemeng.2010.03.012)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/201003012>)

Krambeck, F. J. & Betenbaugh, M. J. (2005), 'A mathematical model of N-linked glycosylation', *Biotechnology and Bioengineering* 92(6), 711–728. CrossRef (<http://dx.doi.org/10.1002/bit.20645>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=A+mathematical+model+of+N-linked+glycosylation&publication+year=2005&author=Krambeck+F.+J.&author=Betenbaugh+M.+J.&journal=Biotechnology+and+Bioengineering&volume=92&doi=10.1002/bit.20645&pages=711-728)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/16247773>)

Krättli, M., Müller-Späh, T. & Morbidelli, M. (2013), 'Multifraction separation in countercurrent chromatography (MCSGP)', *Biotechnology and Bioengineering* 110(9), 2436–2444. CrossRef (<http://dx.doi.org/10.1002/bit.24901>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Multifraction+separation+in+countercurrent+chromatography+\(MCSGP\)&publication+year=2013&author=Krattli+M.&author=Muller-Spach+T.&author=Morbidelli+M.&journal=Biotechnology+and+Bioengineering&volume=110&doi=10.1002/bit.24901](https://scholar.google.com/scholar_lookup?title=Multifraction+separation+in+countercurrent+chromatography+(MCSGP)&publication+year=2013&author=Krattli+M.&author=Muller-Spach+T.&author=Morbidelli+M.&journal=Biotechnology+and+Bioengineering&volume=110&doi=10.1002/bit.24901))
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24901>)

Kratzer, R., Dorn, I., Mcnaull, S., Rode, C., Lilly, E., Shea, L. O., Campbell, C. & Diluzio, W. (2017), *Biomanufacturing Technology Roadmap: Process Technologies, technical report*, BioPhorum Operations Group Ltd.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Biomanufacturing+Technology+Roadmap:+Process+Technologies+technical+report&publication+year=2017&author=Kratzer+R.&author=Dorn+I.&author=Mcnaull+S.&author=Rode+C.&journal=Biomanufacturing+Technology+Roadmap:+Process+Technologies+technical+report)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/201700000>)

Kreye, S., Stahn, R., Nawrath, K., Danielczyk, A., Goletz, S. & Gmbh, G. (2015), 'GlycoExpress: A toolbox for the high yield production of glycooptimized fully human biopharmaceuticals in perfusion bioreactors at different scales', in *ECD Digital Archives*.

Google Scholar (<https://scholar.google.com/scholar?q=Kreye+S.+Stahn+R.+Nawrath+K.+Danielczyk+A.+Goletz+S.+&Gmbh+G.+&publication+year=2015&author=Kreye+S.+Stahn+R.+Nawrath+K.+Danielczyk+A.+Goletz+S.+&Gmbh+G.+&journal=ECD+Digital+Archives>)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/201500000>)

Kumar, N., Bansal, A., Sarma, G. & Rawal, R. K. (2014), 'Chemometrics tools used in analytical chemistry: An overview', *Talanta* 123, 186–199. CrossRef (<http://dx.doi.org/10.1016/j.talanta.2014.02.003>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Chemometrics+tools+used+in+analytical+chemistry:+An+overview&publication+year=2014&author=Kumar+N.&author=Bansal+A.&author=Sarma+G.&author=Rawal+R.+K.&journal=Talanta&volume=123&doi=10.1016/j.talanta.2014.02.003)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24725882>)

Kunkel, J. P., Jan, D. C., Butler, M. & Jamieson, J. C. (2000), 'Comparisons of the glycosylation of a monoclonal antibody produced under nominally identical cell culture conditions in two different bioreactors', *Biotechnology Progress* 16(3), 462–470. CrossRef (<http://dx.doi.org/10.1021/bp000026u>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Comparisons+of+the+glycosylation+of+a+monoclonal+antibody+produced+under+nominally+identical+cell+culture+conditions+in+two+different+bioreactors&publication+year=2000&doi=10.1021/bp000026u)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/10835250>)

Kunkel, J. P., Jan, D. C., Jamieson, J. C. & Butler, M. (1998), 'Dissolved oxygen concentration in serum-free continuous culture affects N-linked glycosylation of a monoclonal antibody', *Journal of Biotechnology* 62, 55–71. CrossRef ([http://dx.doi.org/10.1016/S0168-1656\(98\)00044-3](http://dx.doi.org/10.1016/S0168-1656(98)00044-3)) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Dissolved+oxygen+concentration+in+serum-free+continuous+culture+affects+N-linked+glycosylation+of+a+monoclonal+antibody&publication+year=1998&author=Kunkel+J.+P.&author=Jan+D.+C.&author=Jamieson+J.+C.&author=Butler+M.&journal=Journal+of+Biotechnology&volume=62&doi=10.1016/S0168-1656\(98\)00044-3&pages=55-71](https://scholar.google.com/scholar_lookup?title=Dissolved+oxygen+concentration+in+serum-free+continuous+culture+affects+N-linked+glycosylation+of+a+monoclonal+antibody&publication+year=1998&author=Kunkel+J.+P.&author=Jan+D.+C.&author=Jamieson+J.+C.&author=Butler+M.&journal=Journal+of+Biotechnology&volume=62&doi=10.1016/S0168-1656(98)00044-3&pages=55-71))
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/9684342>)

Lai, T., Yang, Y. & Ng, S. K. (2013), 'Advances in mammalian cell line development technologies', *Pharmaceuticals* 6, 579–603. CrossRef (<http://dx.doi.org/10.3390/ph6050579>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Advances+in+mammalian+cell+line+development+technologies&publication+year=2013&author=Lai+T.&author=Yang+Y.&author=Ng+S.+K.&journal=Pharmaceuticals&volume=6&doi=10.3390/ph6050579)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24276168>)

Lairson, L., Henrissat, B., Davies, G. & Withers, S. (2008), 'Glycosyltransferases: Structures, functions, and mechanisms', *Annual Review of Biochemistry* 77(1), 521–555. CrossRef (<http://dx.doi.org/10.1146/annurev.biochem.76.061005.092322>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Glycosyltransferases:+Structures+functions+and+mechanisms&publication+year=2008&author=Lairson+L.&author=Henrissat+B.&author=Davies+G.&author=Withers+S.&journal=Annual+Review+of+Biochemistry&volume=77&doi=10.1146/annurev.biochem.76.061005.092322)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18518825>)

-
- Lameris, R., de Bruin, R. C., Schneiders, F. L., van, Henegouwen, Bergen en, Verheul, P. M., de Gruij, H. M., van der Vliet, T. D. H. J. (2014), 'Bispecific antibody platforms for cancer immunotherapy', *Critical Reviews in Oncology/Hematology* 92(3), 153–165. CrossRef (<http://dx.doi.org/10.1016/j.critrevonc.2014.08.003>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Bispecific+antibody+platforms+for+cancer+immunotherapy&publication+year=2014&author=Lameris+R.&author=de+Bruin+R.+C.&author=Schneiders+F.+L.&author=van&author=Hen+165) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25195094>)
-
- Langer, E. S. (2011), 'Trends in perfusion bioreactors: The next revolution in bioprocessing?', *BioProcess International* 9(10), 18–22. Google Scholar (https://scholar.google.com/scholar_lookup?title=Trends+in+perfusion+bioreactors:+The+next+revolution+in+bioprocessing?&publication+year=2011&author=Langer+E.+S.&journal=BioProcess+International&volume=9&pages=18-22)
-
- Langer, E. S. (2014), 'Continuous bioprocessing and perfusion: Wider adoption coming as bioprocessing matures', *BioProcessing Journal* 13(1), 43–49. CrossRef (<http://dx.doi.org/10.12665/j131.Langer>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Continuous+bioprocessing+and+perfusion:+Wider+adoption+coming+as+bioprocessing+matures&publication+year=2014&author=Langer+E.+S.&journal=BioProcessing+Journal&volu+49)
-
- Lavery, M. & Nienow, A. W. (1987), 'Oxygen transfer in animal cell culture medium', *Biotechnology and Bioengineering* 30(3), 368–373. CrossRef (<http://dx.doi.org/10.1002/bit.260300307>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Oxygen+transfer+in+animal+cell+culture+medium&publication+year=1987&author=Lavery+M.&author=Nienow+A.+W.&journal=Biotechnology+and+Bioengineering&volume=30&doi=373) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18581370>)
-
- Le, K., Tan, C., Gupta, S., Guhan, T., Barkhordarian, H., Lull, J., Stevens, J. & Munro, T. (2018), 'A novel mammalian cell line development platform utilizing nanofluidics and optoelectro positioning technology', *Biotechnology Progress* 34(6), 1438–1446. CrossRef (<http://dx.doi.org/10.1002/btpr.2690>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=A+novel+mammalian+cell+line+development+platform+utilizing+nanofluidics+and+optoelectro+positioning+technology&publication+year=2018&author=Le+K.&author=Tan+C.&auth+1446) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30009534>)
-
- Li, J., Wong, C. L., Vijayasankaran, N., Hudson, T. & Amanullah, A. (2012), 'Feeding lactate for CHO cell culture processes: Impact on culture metabolism and performance', *Biotechnology and Bioengineering* 109(5), 1173–1186. CrossRef (<http://dx.doi.org/10.1002/bit.24389>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Feeding+lactate+for+CHO+cell+culture+processes:+Impact+on+culture+metabolism+and+performance&publication+year=2012&author=Li+J.&author=Wong+C.+L.&author=Vijayasank+1186) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22124879>)
-
- Lim, Y., Wong, N. S. C., Lee, Y. Y., Ku, S. C. Y., Wong, D. C. F. & Yap, M. G. S. (2010), 'Engineering mammalian cells in bioprocessing: Current achievements and future perspectives', *Biotechnology and Applied Biochemistry* 55(4), 175–189. CrossRef (<http://dx.doi.org/10.1042/BA20090363>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Engineering+mammalian+cells+in+bioprocessing:+Current+achievements+and+future+perspectives&publication+year=2010&author=Lim+Y.&author=Wong+N.+S.+C.&author=Lee+Y.+189) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20392202>)
-
- Lin, H., Leighty, R. W., Godfrey, S. & Wang, S. B. (2017), 'Principles and approach to developing mammalian cell culture media for high cell density perfusion process leveraging established fed-batch media', *Biotechnology Progress* 33(4), 891–901. CrossRef (<http://dx.doi.org/10.1002/btpr.2472>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Principles+and+approach+to+developing+mammalian+cell+culture+media+for+high+cell+density+perfusion+process+leveraging+established+fed-batch+media&publication+year=2017&author=Lin+H.&author=Leighty+R.+W.&author=Godfrey+S.&author=Wang+S.+B.&journal=Biotechnology+Progress&volume=33&doi=10.1002/btpr.24+901) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28371394>)
-
- Ling, W. L. (2015), 'Development of protein-free medium for therapeutic protein production in mammalian cells: Recent advances and perspectives', *Pharmaceutical Bioprocessing* 3, 215–226. CrossRef (<http://dx.doi.org/10.4155/pbp.15.8>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Development+of+protein-free+medium+for+therapeutic+protein+production+in+mammalian+cells:+Recent+advances+and+perspectives&publication+year=2015&author=Ling+W.+L.&journal=Pharmaceutal+Biopr+226)
-
- Lobban, P. (1972), An Enzymatic Method for End-to-End Joining of DNA Molecules, PhD thesis, Stanford University. Google Scholar (<https://scholar.google.com/scholar?q=Lobban+P.+1972+An+Enzymatic+Method+for+End-to-End+Joining+of+DNA+Molecules+PhD+thesis+Stanford+University.>)
-
- Lobban, P. & Kaiser, A. (1973), 'Enzymatic end-to-end joining of DNA molecules', *Journal of Molecular Biology* 78, 483–471. CrossRef ([http://dx.doi.org/10.1016/0022-2836\(73\)90468-3](http://dx.doi.org/10.1016/0022-2836(73)90468-3)) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Enzymatic+end-to-end+joining+of+DNA+molecules&publication+year=1973&author=Lobban+P.&author=Kaiser+A.&journal=Journal+of+Molecular+Biology&volume=78&doi=10.1016/0022-2836\(73\)90468-3&pages=483-471](https://scholar.google.com/scholar_lookup?title=Enzymatic+end-to-end+joining+of+DNA+molecules&publication+year=1973&author=Lobban+P.&author=Kaiser+A.&journal=Journal+of+Molecular+Biology&volume=78&doi=10.1016/0022-2836(73)90468-3&pages=483-471)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/4754844>)
-
- Löffelholz, C., Kaiser, S. C., Kraume, M., Eibl, R. & Eibl, D. (2014), 'Dynamic single-use bioreactors used in modern liter- and m3- scale biotechnological processes: Engineering characteristics and scaling up', *Advances in Biochemical Engineering/Biotechnology* 138, 1–44. Google Scholar (https://scholar.google.com/scholar_lookup?title=Dynamic+single-use+bioreactors+used+in+modern+liter-+and+m3-+scale+biotechnological+processes:+Engineering+characteristics+and+scaling+up&publication+year=2014&author=L%C3%B6ffelholz+C.&author=Kaiser+S.+C.&author=Kraume+M.&author+44)
-
- Long, Q., Liu, X., Yang, Y., Li, L., Harvey, L., McNeil, B. & Bai, Z. (2014), 'The development and application of high throughput cultivation technology in bioprocess development', *Journal of Biotechnology* 192, 323–338. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2014.03.028>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=The+development+and+application+of+high+throughput+cultivation+technology+in+bioprocess+development&publication+year=2014&author=Long+Q.&author=Liu+X.&author=Yang+338) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24698846>)

-
- Losfeld, M.-E., Scibona, E., Lin, C.-W., Villiger, T. K., Gauss, R., Morbidelli, M. & Aebi, M. (2017), 'Influence of protein/glycan interaction on site-specific glycan heterogeneity', *The FASEB Journal* 31(10), 4623–4635. [CrossRef \(http://dx.doi.org/10.1096/fj.201700403R\)](http://dx.doi.org/10.1096/fj.201700403R) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Influence+of+protein/glycan+interaction+on+site-specific+glycan+heterogeneity&publication+year=2017&author=Losfeld+M.-E.&author=Scibona+E.&author=Lin+C.-W.&author=Villiger+T.+K.&author=Gauss+R.&author=Morbidelli+M.&author=Aebi+M.&journal=The+FASEB+Journal&volume=31&doi=10.1096/fj.201700403R&pages=4623-4635\)](https://scholar.google.com/scholar_lookup?title=Influence+of+protein/glycan+interaction+on+site-specific+glycan+heterogeneity&publication+year=2017&author=Losfeld+M.-E.&author=Scibona+E.&author=Lin+C.-W.&author=Villiger+T.+K.&author=Gauss+R.&author=Morbidelli+M.&author=Aebi+M.&journal=The+FASEB+Journal&volume=31&doi=10.1096/fj.201700403R&pages=4623-4635) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/28679530\)](https://www.ncbi.nlm.nih.gov/pubmed/28679530)
-
- Luo, Y. & Chen, G. (2007), 'Combined approach of NMR and chemometrics for screening peptones used in the cell culture medium for the production of a recombinant therapeutic protein', *Biotechnology and Bioengineering* 97(6), 1654–1659. [CrossRef \(http://dx.doi.org/10.1002/bit.21365\)](http://dx.doi.org/10.1002/bit.21365) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Combined+approach+of+NMR+and+chemometrics+for+screening+peptones+used+in+the+cell+culture+medium+for+the+production+of+a+recombinant+therapeutic+protein&public:1659\)](https://scholar.google.com/scholar_lookup?title=Combined+approach+of+NMR+and+chemometrics+for+screening+peptones+used+in+the+cell+culture+medium+for+the+production+of+a+recombinant+therapeutic+protein&public:1659) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/17274067\)](https://www.ncbi.nlm.nih.gov/pubmed/17274067)
-
- Luttmann, R., Borchert, S. O., Mueller, C., Loegering, K., Aupert, F., Weyand, S., Kober, C., Faber, B. & Cornelissen, G. (2015), 'Sequential/parallel production of potential Malaria vaccines: A direct way from single batch to quasi-continuous integrated production', *Journal of Biotechnology* 213, 83–96. [CrossRef \(http://dx.doi.org/10.1016/j.jbiotec.2015.02.022\)](http://dx.doi.org/10.1016/j.jbiotec.2015.02.022) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Sequential/parallel+production+of+potential+Malaria+vaccines:+A+direct+way+from+single+batch+to+quasi-continuous+integrated+production&publication+year=2015&author=Luttmann+R.&author=Borchert+S.O.&author=Mueller+C.&author=Loegering+K.&author=Aupert+F.&author=Weyand+96\)](https://scholar.google.com/scholar_lookup?title=Sequential/parallel+production+of+potential+Malaria+vaccines:+A+direct+way+from+single+batch+to+quasi-continuous+integrated+production&publication+year=2015&author=Luttmann+R.&author=Borchert+S.O.&author=Mueller+C.&author=Loegering+K.&author=Aupert+F.&author=Weyand+96) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/25736485\)](https://www.ncbi.nlm.nih.gov/pubmed/25736485)
-
- Luttmann, R., Bracewell, D. G., Cornelissen, G., Gernaey, K. V., Glassey, J., Hass, V. C., Kaiser, C., Preusse, C., Striedner, G. & Mandenius, C. (2012), 'Soft sensors in bioprocessing: A status report and recommendations', *Biotechnology Journal* 7(8), 1040–1048. [CrossRef \(http://dx.doi.org/10.1002/biot.201100506\)](http://dx.doi.org/10.1002/biot.201100506) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Soft+sensors+in+bioprocessing:+A+status+report+and+recommendations&publication+year=2012&author=Luttmann+R.&author=Bracewell+D.+G.&author=Cornelissen+G.&author=G:1048\)](https://scholar.google.com/scholar_lookup?title=Soft+sensors+in+bioprocessing:+A+status+report+and+recommendations&publication+year=2012&author=Luttmann+R.&author=Bracewell+D.+G.&author=Cornelissen+G.&author=G:1048) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/22489000\)](https://www.ncbi.nlm.nih.gov/pubmed/22489000)
-
- Ma, N., Mollet, M. & Chalmers, J. J. (2003), 'Aeration, mixing and hydrodynamics in bioreactors', in *Encyclopedia of Cell Technology*, John Wiley and Sons, Inc., Hoboken, pp. 225–248. [CrossRef \(http://dx.doi.org/10.1002/0471250570.spi006\)](http://dx.doi.org/10.1002/0471250570.spi006) | [Google Scholar \(https://scholar.google.com/scholar?q=Ma+N.+Mollet+M.+&Chalmers+J.+J.+&publication+year=2003&author=Ma+N.+Mollet+M.+&Chalmers+J.+J.+&journal=Encyclopedia+of+Cell+Technology+John+Wiley+and+Sons+Inc.+Hoboken+pp.+225-248\)](https://scholar.google.com/scholar?q=Ma+N.+Mollet+M.+&Chalmers+J.+J.+&publication+year=2003&author=Ma+N.+Mollet+M.+&Chalmers+J.+J.+&journal=Encyclopedia+of+Cell+Technology+John+Wiley+and+Sons+Inc.+Hoboken+pp.+225-248)
-
- Manchester, K. L. (2007), 'Louis Pasteur, fermentation, and a rival', *South African Journal of Science* 103(9–10), 377–380. [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Louis+Pasteur+fermentation+and+a+rival&publication+year=2007&author=Manchester+K.+L.&journal=South+African+Journal+of+Science&volume=103&pages=377-380\)](https://scholar.google.com/scholar_lookup?title=Louis+Pasteur+fermentation+and+a+rival&publication+year=2007&author=Manchester+K.+L.&journal=South+African+Journal+of+Science&volume=103&pages=377-380)
-
- Marks, D. M. (2003), 'Equipment design considerations for large scale cell culture', *Cytotechnology* 42(1), 21–33. [CrossRef \(http://dx.doi.org/10.1023/A:1026103405618\)](http://dx.doi.org/10.1023/A:1026103405618) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Equipment+design+considerations+for+large+scale+cell+culture&publication+year=2003&author=Marks+D.+M.&journal=Cytotechnology&volume=42&doi=10.1023/A:1026103405618&33\)](https://scholar.google.com/scholar_lookup?title=Equipment+design+considerations+for+large+scale+cell+culture&publication+year=2003&author=Marks+D.+M.&journal=Cytotechnology&volume=42&doi=10.1023/A:1026103405618&33) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/19002925\)](https://www.ncbi.nlm.nih.gov/pubmed/19002925)
-
- Matthews, T. E., Berry, B. N., Smelko, J., Moretto, J., Moore, B. & Wiltberger, K. (2016), 'Closed loop control of lactate concentration in mammalian cell culture by Raman spectroscopy leads to improved cell density, viability, and biopharmaceutical protein production', *Biotechnology and Bioengineering* 113(11), 2416–2424. [CrossRef \(http://dx.doi.org/10.1002/bit.26018\)](http://dx.doi.org/10.1002/bit.26018) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Closed+loop+control+of+lactate+concentration+in+mammalian+cell+culture+by+Raman+spectroscopy+leads+to+improved+cell+density+viability+and+biopharmaceutical+protein+pro:2424\)](https://scholar.google.com/scholar_lookup?title=Closed+loop+control+of+lactate+concentration+in+mammalian+cell+culture+by+Raman+spectroscopy+leads+to+improved+cell+density+viability+and+biopharmaceutical+protein+pro:2424) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/27215441\)](https://www.ncbi.nlm.nih.gov/pubmed/27215441)
-
- McCoy, R. E., Costa, N. a. & Morris, A. E. (2015), 'Factors that determine stability of highly concentrated chemically defined production media', *Biotechnology Progress* 31(2), 493–502. [CrossRef \(http://dx.doi.org/10.1002/btpr.2047\)](http://dx.doi.org/10.1002/btpr.2047) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Factors+that+determine+stability+of+highly+concentrated+chemically+defined+production+media&publication+year=2015&author=McCoy+R.+E.&author=Costa+N.+a.&author=Morris+502\)](https://scholar.google.com/scholar_lookup?title=Factors+that+determine+stability+of+highly+concentrated+chemically+defined+production+media&publication+year=2015&author=McCoy+R.+E.&author=Costa+N.+a.&author=Morris+502) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/25641710\)](https://www.ncbi.nlm.nih.gov/pubmed/25641710)
-
- McCracken, N. A., Kowle, R. & Ouyang, A. (2014), 'Control of galactosylated glycoforms distribution in cell culture system', *Biotechnology Progress* 30(3), 547–553. [CrossRef \(http://dx.doi.org/10.1002/btpr.1906\)](http://dx.doi.org/10.1002/btpr.1906) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Control+of+galactosylated+glycoforms+distribution+in+cell+culture+system&publication+year=2014&author=McCracken+N.+A.&author=Kowle+R.&author=Ouyang+A.&journal=Biotech+553\)](https://scholar.google.com/scholar_lookup?title=Control+of+galactosylated+glycoforms+distribution+in+cell+culture+system&publication+year=2014&author=McCracken+N.+A.&author=Kowle+R.&author=Ouyang+A.&journal=Biotech+553) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/24692242\)](https://www.ncbi.nlm.nih.gov/pubmed/24692242)
-
- McGovern, P. E., Glusker, D. L., Exner, L. J. & Voigt, M. M. (1996), 'Neolithic resinated wine', *Nature* 381, 480–481. [CrossRef \(http://dx.doi.org/10.1038/381480a0\)](http://dx.doi.org/10.1038/381480a0) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Neolithic+resinated+wine&publication+year=1996&author=McGovern+P.+E.&author=Glusker+D.+L.&author=Exner+L.+J.&author=Voigt+M.+M.&journal=Nature&volume=381&doi=10.1038/381480a0\)](https://scholar.google.com/scholar_lookup?title=Neolithic+resinated+wine&publication+year=1996&author=McGovern+P.+E.&author=Glusker+D.+L.&author=Exner+L.+J.&author=Voigt+M.+M.&journal=Nature&volume=381&doi=10.1038/381480a0)
-
- McGovern, P. E., Zhang, J., Tang, J., Zhang, Z., Hall, G. R., Moreau, R. A., Nunez, A., Butrym, E. D., Richards, M. P., Wang, C.-S., Cheng, G., Zhao, Z. & Wang, C. (2004), 'Fermented beverages of pre- and proto-historic China', *Proceedings of the National Academy of Sciences* 101(51), 17593–17598. [CrossRef \(http://dx.doi.org/10.1073/pnas.0407921102\)](http://dx.doi.org/10.1073/pnas.0407921102) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Fermented+beverages+of+pre+and+proto-historic+China&publication+year=2004&author=McGovern+P.+E.&author=Zhang+J.&author=Tang+J.&author=Zhang+Z.&author=Hall+G.+R.&author=Moreau+R.+A.&author=Nunez+A.&author=S.+&author=Cheng+G.&author=Zhao+Z.&author=Wang+C.&journal=Proceedings+of+the+National+Academy+of+Sciences&volume=101&doi=10.1073/pnas.0407921102&pages=17593-17598\)](https://scholar.google.com/scholar_lookup?title=Fermented+beverages+of+pre+and+proto-historic+China&publication+year=2004&author=McGovern+P.+E.&author=Zhang+J.&author=Tang+J.&author=Zhang+Z.&author=Hall+G.+R.&author=Moreau+R.+A.&author=Nunez+A.&author=S.+&author=Cheng+G.&author=Zhao+Z.&author=Wang+C.&journal=Proceedings+of+the+National+Academy+of+Sciences&volume=101&doi=10.1073/pnas.0407921102&pages=17593-17598) | [PubMed \(https://www.ncbi.nlm.nih.gov/pubmed/15590771\)](https://www.ncbi.nlm.nih.gov/pubmed/15590771)
-
- Meglen, R. R. (1992), 'Examining large databases: A chemometric approach using principal component analysis', *Marine Chemistry* 39(1–3), 217–237. [CrossRef \(http://dx.doi.org/10.1016/0304-4203\(92\)90103-H\)](http://dx.doi.org/10.1016/0304-4203(92)90103-H) |

Google Scholar ([https://scholar.google.com/scholar_lookup?](https://scholar.google.com/scholar_lookup?title=Examining+large+databases:+A+chemometric+approach+using+principal+component+analysis&publication+year=1992&author=Meglen+R.+R.&journal=Marine+Chemistry&volume=34203(92)90103-H&pages=217-237)

[title=Examining+large+databases:+A+chemometric+approach+using+principal+component+analysis&publication+year=1992&author=Meglen+R.+R.&journal=Marine+Chemistry&volume=34203\(92\)90103-H&pages=217-237](https://scholar.google.com/scholar_lookup?title=Examining+large+databases:+A+chemometric+approach+using+principal+component+analysis&publication+year=1992&author=Meglen+R.+R.&journal=Marine+Chemistry&volume=34203(92)90103-H&pages=217-237))

Mehdizadeh, H., Lauri, D., Karry, K. M., Moshgbar, M., Procopio-Melino, R. & Drapeau, D. (2015), 'Generic Raman-based calibration models enabling real-time monitoring of cell culture bioreactors', *Biotechnology Progress* 31(4), 1004–1013. [CrossRef \(http://dx.doi.org/10.1002/btpr.2079\)](http://dx.doi.org/10.1002/btpr.2079) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Generic+Raman-based+calibration+models+enabling+real-time+monitoring+of+cell+culture+bioreactors&publication+year=2015&author=Mehdizadeh+H.&author=Lauri+D.&author=Karry+K.+M.&author=Moshgbar+M.&author=Procopio-Melino+R.&author=Drapeau+D.&journal=Biotechnology+Progress&volume=31&doi=10.1002/btpr.2079&pages=1004-1013)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25825868>)

Meier, S. J., Hatton, T. A. & Wang, D. I. (1999), 'Cell death from bursting bubbles: Role of cell attachment to rising bubbles in sparged reactors', *Biotechnology and Bioengineering* 62(4), 468–478.

[CrossRef \(http://dx.doi.org/10.1002/\(SICI\)1097-0290\(19990220\)62:4<468::AID-BIT10>3.0.CO;2-N\)](http://dx.doi.org/10.1002/(SICI)1097-0290(19990220)62:4<468::AID-BIT10>3.0.CO;2-N) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Cell+death+from+bursting+bubbles:+Role+of+cell+attachment+to+rising+bubbles+in+sparged+reactors&publication+year=1999&author=Meier+S.+J.&author=Hatton+T.+A.&author=W0290\(19990220\)62:4%3C468::AID-BIT10%3E3.0.CO;2-N&pages=468-478](https://scholar.google.com/scholar_lookup?title=Cell+death+from+bursting+bubbles:+Role+of+cell+attachment+to+rising+bubbles+in+sparged+reactors&publication+year=1999&author=Meier+S.+J.&author=Hatton+T.+A.&author=W0290(19990220)62:4%3C468::AID-BIT10%3E3.0.CO;2-N&pages=468-478))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/9921156>)

Mercier, S. M., Diepenbroek, B., Wijffels, R. H. & Streefland, M. (2014), 'Multivariate PAT solutions for biopharmaceutical cultivation: current progress and limitations', *Trends in Biotechnology* 32(6), 329–336. [CrossRef \(http://dx.doi.org/10.1016/j.tibtech.2014.03.008\)](http://dx.doi.org/10.1016/j.tibtech.2014.03.008) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Multivariate+PAT+solutions+for+biopharmaceutical+cultivation:+current+progress+and+limitations&publication+year=2014&author=Mercier+S.+M.&author=Diepenbroek+B.&author=336)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24732022>)

Mercier, S. M., Rouel, P. M. & Lebrun, P. (2016), 'Process analytical technology tools for perfusion cell culture', *Engineering in Life Sciences* 16(1), 25–35.

[CrossRef \(http://dx.doi.org/10.1002/elsc.201500035\)](http://dx.doi.org/10.1002/elsc.201500035) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Process+analytical+technology+tools+for+perfusion+cell+culture&publication+year=2016&author=Mercier+S.+M.&author=Rouel+P.+M.&author=Lebrun+P.&journal=Engineering+in+Li35)

Mercille, S., Johnson, M., Lanthier, S., Kamen, A. A. & Messie, B. (2000), 'Understanding factors that limit the productivity of suspension-based perfusion cultures operated at high medium renewal rates', *Biotechnology and Bioengineering* 67(4), 435–450. [CrossRef \(http://dx.doi.org/10.1002/\(SICI\)1097-0290\(20000220\)67:4<435::AID-BIT7>3.0.CO;2-Y\)](http://dx.doi.org/10.1002/(SICI)1097-0290(20000220)67:4<435::AID-BIT7>3.0.CO;2-Y) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Understanding+factors+that+limit+the+productivity+of+suspension-based+perfusion+cultures+operated+at+high+medium+renewal+rates&publication+year=2000&author=Mercille+S.&author=Johnson+M.&author=Lanthier+S.&author=Kamen+A.+A.&author0290\(20000220\)67:4%3C435::AID-BIT7%3E3.0.CO;2-Y&pages=435-450](https://scholar.google.com/scholar_lookup?title=Understanding+factors+that+limit+the+productivity+of+suspension-based+perfusion+cultures+operated+at+high+medium+renewal+rates&publication+year=2000&author=Mercille+S.&author=Johnson+M.&author=Lanthier+S.&author=Kamen+A.+A.&author0290(20000220)67:4%3C435::AID-BIT7%3E3.0.CO;2-Y&pages=435-450))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/10620759>)

Mertz, J. E. & Davis, R. (1972), 'Cleavage of DNA by RI restriction endonuclease generates cohesive ends', *Proceedings of the National Academy of Sciences of the United States of America* 69(11), 3370–3374. [CrossRef \(http://dx.doi.org/10.1073/pnas.69.11.3370\)](http://dx.doi.org/10.1073/pnas.69.11.3370) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Cleavage+of+DNA+by+RI+restriction+endonuclease+generates+cohesive+ends&publication+year=1972&author=Mertz+J.+E.&author=Davis+R.&journal=Proceedings+of+the+National+3374)

Meuwly, F., Weber, U., Ziegler, T., Gervais, A., Mastrangeli, R., Crisci, C., Rossi, M., Bernard, A., von Stockar, U. & Kadouri, A. (2006), 'Conversion of a CHO cell culture process from perfusion to fed-batch technology without altering product quality', *Journal of Biotechnology* 123(1), 106–116. [CrossRef \(http://dx.doi.org/10.1016/j.jbiotec.2005.10.013\)](http://dx.doi.org/10.1016/j.jbiotec.2005.10.013) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Conversion+of+a+CHO+cell+culture+process+from+perfusion+to+fed-batch+technology+without+altering+product+quality&publication+year=2006&author=Meuwly+F.&author=Weber+U.&author=Ziegler+T.&author=Gervais+A.&author=Mastrangeli+R.&author116)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/16324762>)

Miller, W. M., Blanch, H. W. & Wilke, C. R. (1988), 'A kinetic analysis of hybridoma growth and metabolism in batch and continuous suspension culture: effect of nutrient concentration, dilution rate, and pH', *Biotechnology and Bioengineering* 32(8), 947–965. [CrossRef \(http://dx.doi.org/10.1002/bit.260320803\)](http://dx.doi.org/10.1002/bit.260320803) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=A+kinetic+analysis+of+hybridoma+growth+and+metabolism+in+batch+and+continuous+suspension+culture:+effect+of+nutrient+concentration+dilution+rate+and+pH&publication+year=1988&author=Miller+W.+M.&author=Blanch+H.+W.&author=Wilke+C.+R.&author965)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18587813>)

Miller, W. M., Wilke, C. R. & Blanch, H. W. (1987), 'Effects of dissolved oxygen concentration on hybridoma growth and metabolism in continuous culture', *Journal of Cellular Physiology* 132(3), 524–530. [CrossRef \(http://dx.doi.org/10.1002/jcp.1041320315\)](http://dx.doi.org/10.1002/jcp.1041320315) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Effects+of+dissolved+oxygen+concentration+on+hybridoma+growth+and+metabolism+in+continuous+culture&publication+year=1987&author=Miller+W.+M.&author=Wilke+C.+R.&author530)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/3654764>)

Mokuolu, S. (2018), 'New standards define single-use materials qualification', *Pharmaceutical Technology* 42(2), 52–53.

Google Scholar (https://scholar.google.com/scholar_lookup?title=New+standards+define+single-use+materials+qualification&publication+year=2018&author=Mokuolu+S.&journal=Pharmaceutical+Technology&volume=42&pages=52-53)

Monteil, D. T., Juvet, V., Paz, J., Moniatte, M., Baldi, L., Hacker, D. L. & Wurm, F. M. (2016), 'A comparison of orbitally-shaken and stirred-tank bioreactors: pH modulation and bioreactor type affect CHO cell growth and protein glycosylation', *Biotechnology* [CrossRef \(http://dx.doi.org/10.1002/btpr.2328\)](http://dx.doi.org/10.1002/btpr.2328) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=A+comparison+of+orbitally-shaken+and+stirred-tank+bioreactors:+pH+modulation+and+bioreactor+type+effect+CHO+cell+growth+and+protein+glycosylation&publication+year=2016&author=Monteil+D.+T.&author=Juvet+V.&author=P0290\(20160220\)38:2%3C100::AID-BT10%3E3.0.CO;2-Y&pages=99-107](https://scholar.google.com/scholar_lookup?title=A+comparison+of+orbitally-shaken+and+stirred-tank+bioreactors:+pH+modulation+and+bioreactor+type+effect+CHO+cell+growth+and+protein+glycosylation&publication+year=2016&author=Monteil+D.+T.&author=Juvet+V.&author=P0290(20160220)38:2%3C100::AID-BT10%3E3.0.CO;2-Y&pages=99-107))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27051411>)

Morari, M. & Zafiriou, E. (1989), *Robust Process Control*. Prentice Hall.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Robust+Process+Control&publication+year=1989&author=Morari+M.&author=Zafiriou+E.)

Morrow, J., Cohen, S. N., Chang, A. C., Boyer, H. W. & Goodman, H. M. (1974), 'Replication and transcription of eukaryotic DNA in *Escherichia coli*', *Proceedings of the National Academy of Sciences of the United States of America* 71(5), 1743–1747. [CrossRef \(http://dx.doi.org/10.1073/pnas.71.5.1743\)](http://dx.doi.org/10.1073/pnas.71.5.1743) |

- Google Scholar (<https://scholar.google.com/scholar?q=Morrow+J.+Cohen+S.+N.+Chang+A.+C.+Boyer+H.+W.+&+Goodman+H.+M.+>
(1974)+%E2%80%98Replication+and+transcription+of+eukaryotic+DNA+in+Escherichia+coli%E2%80%99+Proceedings+of+the+National+Academy+of+the+United+States+of+Am
-
- Moussa, A. S., Soos, M., Sefcik, J. & Morbidelli, M. (2007), 'Effect of solid volume fraction on aggregation and breakage in colloidal suspensions in batch and continuous stirred tanks', *Langmuir* 23(4), 1664–1673. CrossRef (<http://dx.doi.org/10.1021/la062138m>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Effect+of+solid+volume+fraction+on+aggregation+and+breakage+in+colloidal+suspensions+in+batch+and+continuous+stirred+tanks&publication+year=2007&author=Moussa+A.+S.&1673)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/17279643>)
-
- Moyle, D. (2017), *Biomanufacturing Technology Roadmap: Modular and Mobile*, Technical report, BioPhorum Operations Group Ltd.
Google Scholar (<https://scholar.google.com/scholar?q=Moyle+D.+>
(2017)+Biomanufacturing+Technology+Roadmap:+Modular+and+Mobile+Technical+report+BioPhorum+Operations+Group+Ltd.)
-
- Mulukutla, B. C., Gramer, M. & Hu, W. S. (2012), 'On metabolic shift to lactate consumption in fed-batch culture of mammalian cells', *Metabolic Engineering* 14(2), 138–149.
CrossRef (<http://dx.doi.org/10.1016/j.ymben.2011.12.006>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=On+metabolic+shift+to+lactate+consumption+in+fed-batch+culture+of+mammalian+cells&publication+year=2012&author=Mulukutla+B.+C.&author=Gramer+M.&author=Hu+W.+S.&journal=Metabolic+Engineering&volume=14&doi=10.1016/j.ymben.2011.12.006)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22244936>)
-
- Narayanan, H., Luna, M., von Stoch, M., Cruz Bournazou, M., Polotti, G., Morbidelli, M., Butté, A. & Sokolov, M. (2019a), 'Bioprocess in the digital age: The role of process models', *Biotechnology Journal* 15(1), <https://doi.org/10.1002/biot.201900172> (<https://doi.org/10.1002/biot.201900172>).
Google Scholar (https://scholar.google.com/scholar_lookup?title=Bioprocess+in+the+digital+age:+The+role+of+process+models&publication+year=2019&author=Narayanan+H.&author=Luna+M.&author=Stoch+M.&author=Cruz+Bournazou+M.&author=Sokolov+M.&author=Butte+A.+
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/31486583>)
-
- Narayanan, H., Sokolov, M., Butté, A. & Morbidelli, M. (2019b), 'Decision Tree-PLS (DT-PLS) algorithm for the development of process: Specific local prediction models', *Biotechnology Progress* 35(4), e2818. CrossRef (<http://dx.doi.org/10.1002/btpr.2818>) |
Google Scholar ([https://scholar.google.com/scholar_lookup?title=Decision+Tree-PLS+\(DT-PLS\)+algorithm+for+the+development+of+process:+Specific+local+prediction+models&publication+year=2019&author=Narayanan+H.&author=Sokolov+M.&author=Butte+A.+&author=Morbidelli+M.](https://scholar.google.com/scholar_lookup?title=Decision+Tree-PLS+(DT-PLS)+algorithm+for+the+development+of+process:+Specific+local+prediction+models&publication+year=2019&author=Narayanan+H.&author=Sokolov+M.&author=Butte+A.+&author=Morbidelli+M.)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30969466>)
-
- Narayanan, H., Sokolov, M., Morbidelli, M. & Butté, A. (2019c), 'A new generation of predictive models: The added value of hybrid models for manufacturing processes of therapeutic proteins', *Biotechnology and Bioengineering* 116(10), 2540–2549. CrossRef (<http://dx.doi.org/10.1002/bit.27097>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=A+new+generation+of+predictive+models:+The+added+value+of+hybrid+models+for+manufacturing+processes+of+therapeutic+proteins&publication+year=2019&author=Narayanan+H.&author=Sokolov+M.&author=Morbidelli+M.&author=Butte+A.
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/31237678>)
-
- Nasr, M. M., Krumme, M., Matsuda, Y., Trout, B. L., Badman, C., Mascia, S., Cooney, C. L., Jensen, K. D., Florence, A., Johnston, C., Konstantinov, K. & Lee, S. L. (2017), 'Regulatory perspectives on continuous pharmaceutical manufacturing: Moving from theory to practice, September 26–27, 2016, International Symposium on the Continuous Manufacturing of Pharmaceuticals', *Journal of Pharmaceutical Sciences* 106(11), 3199–3206. CrossRef (<http://dx.doi.org/10.1016/j.xphs.2017.06.015>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Regulatory+perspectives+on+continuous+pharmaceutical+manufacturing:+Moving+from+theory+to+practice+September+26-27+2016+International+Symposium+on+the+Continuous+Manufacturing+of+Pharmaceuticals&publication+year=2017&author=Nasr+M.+M.&author=Krumme+M.&author=Matsuda+Y.&author=Trout+B.+L.&author=Badman+C.&author=Mascia+S.&author=Cooney+C.+L.&author=Jensen+K.+D.&author=Florence+A.&author=Johnston+C.&author=Konstantinov+K.&author=Lee+S.+L.
3206)
-
- Neunstoecklin, B., Stettler, M., Solacroup, T., Broly, H., Morbidelli, M. & Soos, M. (2015), 'Determination of the maximum operating range of hydrodynamic stress in mammalian cell culture', *Journal of Biotechnology* 194, 100–109. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2014.12.003>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Determination+of+the+maximum+operating+range+of+hydrodynamic+stress+in+mammalian+cell+culture&publication+year=2015&author=Neunstoecklin+B.&author=Stettler+M.&author=Solacroup+T.&author=Broly+H.&author=Morbidelli+M.&author=Soos+M.
109) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25529344>)
-
- Neunstoecklin, B., Villiger, T. K., Lucas, E., Stettler, M., Broly, H., Morbidelli, M. & Soos, M. (2016), 'Pilot-scale verification of maximum tolerable hydrodynamic stress for mammalian cell culture', *Applied Microbiology and Biotechnology* 100(8), 3489–3498. CrossRef (<http://dx.doi.org/10.1007/s00253-015-7193-x>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Pilot-scale+verification+of+maximum+tolerable+hydrodynamic+stress+for+mammalian+cell+culture&publication+year=2016&author=Neunstoecklin+B.&author=Villiger+T.+K.&author=Lucas+E.+&author=Stettler+M.&author=Broly+H.&author=Morbidelli+M.&author=Soos+M.
015-7193-x&pages=3489-3498)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26637424>)
-
- Nienow, A. W. (1997), 'On impeller circulation and mixing effectiveness in the turbulent flow regime', *Chemical Engineering Science* 52(15), 2557–2565.
CrossRef ([http://dx.doi.org/10.1016/S0009-2509\(97\)00072-9](http://dx.doi.org/10.1016/S0009-2509(97)00072-9)) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=On+impeller+circulation+and+mixing+effectiveness+in+the+turbulent+flow+regime&publication+year=1997&author=Nienow+A.+W.&journal=Chemical+Engineering+Science&volume=52&issue=15&pages=2557-2565)
-
- Nienow, A. W. (1998), 'Hydrodynamics of stirred bioreactors', *Applied Mechanics Reviews* 51(1), 3–32. CrossRef (<http://dx.doi.org/10.1115/1.3098990>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Hydrodynamics+of+stirred+bioreactors&publication+year=1998&author=Nienow+A.+W.&journal=Applied+Mechanics+Reviews&volume=51&doi=10.1115/1.3098990&pages=3-32)
-
- Nienow, A. W. (2006), 'Reactor engineering in large scale animal cell culture', *Cytotechnology* 50(1–3), 9–33. CrossRef (<http://dx.doi.org/10.1007/s10616-006-9005-8>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Reactor+engineering+in+large+scale+animal+cell+culture&publication+year=2006&author=Nienow+A.+W.&journal=Cytotechnology&volume=50&doi=10.1007/s10616-006-9005-8&pages=9-33)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19003068>)
-
- Nienow, A. W. (2010), 'Impeller selection for animal cell culture', in *Encyclopedia of Industrial Biotechnology*, American Cancer Society, pp. 1–25.
CrossRef (<http://dx.doi.org/10.1002/9780470054581.eib636>) |

Google Scholar ([https://scholar.google.com/scholar?q=Nienow+A.+W.+\(2010\)+%E2%80%98Impeller+selection+for+animal+cell+culture%E2%80%99+in+Encyclopedia+of+Industrial+Biotechnology+American+Cancer+Society+pp.+1%E2%80%9325.](https://scholar.google.com/scholar?q=Nienow+A.+W.+(2010)+%E2%80%98Impeller+selection+for+animal+cell+culture%E2%80%99+in+Encyclopedia+of+Industrial+Biotechnology+American+Cancer+Society+pp.+1%E2%80%9325.))

Nienow, A. W., Rielly, C. D., Brosnan, K., Bargh, N., Lee, K., Coopman, K. & Hewitt, C. J. (2013), 'The physical characterisation of a microscale parallel bioreactor platform with an industrial CHO cell line expressing an IgG4', *Biochemical Engineering Journal* 76, 25–36. CrossRef (<http://dx.doi.org/10.1016/j.bej.2013.04.011>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=The+physical+characterisation+of+a+microscale+parallel+bioreactor+platform+with+an+industrial+CHO+cell+line+expressing+an+IgG4&publication+year=2013&author=Nienow+A.+W.36)

Opel, C. F., Li, J. & Amanullah, A. (2010), 'Quantitative modeling of viable cell density, cell size, intracellular conductivity, and membrane capacitance in batch and fed-batch CHO processes using dielectric spectroscopy', *Biotechnology Progress* 26(4), 1187–1199.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Quantitative+modeling+of+viable+cell+density+cell+size+intracellular+conductivity+and+membrane+capacitance+in+batch+and+fed-batch+CHO+processes+using+dielectric+spectroscopy&publication+year=2010&author=Opel+C.+F.+author=Li+J.+author=Amanullah+A.+journal=Biotechnology+Progress&volume=26&pa1199)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20730773>)

Ozturk, S. S. (1996), 'Engineering challenges in high density cell culture systems', *Cytotechnology* 22, 3–16. CrossRef (<http://dx.doi.org/10.1007/BF00353919>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Engineering+challenges+in+high+density+cell+culture+systems&publication+year=1996&author=Ozturk+S.+S.+journal=Cytotechnology&volume=22&doi=10.1007/BF00353919&pages=16)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22358910>)

Ozturk, S. S. (2014), 'Opportunities and challenges for the implementation of continuous processing in biomanufacturing', in Subramanian, G., ed., *Continuous Processing in Pharmaceutical Manufacturing*, Wiley-Blackwell, chapter 18, pp. 457–478.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Continuous+Processing+in+Pharmaceutical+Manufacturing&publication+year=2014&author=Ozturk+S.+S.+author=Subramanian+G.+pages=457-478)

Ozturk, S. S. & Hu, W.-S. (2006), *Cell Culture Technology for Pharmaceutical and Cell-Based Therapies*, Taylor and Francis.

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Cell+Culture+Technology+for+Pharmaceutical+and+Cell-Based+Therapies&publication+year=2006&author=Ozturk+S.+S.+author=Hu+W.-S.\)](https://scholar.google.com/scholar_lookup?title=Cell+Culture+Technology+for+Pharmaceutical+and+Cell-Based+Therapies&publication+year=2006&author=Ozturk+S.+S.+author=Hu+W.-S.)

Ozturk, S. S. & Kompala, D. S. (2006), 'Optimization of high cell density perfusion bioreactors', in *Cell Culture Technology for Pharmaceutical and Cell-Based Therapies*, Taylor and Francis Group, Boca Raton, pp. 387–416.

Google Scholar ([https://scholar.google.com/scholar?q=Ozturk+S.+S.+&Kompala+D.+S.+\(2006\)+%E2%80%98Optimization+of+high+cell+density+perfusion+bioreactors%E2%80%99+in+Cell+Culture+Technology+for+Pharmaceutical+and+Cell-Based+Therapies+Taylor+and+Francis+Group+Boca+Raton+pp.+387%E2%80%93416.\)](https://scholar.google.com/scholar?q=Ozturk+S.+S.+&Kompala+D.+S.+(2006)+%E2%80%98Optimization+of+high+cell+density+perfusion+bioreactors%E2%80%99+in+Cell+Culture+Technology+for+Pharmaceutical+and+Cell-Based+Therapies+Taylor+and+Francis+Group+Boca+Raton+pp.+387%E2%80%93416.))

Ozturk, S. S. & Palsson, B. O. (1990), 'Effects of dissolved oxygen on hybridoma cell growth, metabolism, and antibody production kinetics in continuous culture', *Biotechnology Progress* 6(6), 437–446. CrossRef (<http://dx.doi.org/10.1021/bp00006a006>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Effects+of+dissolved+oxygen+on+hybridoma+cell+growth+metabolism+and+antibody+production+kinetics+in+continuous+culture&publication+year=1990&author=Ozturk+S.+S.+auti446)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/1366834>)

Ozturk, S. S. & Palsson, B. O. (1991), 'Growth, metabolic, and antibody production kinetics of hybridoma cell culture: 2. Effects of serum concentration, dissolved oxygen concentration, and medium pH in a batch reactor', *Biotechnology Progress* 7(6), 481–494. CrossRef (<http://dx.doi.org/10.1021/bp00012a002>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Growth+metabolic+and+antibody+production+kinetics+of+hybridoma+cell+culture:+2.+Effects+of+serum+concentration+dissolved+oxygen+concentration+and+medium+pH+in+a+batch494)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/1367750>)

Pardee, A. B. (1989), 'G1 events and regulation of cell proliferation', *Science* 246(4930), 603–608. CrossRef (<http://dx.doi.org/10.1126/science.2683075>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=G1+events+and+regulation+of+cell+proliferation&publication+year=1989&author=Pardee+A.+B.+journal=Science&volume=246&doi=10.1126/science.2683075&pages=603-608)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/2683075>)

Park, J. H., Noh, S. M., Woo, J. R., Kim, J. W. & Lee, G. M. (2015), 'Valeric acid induces cell cycle arrest at G1 phase in CHO cell cultures and improves recombinant antibody productivity', *Biotechnology Journal* 11(4), 487–496. CrossRef (<http://dx.doi.org/10.1002/biot.201500327>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Valeric+acid+induces+cell+cycle+arrest+at+G1+phase+in+CHO+cell+cultures+and+improves+recombinant+antibody+productivity&publication+year=2015&author=Park+J.+H.+author=496)

Pasteur, L. (1885), *Mémoire sur la fermentation alcoolique*, PhD thesis, Académie des sciences.

Google Scholar ([https://scholar.google.com/scholar?q=Pasteur+L.+\(1885\)+M%C3%A9moire+sur+la+fermentation+alcoolique+PhD+thesis+Acad%C3%A9mie+des+sciences.\)](https://scholar.google.com/scholar?q=Pasteur+L.+(1885)+M%C3%A9moire+sur+la+fermentation+alcoolique+PhD+thesis+Acad%C3%A9mie+des+sciences.)

Pattison, R., Swamy, J., Mendenhall, B., Hwang, C. & Frohlich, B. (2000), 'Measurement and control of dissolved carbon dioxide in mammalian cell culture processes using an in situ fiber optic chemical sensor', *Biotechnology Progress* 16(5), 769–774. CrossRef (<http://dx.doi.org/10.1021/bp000089c>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Measurement+and+control+of+dissolved+carbon+dioxide+in+mammalian+cell+culture+processes+using+an+in+situ+fiber+optic+chemical+sensor&publication+year=2000&author=Pa774)

Pfister, D., Nicoud, L. & Morbidelli, M. (2018), *Continuous Biopharmaceutical Processes*, Cambridge University Press. CrossRef (<http://dx.doi.org/10.1017/9781108332897>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=Continuous+Biopharmaceutical+Processes&publication+year=2018&author=Pfister+D.+author=Nicoud+L.+author=Morbidelli+M.\)](https://scholar.google.com/scholar_lookup?title=Continuous+Biopharmaceutical+Processes&publication+year=2018&author=Pfister+D.+author=Nicoud+L.+author=Morbidelli+M.)

Pham, P. L., Kamen, A. & Durocher, Y. (2006), 'Large-scale transfection of mammalian cells for the fast production of recombinant protein', *Molecular Biotechnology* 34(2), 225–237.

CrossRef (<http://dx.doi.org/10.1385/MB:34:2:225>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Large-scale+transfection+of+mammalian+cells+for+the+fast+production+of+recombinant+protein&publication+year=2006&author=Pham+P.+L.&author=Kamen+A.&author=Durocher+Y.&journal=237) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/17172668>)

Pilkington, P. H., Margaritis, A., Mensour, N. A. & Russell, I. (1998), 'Fundamentals of immobilised yeast cells for continuous beer fermentation: A review', *Journal of the Institute of Brewing* 104(1), 19–31. CrossRef (<http://dx.doi.org/10.1002/j.2050-0416.1998.tb00970.x>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Fundamentals+of+immobilised+yeast+cells+for+continuous+beer+fermentation:+A+review&publication+year=1998&author=Pilkington+P.+H.&author=Margaritis+A.&author=Mensour+0416.1998.tb00970.x&pages=19-31)

Pohlscheidt, M., Jacobs, M., Wolf, S., Thiele, J., Jockwer, A., Gabelsberger, J., Jenzsch, M., Tebbe, H. & Burg, J. (2013), 'Optimizing capacity utilization by large scale 3000 L perfusion in seed train bioreactors', *Biotechnology Progress* 29(1), 222–229. CrossRef (<http://dx.doi.org/10.1002/btpr.1672>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Optimizing+capacity+utilization+by+large+scale+3000+L+perfusion+in+seed+train+bioreactors&publication+year=2013&author=Pohlscheidt+M.&author=Jacobs+M.&author=Wolf+S.&229) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23225663>)

Politis, S., Colombo, P., Colombo, G. & Rekkas, D. (2017), 'Design of experiments (DoE) in pharmaceutical development', *Drug Development and Industrial Pharmacy* 43(6), 889–901. CrossRef (<http://dx.doi.org/10.1080/03639045.2017.1291672>) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Design+of+experiments+\(DoE\)+in+pharmaceutical+development&publication+year=2017&author=Politis+S.&author=Colombo+P.&author=Colombo+G.&author=Rekkas+D.&journal=Drug+Development+and+Indus+901](https://scholar.google.com/scholar_lookup?title=Design+of+experiments+(DoE)+in+pharmaceutical+development&publication+year=2017&author=Politis+S.&author=Colombo+P.&author=Colombo+G.&author=Rekkas+D.&journal=Drug+Development+and+Indus+901))

Pollock, J., Coffman, J., Ho, S. V. & Farid, S. S. (2017), 'Integrated continuous bioprocessing: Economic, operational, and environmental feasibility for clinical and commercial antibody manufacture', *Biotechnology Progress* 33(4), 854–866. CrossRef (<http://dx.doi.org/10.1002/btpr.2492>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Integrated+continuous+bioprocessing:+Economic+operational+and+environmental+feasibility+for+clinical+and+commercial+antibody+manufacture&publication+year=2017&author=866) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28480535>)

Pollock, J., Ho, S. V. & Farid, S. S. (2013), 'Fed-batch and perfusion culture processes: Economic, environmental, and operational feasibility under uncertainty', *Biotechnology and Bioengineering* 110(1), 206–219. CrossRef (<http://dx.doi.org/10.1002/bit.24608>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Fed-batch+and+perfusion+culture+processes:+Economic+environmental+and+operational+feasibility+under+uncertainty&publication+year=2013&author=Pollock+J.&author=Ho+S.+V.&author=219) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22806692>)

Pörtner, R. (2015), 'Bioreactors for mammalian cells', in Al-Rubeai, M., ed., *Animal Cell Culture*, Springer International Publishing, pp. 89–135. CrossRef (http://dx.doi.org/10.1007/978-3-319-10320-4_4) | Google Scholar ([https://scholar.google.com/scholar?q=P%C3%B6rtner+R.+\(2015\)+%E2%80%98Bioreactors+for+mammalian+cells%E2%80%99+in+Al-Rubeai+M.+ed.+Animal+Cell+Culture+Springer+International+Publishing+pp.+89%E2%80%93135](https://scholar.google.com/scholar?q=P%C3%B6rtner+R.+(2015)+%E2%80%98Bioreactors+for+mammalian+cells%E2%80%99+in+Al-Rubeai+M.+ed.+Animal+Cell+Culture+Springer+International+Publishing+pp.+89%E2%80%93135))

Pörtner, R. & Schäfer, T. (1996), 'Modelling hybridoma cell growth and metabolism a comparison of selected models and data', *Journal of Biotechnology* 49(1–3), 119–135. CrossRef ([http://dx.doi.org/10.1016/0168-1656\(96\)01535-0](http://dx.doi.org/10.1016/0168-1656(96)01535-0)) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Modelling+hybridoma+cell+growth+and+metabolism+a+comparison+of+selected+models+and+data&publication+year=1996&author=P%C3%B6rtner+R.&author=Sch%C3%A4fer+T.&journal=1656\(96\)01535-0&pages=119-135](https://scholar.google.com/scholar_lookup?title=Modelling+hybridoma+cell+growth+and+metabolism+a+comparison+of+selected+models+and+data&publication+year=1996&author=P%C3%B6rtner+R.&author=Sch%C3%A4fer+T.&journal=1656(96)01535-0&pages=119-135)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/8879168>)

Radoniqi, F., Zhang, H., Bardliving, C. L., Shamlou, P. & Coffman, J. (2018), 'Computational fluid dynamic modeling of alternating tangential flow filtration for perfusion cell culture', *Biotechnology and Bioengineering* 115(11), 2751–2759. CrossRef (<http://dx.doi.org/10.1002/bit.26813>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Computational+fluid+dynamic+modeling+of+alternating+tangential+flow+filtration+for+perfusion+cell+culture&publication+year=2018&author=Radoniqi+F.&author=Zhang+H.&author=2759) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30080936>)

Raghunath, B., Bin, W., Pattnaik, P. & Janssens, J. (2013), 'Best practices for optimization and scale-up of microfiltration TFF processes', *BioProcessing Journal* 11(1), 30–40. CrossRef (<http://dx.doi.org/10.12665/j111.Raghunath>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Best+practices+for+optimization+and+scale-up+of+microfiltration+TFF+processes&publication+year=2013&author=Raghunath+B.&author=Bin+W.&author=Pattnaik+P.&author=Janssens+J.&journal=BioProcessing+Journal+volume=1+40)

Ramakrishnan, B., Boeggeman, E., Ramasamy, V. & Qasba, P. K. (2004), 'Structure and catalytic cycle of β -1,4-galactosyltransferase', *Current Opinion in Structural Biology* 14(5), 593–600. CrossRef (<http://dx.doi.org/10.1016/j.sbi.2004.09.006>) | Google Scholar ([https://scholar.google.com/scholar?q=Ramakrishnan+B.+Boeggeman+E.+Ramasamy+V.+&Qasba+P.+K.+\(2004\)+%E2%80%98Structure+and+catalytic+cycle+of+%CE%B2-1+4-galactosyltransferase%E2%80%99+Current+Opinion+in+Structural+Biology+14\(5\)+593%E2%80%93600](https://scholar.google.com/scholar?q=Ramakrishnan+B.+Boeggeman+E.+Ramasamy+V.+&Qasba+P.+K.+(2004)+%E2%80%98Structure+and+catalytic+cycle+of+%CE%B2-1+4-galactosyltransferase%E2%80%99+Current+Opinion+in+Structural+Biology+14(5)+593%E2%80%93600))

Ranganathan, P. & Sivaraman, S. (2011), 'Investigations on hydrodynamics and mass transfer in gas–liquid stirred reactor using computational fluid dynamics', *Chemical Engineering Science* 66(14), 3108–3124. CrossRef (<http://dx.doi.org/10.1016/j.ces.2011.03.007>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Investigations+on+hydrodynamics+and+mass+transfer+in+gas%E2%80%93liquid+stirred+reactor+using+computational+fluid+dynamics&publication+year=2011&author=Ranganathan+3124)

Rathore, A. S. (2009), 'Roadmap for implementation of quality by design (QbD) for biotechnology products'. *Trend in Biotechnology*, 27(9), 546–553, <https://doi.org/10.1016> (<https://doi.org/10.1016/j.tibtech.2009.06.006>) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Roadmap+for+implementation+of+quality+by+design+\(QbD\)+for+biotechnology+products&publication+year=2009&author=Rathore+A.+S.&journal=Trend+in+Biotechnology&volume=27&doi=10.1016/j.tibtech.2009.06.006&pages=546-553](https://scholar.google.com/scholar_lookup?title=Roadmap+for+implementation+of+quality+by+design+(QbD)+for+biotechnology+products&publication+year=2009&author=Rathore+A.+S.&journal=Trend+in+Biotechnology&volume=27&doi=10.1016/j.tibtech.2009.06.006&pages=546-553)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19647883>)

- Rathore, A. S. (2014), 'QbD/PAT for bioprocessing: Moving from theory to implementation', *Current Opinion in Chemical Engineering*, 6, 1–8. <https://doi.org/10.1016/j.coche.2014.05> (<https://doi.org/10.1016/j.coche.2014.05>) .006. [CrossRef \(http://dx.doi.org/10.1016/j.coche.2014.05.006\)](https://doi.org/10.1016/j.coche.2014.05.006) | Google Scholar (https://scholar.google.com/scholar_lookup?title=QbD/PAT+for+bioprocessing:+Moving+from+theory+to+implementation&publication+year=2014&author=Rathore+A.+S.&journal=Current+Opinion+in+Chemical+Engineering&volume=8)
-
- Rathore, A. S., Kateja, N. & Kumar, D. (2018), 'Process integration and control in continuous bioprocessing', *Current Opinion in Chemical Engineering* 22, 18–25. [CrossRef \(http://dx.doi.org/10.1016/j.coche.2018.08.005\)](https://doi.org/10.1016/j.coche.2018.08.005) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Process+integration+and+control+in+continuous+bioprocessing&publication+year=2018&author=Rathore+A.+S.&author=Kateja+N.&author=Kumar+D.&journal=Current+Opinion+in+Chemical+Engineering&volume=25)
-
- Rathore, A. S., Pathak, M. & Godara, A. (2016), 'Process development in the QbD paradigm: Role of process integration in process optimization for production of biotherapeutics', *Biotechnology Progress* 32(2), 355–362. [CrossRef \(http://dx.doi.org/10.1002/btpr.2209\)](https://doi.org/10.1002/btpr.2209) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Process+development+in+the+QbD+paradigm:+Role+of+process+integration+in+process+optimization+for+production+of+biotherapeutics&publication+year=2016&author=Rathore+A.+S.&author=Pathak+M.&author=Godara+A.&journal=Biotechnology+Progress&volume=32) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26588604>)
-
- Rathore, A. S. & Winkle, H. (2009), 'Quality by design for biopharmaceuticals', *Nature* 27(1), 26–34. Google Scholar (https://scholar.google.com/scholar_lookup?title=Quality+by+design+for+biopharmaceuticals&publication+year=2009&author=Rathore+A.+S.&author=Winkle+H.&journal=Nature&volume=27&pages=26-34) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19131992>)
-
- Read, E., Park, J., Shah, R., Riley, B., Brorson, K. & Rathore, A. (2010a), 'Process analytical technology (PAT) for biopharmaceutical products: Part I. Concepts and applications', *Biotechnology and Bioengineering* 105(2), 276–284. [CrossRef \(http://dx.doi.org/10.1002/bit.22528\)](https://doi.org/10.1002/bit.22528) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Process+analytical+technology+\(PAT\)+for+biopharmaceutical+products:+Part+I.+Concepts+and+applications&publication+year=2010&author=Read+E.&author=Park+J.&author=Shah+R.&author=Riley+B.&author=Brorson+K.&author=Rathore+A.&journal=Biotechnology+and+Bioengineering&volume=105](https://scholar.google.com/scholar_lookup?title=Process+analytical+technology+(PAT)+for+biopharmaceutical+products:+Part+I.+Concepts+and+applications&publication+year=2010&author=Read+E.&author=Park+J.&author=Shah+R.&author=Riley+B.&author=Brorson+K.&author=Rathore+A.&journal=Biotechnology+and+Bioengineering&volume=105)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19731253>)
-
- Read, E., Shah, R., Riley, B., Park, J., Brorson, K. & Rathore, A. (2010b), 'Process analytical technology (PAT) for biopharmaceutical products: Part II. Concepts and applications', *Biotechnology and Bioengineering* 105(2), 285–295. [CrossRef \(http://dx.doi.org/10.1002/bit.22529\)](https://doi.org/10.1002/bit.22529) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Process+analytical+technology+\(PAT\)+for+biopharmaceutical+products:+Part+II.+Concepts+and+applications&publication+year=2010&author=Read+E.&author=Shah+R.&author=Riley+B.&author=Park+J.&author=Brorson+K.&author=Rathore+A.&journal=Biotechnology+and+Bioengineering&volume=105](https://scholar.google.com/scholar_lookup?title=Process+analytical+technology+(PAT)+for+biopharmaceutical+products:+Part+II.+Concepts+and+applications&publication+year=2010&author=Read+E.&author=Shah+R.&author=Riley+B.&author=Park+J.&author=Brorson+K.&author=Rathore+A.&journal=Biotechnology+and+Bioengineering&volume=105)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19731253>)
-
- Reinhart, D., Damjanovic, L., Kaisermayer, C. & Kunert, R. (2015), 'Benchmarking of commercially available CHO cell culture media for antibody production', *Applied Microbiology and Biotechnology* 99(11), 4645–4657. [CrossRef \(http://dx.doi.org/10.1007/s00253-015-6514-4\)](https://doi.org/10.1007/s00253-015-6514-4) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Benchmarking+of+commercially+available+CHO+cell+culture+media+for+antibody+production&publication+year=2015&author=Reinhart+D.&author=Damjanovic+L.&author=Kaisermayer+C.&author=Kunert+R.&journal=Applied+Microbiology+and+Biotechnology&volume=99) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25846330>)
-
- Rivinoja, A., Hassinen, A., Kokkonen, N., Kauppi, A. & Kellokumpu, S. (2009), 'Elevated Golgi pH impairs terminal N-glycosylation by inducing mislocalization of Golgi glycosyltransferases', *Journal of Cellular Physiology* 220(1), 144–154. [CrossRef \(http://dx.doi.org/10.1002/jcp.21744\)](https://doi.org/10.1002/jcp.21744) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Elevated+Golgi+pH+impairs+terminal+N-glycosylation+by+inducing+mislocalization+of+Golgi+glycosyltransferases&publication+year=2009&author=Rivinoja+A.&author=Hassinen+A.&author=Kokkonen+N.&author=Kauppi+A.&author=Kellokumpu+S.&journal=Journal+of+Cellular+Physiology&volume=220) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19277980>)
-
- Rodrigues, M. E., Costa, A. R., Henriques, M., Azeredo, J. & Oliveira, R. (2010), 'Technological progresses in monoclonal antibody production systems', *Biotechnology Progress* 26(2), 332–351. Google Scholar (https://scholar.google.com/scholar_lookup?title=Technological+progresses+in+monoclonal+antibody+production+systems&publication+year=2010&author=Rodrigues+M.+E.&author=Costa+A.+R.&author=Henriques+M.&author=Azeredo+J.&author=Oliveira+R.&journal=Biotechnology+Progress&volume=26) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20043321>)
-
- Rouiller, Y., Bielser, J.-M., Brühlmann, D., Jordan, M., Broly, H. & Stettler, M. (2016), 'Screening and assessment of performance and molecule quality attributes of industrial cell lines across different fed-batch systems', *Biotechnology Progress* 32(1), 160–170. [CrossRef \(http://dx.doi.org/10.1002/btpr.2186\)](https://doi.org/10.1002/btpr.2186) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Screening+and+assessment+of+performance+and+molecule+quality+attributes+of+industrial+cell+lines+across+different+fed-batch+systems&publication+year=2016&author=Rouiller+Y.&author=Bielser+J.-M.&author=Bruehlmann+D.&author=Jordan+M.&author=Broly+H.&author=Stettler+M.&journal=Biotechnology+Progress&volume=32) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26489036>)
-
- Rouiller, Y., Périlleux, A., Collet, N., Jordan, M., Stettler, M. & Broly, H. (2013), 'A high-throughput media design approach for high performance mammalian fed-batch cultures', *mAbs* 5(3), 501–511. [CrossRef \(http://dx.doi.org/10.4161/mabs.23942\)](https://doi.org/10.4161/mabs.23942) | Google Scholar (

Google Scholar (https://scholar.google.com/scholar_lookup?title=Beyond+de-foaming:+The+effects+of+antifoams+on+bioprocess+productivity&publication+year=2012&author=Routledge+S.+J.&journal=Computational+and+Structural+Biotechnology+Journal&volum) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24688674>)

Roy, J. (2009), 'Glycosylation of antibody therapeutics: Optimisation for purpose', in *Methods in Molecular Biology*, Vol. 483, pp. 223–238.

CrossRef (http://dx.doi.org/10.1007/978-1-59745-407-0_13) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Glycosylation+of+antibody+therapeutics:+Optimisation+for+purpose&publication+year=2009&author=Roy+J.&journal=Methods+in+Molecular+Biology&volume=483&doi=10.1007/978-1-59745-407-0+13&pages=223-238)

Roychoudhury, P., O'Kennedy, R., McNeil, B. & Harvey, L. M. (2007), *Analytica Chimica Acta* 590(1), 110–117. CrossRef (<http://dx.doi.org/10.1016/j.aca.2007.03.011>) |

Google Scholar (<https://scholar.google.com/scholar?q=Roychoudhury+P.+O%E2%80%99Kennedy+R.+McNeil+B.+&+Harvey+L.+M.+>)
(2007)+Analytica+Chimica+Acta+590(1)+110%E2%80%93117.)

Running, J. A. & Bansal, K. (2016), 'Oxygen transfer rates in shaken culture vessels from Fernbach flasks to microtiter plates', *Biotechnology and Bioengineering* 113(8), 1729–1735.

CrossRef (<http://dx.doi.org/10.1002/bit.25938>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Oxygen+transfer+rates+in+shaken+culture+vessels+from+Fernbach+flasks+to+microtiter+plates&publication+year=2016&author=Running+J.+A.&author=Bansal+K.&journal=Biotechn
1735)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26806816>)

Rutherford, K., Mahmoudi, S. M. S., Lee, K. C. & Yianneskis, M. (1996), 'The influence of Rushton impeller blade and disk thickness on the mixing characteristics of stirred vessels', *Chemical Engineering Research and Design* 74(3), 369–378.

Google Scholar (https://scholar.google.com/scholar_lookup?title=The+influence+of+Rushton+impeller+blade+and+disk+thickness+on+the+mixing+characteristics+of+stirred+vessels&publication+year=1996&author=Rutherford+K.&author=Mahmou
378)

Ryan, P. W., Li, B., Shanahan, M., Leister, K. J. & Ryder, A. G. (2010), 'Prediction of cell culture media performance using fluorescence spectroscopy', *Analytical Chemistry* 82(4), 1311–1317.

CrossRef (<http://dx.doi.org/10.1021/ac902337c>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Prediction+of+cell+culture+media+performance+using+fluorescence+spectroscopy&publication+year=2010&author=Ryan+P.+W.&author=Li+B.&author=Shanahan+M.&author=Leister
1317)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20088547>)

Saha, D., Soos, M., Lüthi, B., Holzner, M., Liberzon, A., Babler, M. U. & Kinzelbach, W. (2014), 'Experimental characterization of breakage rate of colloidal aggregates in axisymmetric extensional flow', *Langmuir* 30(48), 14385–14395. CrossRef (<http://dx.doi.org/10.1021/la502686b>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Experimental+characterization+of+breakage+rate+of+colloidal+aggregates+in+axisymmetric+extensional+flow&publication+year=2014&author=Saha+D.&author=Soos+M.&author=L
14395)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25375364>)

Sajjadi, S. & Yianneskis, M. (2003), 'Semibatch emulsion polymerization of methyl methacrylate with a neat monomer feed', *Polymer Reaction Engineering* 11(4), 715–736.

CrossRef (<http://dx.doi.org/10.1081/PRE-120026371>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Semibatch+emulsion+polymerization+of+methyl+methacrylate+with+a+neat+monomer+feed&publication+year=2003&author=Sajjadi+S.&author=Yianneskis+M.&journal=Polymer+Re
120026371&pages=715-736)

Sano, C. (2009), 'History of glutamate production', *American Journal of Clinical Nutrition* 90(3), 728–732. CrossRef (<http://dx.doi.org/10.3945/ajcn.2009.27462F>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=History+of+glutamate+production&publication+year=2009&author=Sano+C.&journal=American+Journal+of+Clinical+Nutrition&volume=90&doi=10.3945/ajcn.2009.27462F&pages=728-732)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19640955>)

Sawyer, D., Sanderson, K., Lu, R., Daszkowski, T., Clark, E., Mcduff, P., Astrom, J., Heffernan, C., Duffy, L., Poole, S., Ryll, T., Sheehy, P., Strachan, D., Souquet, J., Beattie, D., Pollard, D., Stauch, O., Bezy, P., Sauer, T., Boettcher, L., Simpson, C., Dakin, J., Pitt, S. & Boyle, A. (2017a), *Biomanufacturing Technology Roadmap: Overview*, Technical report, BioPhorum Operations Group Ltd.

Google Scholar (<https://scholar.google.com/scholar?q=Sawyer+D.+Sanderson+K.+Lu+R.+Daszkowski+T.+Clark+E.+Mcduff+P.+Astrom+J.+Heffernan+C.+Duffy+L.+Poole+S.+Ryll+T.+Sheehy+P.+Strachan+D.+Souquet+J.+Beattie+D.+Pollard+D.+St>
(2017a)+Biomanufacturing+Technology+Roadmap:+Overview+Technical+report+BioPhorum+Operations+Group+Ltd.)

Sawyer, D., Sanderson, K., Lu, R., Daszkowski, T., Clark, E., Mcduff, P., Heffernan, C., Duffy, L., Poole, S., Ryll, T., Sheehy, P., Strachan, D., Beattie, D., Souquet, J., Pollard, D., Stauch, O., Bezy, P., Sauer, T., Boettcher, L., Simpson, C., Dakin, J., Pitt, S. & Boyle, A. (2017b), *Biomanufacturing Technology Roadmap: Executive Summary*, Technical report, BioPhorum Operations Group Ltd.

Google Scholar (<https://scholar.google.com/scholar?q=Sawyer+D.+Sanderson+K.+Lu+R.+Daszkowski+T.+Clark+E.+Mcduff+P.+Heffernan+C.+Duffy+L.+Poole+S.+Ryll+T.+Sheehy+P.+Strachan+D.+Beattie+D.+Souquet+J.+Pollard+D.+Stauch+O.+Bo>
(2017b)+Biomanufacturing+Technology+Roadmap:+Executive+Summary+Technical+report+BioPhorum+Operations+Group+Ltd.)

Scarff, M., Arnold, S. A., Harvey, L. M. & McNeil, B. (2006), 'Near infrared spectroscopy for bioprocess monitoring and control: Current status and future trends', *Critical Reviews in Biotechnology* 26(1), 17–39. CrossRef (<http://dx.doi.org/10.1080/07388550500513677>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Near+infrared+spectroscopy+for+bioprocess+monitoring+and+control:+Current+status+and+future+trends&publication+year=2006&author=Scarff+M.&author=Arnold+S.+A.&author=39)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/16594523>)

Seth, G., Hamilton, R. W., Stapp, T. R., Zheng, L., Meier, A., Petty, K., Leung, S. & Chary, S. (2013), 'Development of a new bioprocess scheme using frozen seed train intermediates to initiate CHO cell culture manufacturing campaigns', *Biotechnology and Bioengineering* 110(5), 1376–1385. CrossRef (<http://dx.doi.org/10.1002/bit.24808>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Development+of+a+new+bioprocess+scheme+using+frozen+seed+train+intermediates+to+initiate+CHO+cell+culture+manufacturing+campaigns&publication+year=2013&author=Set
1385)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23242970>)

Shah, Y. T. (1979), *Gas Liquid Solid Reactor Design*, Vol. 327, McGraw-Hill International.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Gas+Liquid+Solid+Reactor+Design&publication+year=1979&author=Shah+Y.+T.)

Sharma, C., Malhotra, D. & Rathore, A. S. (2011), 'Review of computational fluid dynamics applications in biotechnology processes', *Biotechnology Progress* 27(6), 1497–1510.

CrossRef (<http://dx.doi.org/10.1002/btpr.689>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Review+of+computational+fluid+dynamics+applications+in+biotechnology+processes&publication+year=2011&author=Sharma+C.&author=Malhotra+D.&author=Rathore+A.+S.&journal=1510)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22235483>)

Sherr, C. J. & Roberts, J. M. (1999), 'CDK inhibitors: Positive and negative regulators of G1-phase progression', *Genes & Development* 13(12), 1501–1512.

CrossRef (<http://dx.doi.org/10.1101/gad.13.12.1501>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=CDK+inhibitors:+Positive+and+negative+regulators+of+G1-phase+progression&publication+year=1999&author=Sherr+C.+J.&author=Roberts+J.+M.&journal=Genes+&+Development&volume=13&doi=10.1101/gad.13.12.1501&pages=1501-1512)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/10385618>)

Shukla, A. A. & Gottschalk, U. (2013), 'Single-use disposable technologies for biopharmaceutical manufacturing', *Trends in Biotechnology* 31(3), 147–154.

CrossRef (<http://dx.doi.org/10.1016/j.tibtech.2012.10.004>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Single-use+disposable+technologies+for+biopharmaceutical+manufacturing&publication+year=2013&author=Shukla+A.+A.&author=Gottschalk+U.&journal=Trends+in+Biotechnology&volume=31)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23178074>)

Shukla, A. A. & Thömmes, J. (2010), 'Recent advances in large-scale production of monoclonal antibodies and related proteins', *Trends in Biotechnology* 28(5), 253–261.

CrossRef (<http://dx.doi.org/10.1016/j.tibtech.2010.02.001>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Recent+advances+in+large-scale+production+of+monoclonal+antibodies+and+related+proteins&publication+year=2010&author=Shukla+A.+A.&author=Th%C3%B6mmes+J.&journal=Trends+in+Biotechnology&volume=261)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20304511>)

Sidoli, F. R., Asprey, S. P. & Mantalaris, A. (2006), 'A coupled single cell-population-balance model for mammalian cell cultures', *Industrial and Engineering Chemistry Research* 45(16), 5801–5811.

CrossRef (<http://dx.doi.org/10.1021/ie0511581>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=A+coupled+single+cell-population-balance+model+for+mammalian+cell+cultures&publication+year=2006&author=Sidoli+F.+R.&author=Asprey+S.+P.&author=Mantalaris+A.&journal=Industrial+and+Engineering+Chemistry+5811)

Sidoli, F. R., Mantalaris, A. & Asprey, S. P. (2004), 'Modelling of mammalian cells and cell culture processes', *Cytotechnology* 44(1–2), 27–46.

CrossRef (<http://dx.doi.org/10.1023/B:CYTO.0000043397.94527.84>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Modelling+of+mammalian+cells+and+cell+culture+processes&publication+year=2004&author=Sidoli+F.+R.&author=Mantalaris+A.&author=Asprey+S.+P.&journal=Cytotechnology&volume=46)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19003227>)

Sieblist, C., Jenzsch, M. & Pohlscheidt, M. (2013), 'Influence of pluronic F68 on oxygen mass transfer', *Biotechnology Progress* 29(5), 1278–1288. CrossRef (<http://dx.doi.org/10.1002/btpr.1770>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Influence+of+pluronic+F68+on+oxygen+mass+transfer&publication+year=2013&author=Sieblist+C.&author=Jenzsch+M.&author=Pohlscheidt+M.&journal=Biotechnology+Progress&volume=1288)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/23843368>)

Sieblist, C., Jenzsch, M. & Pohlscheidt, M. (2016), 'Equipment characterization to mitigate risks during transfers of cell culture manufacturing processes', *Cytotechnology* 68(4), 1381–1401.

CrossRef (<http://dx.doi.org/10.1007/s10616-015-9899-0>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Equipment+characterization+to+mitigate+risks+during+transfers+of+cell+culture+manufacturing+processes&publication+year=2016&author=Sieblist+C.&author=Jenzsch+M.&author=015-9899-0&pages=1381-1401)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26231834>)

Sieblist, C., Jenzsch, M., Pohlscheidt, M. & Lübbert, A. (2011), 'Insights into large-scale cell-culture reactors: I. Liquid mixing and oxygen supply', *Biotechnology Journal* 6(12), 1532–1546.

CrossRef (<http://dx.doi.org/10.1002/biot.201000408>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Insights+into+large-scale+cell-culture+reactors:+I.+Liquid+mixing+and+oxygen+supply&publication+year=2011&author=Sieblist+C.&author=Jenzsch+M.&author=Pohlscheidt+M.&author=L%C3%BCbbert+A.&journal=Biotechnology+1546)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21818860>)

Sieck, J. B., Budach, W. E., Suemeghy, Z., Leist, C., Villiger, T. K., Morbidelli, M. & Soos, M. (2014), 'Adaptation for survival: Phenotype and transcriptome response of CHO cells to elevated stress induced by agitation and sparging', *Journal of Biotechnology* 189, 94–103. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2014.08.042>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Adaptation+for+survival:+Phenotype+and+transcriptome+response+of+CHO+cells+to+elevated+stress+induced+by+agitation+and+sparging&publication+year=2014&author=Sieck+J.+103)

PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25218361>)

Sieck, J. B., Cordes, T., Budach, W. E., Rhiel, M. H., Suemeghy, Z., Leist, C., Villiger, T. K., Morbidelli, M. & Soos, M. (2013), 'Development of a scale-down model of hydrodynamic stress to study the performance of an industrial CHO cell line under simulated production scale bioreactor conditions', *Journal of Biotechnology* 164(1), 41–49.

CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2012.11.012>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Development+of+a+scale-down+model+of+hydrodynamic+stress+to+study+the+performance+of+an+industrial+CHO+cell+line+under+simulated+production+scale+bioreactor+conditions&publication+year=2013&volume=49)

Siganporia, C. C., Ghosh, S., Daszkowski, T., Papageorgiou, L. G. & Farid, S. S. (2014), 'Capacity planning for batch and perfusion bioprocesses across multiple biopharmaceutical facilities', *Biotechnology Progress* 30, 594–606. CrossRef (<http://dx.doi.org/10.1002/btpr.1860>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Capacity+planning+for+batch+and+perfusion+bioprocesses+across+multiple+biopharmaceutical+facilities&publication+year=2014&author=Sigani+Poria+C.+C.&author=Ghosh+S.&author=606)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24376262>)

Singer, M. & Soll, D. (1973), 'Guidelines for DNA hybrid molecules', *Science* 181, 1114. CrossRef (<http://dx.doi.org/10.1126/science.181.4105.1114>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Guidelines+for+DNA+hybrid+molecules&publication+year=1973&author=Singer+M.&author=Soll+D.&journal=Science&volume=181&doi=10.1126/science.181.4105.1114)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/11663279>)

Smelko, P. J., Wiltberger, R. K., Hickman, F. E., et al. (2011), 'Performance of high intensity fed-batch mammalian cell cultures in disposable bioreactor systems', *Biotechnology Progress* 27(5), 1358–

1364. CrossRef (<http://dx.doi.org/10.1002/btpr.634>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Performance+of+high+intensity+fed-

batch+mammalian+cell+cultures+in+disposable+bioreactor+systems&publication+year=2011&author=Smelko+P.+J.&author=Wiltberger+R.+K.&author=Hickman+F.+E.&journal=Biotechnolo

1364)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21626722>)

Sokolov, M., Morbidelli, M., Butté, A., Souquet, J., Broly, H. (2018), 'Sequential multivariate cell culture modeling at multiple scales supports systematic shaping of a monoclonal antibody toward a

quality target', *Biotechnology Journal* 13(4), 1700461. CrossRef (<http://dx.doi.org/10.1002/biot.201700461>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Sequential+multivariate+cell+culture+modeling+at+multiple+scales+supports+systematic+shaping+of+a+monoclonal+antibody+toward+a+quality+target&publication+year=2018&aut

Sokolov, M., Ritscher, J., MacKinnon, N., Bielser, J.-M., Brühlmann, D., Rothenhäusler, D., Thanei, G., Soos, M., Stettler, M., Souquet, J., Broly, H., Morbidelli, M. & Butté, A. (2017a), 'Robust factor selection in early cell culture process development for the production of a biosimilar monoclonal antibody', *Biotechnology Progress* 33(1), 181–191.

CrossRef (<http://dx.doi.org/10.1002/btpr.2374>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Robust+factor+selection+in+early+cell+culture+process+development+for+the+production+of+a+biosimilar+monoclonal+antibody&publication+year=2017&author=Sokolov+M.&author=M.&author=Br%C3%BChlmann+D.&author=Rothenh%C3%A4usler+D.&author=Thanei+G.&author=Soos+M.&author=Stettler+M.&author=Souquet+J.&author=Broly+H.&author=Morbidelli+M

191)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27689949>)

Sokolov, M., Ritscher, J., MacKinnon, N., Bielser, J.-M., Brühlmann, D., Rothenhäusler, D., Thanei, G., Soos, M., Stettler, M., Souquet, J., Broly, H., Morbidelli, M. & Butté, A. (2017b), 'Robust factor selection in early cell culture process development for the production of a biosimilar monoclonal antibody', *Biotech Progress Journal* 33(1), 181–191.

CrossRef (<http://dx.doi.org/10.1002/btpr.2374>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Robust+factor+selection+in+early+cell+culture+process+development+for+the+production+of+a+biosimilar+monoclonal+antibody&publication+year=2017&author=Sokolov+M.&author=M.&author=Br%C3%BChlmann+D.&author=Rothenh%C3%A4usler+D.&author=Thanei+G.&author=Soos+M.&author=Stettler+M.&author=Souquet+J.&author=Broly+H.&author=Morbidelli+M

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27689949>)

Sokolov, M., Ritscher, J., Mackinnon, N., Souquet, J., Broly, H., Morbidelli, M. & Butté, A. (2017c), 'Enhanced process understanding and multivariate prediction of the relationship between cell culture process and monoclonal antibody quality', *Biotechnology Progress* pp. 1–13.

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Enhanced+process+understanding+and+multivariate+prediction+of+the+relationship+between+cell+culture+process+and+monoclonal+antibody+quality&publication+year=2017&aut

Sokolov, M., Soos, M., Neunstoecklin, B., Morbidelli, M., Butté, A., Leardi, R., Solacroup, T., Stettler, M. & Broly, H. (2015), 'Fingerprint detection and process prediction by multivariate analysis of fed-batch monoclonal antibody cell culture data', *Biotechnology Progress* 31(6), 1633–1644. CrossRef (<http://dx.doi.org/10.1002/btpr.2174>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Fingerprint+detection+and+process+prediction+by+multivariate+analysis+of+fed-

batch+monoclonal+antibody+cell+culture+data&publication+year=2015&author=Sokolov+M.&author=Soos+M.&author=Neunstoecklin+B.&author=Morbidelli+M.&author=Butt%C3%A9+A

1644)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26399784>)

Soleas, G. J., Diamandis, E. P. & Goldberg, D. M. (1997), 'Wine as a biological fluid: History, production, and role in disease prevention', *Journal of Clinical Laboratory Analysis* 11(5), 287–313.

CrossRef ([http://dx.doi.org/10.1002/\(SICI\)1098-2825\(1997\)11:5<287::AID-JCLA6>3.0.CO;2-4](http://dx.doi.org/10.1002/(SICI)1098-2825(1997)11:5<287::AID-JCLA6>3.0.CO;2-4)) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Wine+as+a+biological+fluid:+History+production+and+role+in+disease+prevention&publication+year=1997&author=Soleas+G.+J.&author=Diamandis+E.+P.&author=Goldberg+D.+M.&2825(1997)11:5%3C287::AID-JCLA6%3E3.0.CO;2-4&pages=287-313)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/9292395>)

Solomon, B. L. & Garrido-Laguna, I. (2018), 'TIGIT: A novel immunotherapy target moving from bench to bedside', *Cancer Immunology, Immunotherapy* 67(11), 1659–1667.

Google Scholar (<https://scholar.google.com/scholar?q=Solomon+B.+L.+&+Garrido-Laguna+I.+>

(2018)+%E2%80%98TIGIT:+A+novel+immunotherapy+target+moving+from+bench+to+bedside%E2%80%99+Cancer+Immunology+Immunotherapy+67(11)+1659%E2%80%931667.)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30232519>)

Sommeregger, W., Sissolak, B., Kandra, K., von Stosch, M., Mayer, M. & Striedner, G. (2017), 'Quality by control: Towards model predictive control of mammalian cell culture bioprocesses',

Biotechnology Journal 12(7), 1–7. CrossRef (<http://dx.doi.org/10.1002/biot.201600546>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Quality+by+control:+Towards+model+predictive+control+of+mammalian+cell+culture+bioprocesses&publication+year=2017&author=Sommeregger+W.&author=Sissolak+B.&author=I

7)

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28371494>)

Soos, M., Ehrl, L., Baßler, M. U. & Morbidelli, M. (2010), 'Aggregate breakup in a contracting nozzle', *Langmuir* 26(1), 10–18. CrossRef (<http://dx.doi.org/10.1021/la903982n>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Aggregate+breakup+in+a+contracting+nozzle&publication+year=2010&author=Soos+M.&author=Ehrl+L.&author=Ba%C3%8C%CB%86bler+M.+U.&author=Morbidelli+M.&journal=Lang

18)

Soos, M., Kaufmann, R., Winteler, R., Kroupa, M. & Lüthi, B. (2013), 'Determination of maximum turbulent energy dissipation rate generated by a rushton impeller through large eddy simulation',

AIChE Journal 59(10), 3642–3658. CrossRef (<http://dx.doi.org/10.1002/aic.14206>) |

Google Scholar (https://scholar.google.com/scholar_lookup?

title=Determination+of+maximum+turbulent+energy+dissipation+rate+generated+by+a+rushton+impeller+through+large+eddy+simulation&publication+year=2013&author=Soos+M.&author=

3658)

-
- Sou, S. N., Jedrzejewski, P. M., Lee, K., Sellick, C., Polizzi, K. M. & Kontoravdi, C. (2017), 'Model-based investigation of intracellular processes determining antibody Fc-glycosylation under mild hypothermia', *Biotechnology and Bioengineering* 114(7), 1570–1582. [CrossRef \(http://dx.doi.org/10.1002/bit.26225\)](http://dx.doi.org/10.1002/bit.26225) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Model-based+investigation+of+intracellular+processes+determining+antibody+Fc-glycosylation+under+mild+hypothermia&publication+year=2017&author=Sou+S.+N.&author=Jedrzejewski+P.+M.&author=Lee+K.&author=Sellick+C.&author=Polizzi+K.+M.&author=Kontoravdi+C.\)](https://scholar.google.com/scholar_lookup?title=Model-based+investigation+of+intracellular+processes+determining+antibody+Fc-glycosylation+under+mild+hypothermia&publication+year=2017&author=Sou+S.+N.&author=Jedrzejewski+P.+M.&author=Lee+K.&author=Sellick+C.&author=Polizzi+K.+M.&author=Kontoravdi+C.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27869292>)
-
- Stahmann, K. P., Revuelta, J. L. & Seulberger, H. (2000), 'Three biotechnical processes using *Ashbya gossypii*, *Candida famata*, or *Bacillus subtilis* compete with chemical riboflavin production', *Applied Microbiology and Biotechnology* 53(5), 509–516. [CrossRef \(http://dx.doi.org/10.1007/s002530051649\)](http://dx.doi.org/10.1007/s002530051649) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Three+biotechnical+processes+using+Ashbya+gossypii+Candida+famata+or+Bacillus+subtilis+compete+with+chemical+riboflavin+production&publication+year=2000&author=Stahmann+K.+P.\)](https://scholar.google.com/scholar_lookup?title=Three+biotechnical+processes+using+Ashbya+gossypii+Candida+famata+or+Bacillus+subtilis+compete+with+chemical+riboflavin+production&publication+year=2000&author=Stahmann+K.+P.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/10855708>)
-
- Steeno, G. S. (2010), 'Experimental design for pharmaceutical development', in David, J., ed., *Chemical Engineering in the Pharmaceutical Industry: R&D to Manufacturing*, John Wiley and Sons, pp. 597–620. [CrossRef \(http://dx.doi.org/10.1002/9780470882221.ch32\)](http://dx.doi.org/10.1002/9780470882221.ch32) | [Google Scholar \(https://scholar.google.com/scholar?q=Steen+G.+S.+&publication+year=2010&title=Experimental+design+for+pharmaceutical+development&author=Steen+G.+S.\)](https://scholar.google.com/scholar?q=Steen+G.+S.+&publication+year=2010&title=Experimental+design+for+pharmaceutical+development&author=Steen+G.+S.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27376629>)
-
- Steinebach, F., Angarita, M., Karst, D. J., Müller-Späh, T. & Morbidelli, M. (2016a), 'Model based adaptive control of a continuous capture process for monoclonal antibodies production', *Journal of Chromatography A* 1444, 50–56. [CrossRef \(http://dx.doi.org/10.1016/j.chroma.2016.03.014\)](http://dx.doi.org/10.1016/j.chroma.2016.03.014) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Model+based+adaptive+control+of+a+continuous+capture+process+for+monoclonal+antibodies+production&publication+year=2016&author=Steinebach+F.&author=Angarita+M.&author=Müller-Späh+T.&author=Morbidelli+M.\)](https://scholar.google.com/scholar_lookup?title=Model+based+adaptive+control+of+a+continuous+capture+process+for+monoclonal+antibodies+production&publication+year=2016&author=Steinebach+F.&author=Angarita+M.&author=Müller-Späh+T.&author=Morbidelli+M.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27046002>)
-
- Steinebach, F., Müller-Späh, T. & Morbidelli, M. (2016b), 'Continuous counter-current chromatography for capture and polishing steps in biopharmaceutical production', *Biotechnology Journal* 11(9), 1126–1141. [CrossRef \(http://dx.doi.org/10.1002/biot.201500354\)](http://dx.doi.org/10.1002/biot.201500354) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Continuous+counter-current+chromatography+for+capture+and+polishing+steps+in+biopharmaceutical+production&publication+year=2016&author=Steinebach+F.&author=Müller-Späh+T.&author=Morbidelli+M.\)](https://scholar.google.com/scholar_lookup?title=Continuous+counter-current+chromatography+for+capture+and+polishing+steps+in+biopharmaceutical+production&publication+year=2016&author=Steinebach+F.&author=Müller-Späh+T.&author=Morbidelli+M.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27376629>)
-
- Steinebach, F., Ulmer, N., Wolf, M., Decker, L., Schneider, V., Wälchli, R., Karst, D., Souquet, J. & Morbidelli, M. (2017), 'Design and operation of a continuous integrated monoclonal antibody production process', *Biotechnology Progress* 33(5), 1303–1313. [CrossRef \(http://dx.doi.org/10.1002/btpr.2522\)](http://dx.doi.org/10.1002/btpr.2522) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Design+and+operation+of+a+continuous+integrated+monoclonal+antibody+production+process&publication+year=2017&author=Steinebach+F.&author=Ulmer+N.&author=Wolf+M.&author=Decker+L.&author=Schneider+V.&author=Wälchli+R.&author=Karst+D.&author=Souquet+J.&author=Morbidelli+M.\)](https://scholar.google.com/scholar_lookup?title=Design+and+operation+of+a+continuous+integrated+monoclonal+antibody+production+process&publication+year=2017&author=Steinebach+F.&author=Ulmer+N.&author=Wolf+M.&author=Decker+L.&author=Schneider+V.&author=Wälchli+R.&author=Karst+D.&author=Souquet+J.&author=Morbidelli+M.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28691347>)
-
- Swann, P., Brophy, L., Strachan, D., Lilly, E. & Jeffers, P. (2017), Biomanufacturing Technology Roadmap: In-line monitoring and real-time release, Technical report, BioPhorum Operations Group Ltd. [Google Scholar \(https://scholar.google.com/scholar?q=Swann+P.+Brophy+L.+Strachan+D.+Lilly+E.+&Jeffers+P.+&publication+year=2017&title=Biomanufacturing+Technology+Roadmap+In-line+monitoring+and+real-time+release+Technical+report+BioPhorum+Operations+Group+Ltd.\)](https://scholar.google.com/scholar?q=Swann+P.+Brophy+L.+Strachan+D.+Lilly+E.+&Jeffers+P.+&publication+year=2017&title=Biomanufacturing+Technology+Roadmap+In-line+monitoring+and+real-time+release+Technical+report+BioPhorum+Operations+Group+Ltd.)
-
- Tabas, I. & Kornfeld, S. (1979), 'Purification and characterization of a rat liver Golgi alpha-mannosidase capable of processing asparagine-linked oligosaccharides', *The Journal of Biological Chemistry* 254(22), 11655–63. [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Purification+and+characterization+of+a+rat+liver+Golgi+alpha-mannosidase+capable+of+processing+asparagine-linked+oligosaccharides&publication+year=1979&author=Tabas+I.&author=Kornfeld+S.\)](https://scholar.google.com/scholar_lookup?title=Purification+and+characterization+of+a+rat+liver+Golgi+alpha-mannosidase+capable+of+processing+asparagine-linked+oligosaccharides&publication+year=1979&author=Tabas+I.&author=Kornfeld+S.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/500665>)
-
- Takesono, S., Onodera, M., Toda, K., Yoshida, M., Yamagiwa, K. & Ohkawa, A. (2006), 'Improvement of foam breaking and oxygen-transfer performance in a stirred-tank fermenter', *Bioprocess and Biosystems Engineering* 28(4), 235–242. [CrossRef \(http://dx.doi.org/10.1007/s00449-005-0028-x\)](http://dx.doi.org/10.1007/s00449-005-0028-x) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Improvement+of+foam+breaking+and+oxygen-transfer+performance+in+a+stirred-tank+fermenter&publication+year=2006&author=Takesono+S.&author=Onodera+M.&author=Toda+K.&author=Yoshida+M.&author=Yamagiwa+K.&author=Ohkawa+A.\)](https://scholar.google.com/scholar_lookup?title=Improvement+of+foam+breaking+and+oxygen-transfer+performance+in+a+stirred-tank+fermenter&publication+year=2006&author=Takesono+S.&author=Onodera+M.&author=Toda+K.&author=Yoshida+M.&author=Yamagiwa+K.&author=Ohkawa+A.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/16009442>)
-
- Tanzeglock, T., Soos, M., Stephanopoulos, G. & Morbidelli, M. (2009), 'Induction of mammalian cell death by simple shear and extensional flows', *Biotechnology and Bioengineering* 104(2), 360–370. [CrossRef \(http://dx.doi.org/10.1002/bit.22405\)](http://dx.doi.org/10.1002/bit.22405) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Induction+of+mammalian+cell+death+by+simple+shear+and+extensional+flows&publication+year=2009&author=Tanzeglock+T.&author=Soos+M.&author=Stephanopoulos+G.&author=Morbidelli+M.\)](https://scholar.google.com/scholar_lookup?title=Induction+of+mammalian+cell+death+by+simple+shear+and+extensional+flows&publication+year=2009&author=Tanzeglock+T.&author=Soos+M.&author=Stephanopoulos+G.&author=Morbidelli+M.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/19575444>)
-
- Tao, Y., Shih, J., Sinacore, M., Ryll, T. & Yusuf-Makagiansar, H. (2011), 'Development and implementation of a perfusion-based high cell density cell banking process', *Biotechnology Progress* 27(3), 824–829. [CrossRef \(http://dx.doi.org/10.1002/btpr.599\)](http://dx.doi.org/10.1002/btpr.599) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Development+and+implementation+of+a+perfusion-based+high+cell+density+cell+banking+process&publication+year=2011&author=Tao+Y.&author=Shih+J.&author=Sinacore+M.&author=Ryll+T.&author=Yusuf-Makagiansar+H.\)](https://scholar.google.com/scholar_lookup?title=Development+and+implementation+of+a+perfusion-based+high+cell+density+cell+banking+process&publication+year=2011&author=Tao+Y.&author=Shih+J.&author=Sinacore+M.&author=Ryll+T.&author=Yusuf-Makagiansar+H.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/21538974>)
-
- Teixeira, A., Cunha, A., Clemente, J., Moreira, J., Cruz, H., Alves, P., Carrondo, M. & Oliveira, R. (2005), 'Modelling and optimization of a recombinant BHK-21 cultivation process using hybrid grey-box systems', *Journal of Biotechnology* 118(3), 290–303. [CrossRef \(http://dx.doi.org/10.1016/j.jbiotec.2005.04.024\)](http://dx.doi.org/10.1016/j.jbiotec.2005.04.024) | [Google Scholar \(https://scholar.google.com/scholar_lookup?title=Modelling+and+optimization+of+a+recombinant+BHK-21+cultivation+process+using+hybrid+grey-box+systems&publication+year=2005&author=Teixeira+A.&author=Cunha+A.&author=Clemente+J.&author=Moreira+J.&author=Cruz+H.&author=Alves+P.&author=Carrondo+M.&author=Oliveira+R.\)](https://scholar.google.com/scholar_lookup?title=Modelling+and+optimization+of+a+recombinant+BHK-21+cultivation+process+using+hybrid+grey-box+systems&publication+year=2005&author=Teixeira+A.&author=Cunha+A.&author=Clemente+J.&author=Moreira+J.&author=Cruz+H.&author=Alves+P.&author=Carrondo+M.&author=Oliveira+R.) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/16009442>)
-
- Thomas, T. N. (2017), 'Are we prepared to meet the demands of a challenging, but promising future?', in *Integrated Continuous Biomanufacturing III*, Suzanne Farid, University College London, United Kingdom Chetan Goudar, Amgen, USA Paula Alves, IBET, Portugal Veena Warikoo, Axcella Health, Inc., USA Eds, ECI Symposium Series'.

Google Scholar ([https://scholar.google.com/scholar?q=Thomas+T.+N.+\(2017\)+%E2%80%98Are+we+prepared+to+meet+the+demand+of+a+challenging+but+promising+future+%E2%80%99+in+Integrated+Continuous+Biomanufacturing+III+Suzanne+Farid+University+College+London+United+Kingdom+Chetan+Goudar+Amgen+USA+Paula+Alves+IBET+Portugal+](https://scholar.google.com/scholar?q=Thomas+T.+N.+(2017)+%E2%80%98Are+we+prepared+to+meet+the+demand+of+a+challenging+but+promising+future+%E2%80%99+in+Integrated+Continuous+Biomanufacturing+III+Suzanne+Farid+University+College+London+United+Kingdom+Chetan+Goudar+Amgen+USA+Paula+Alves+IBET+Portugal+))

Tribe, L. A., Briens, C. L. & Margaritis, A. (1995), 'Determination of the volumetric mass transfer coefficient (k_{la}) using the dynamic "gas out-gas in" method', *Biotechnology and Bioengineering* 46, 388–392. CrossRef (<http://dx.doi.org/10.1002/bit.260460412>) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Determination+of+the+volumetric+mass+transfer+coefficient+\(k_{la}\)+using+the+dynamic+%E2%80%9Cgas+out%E2%80%9Cgas+in%E2%80%9D+method&publication+year=1995&author=Tribe+L.+A.&author=Briens+C.+L.&author=Margaritis+A.&journal=392](https://scholar.google.com/scholar_lookup?title=Determination+of+the+volumetric+mass+transfer+coefficient+(k%28la%29)+using+the+dynamic+%E2%80%9Cgas+out%E2%80%9Cgas+in%E2%80%9D+method&publication+year=1995&author=Tribe+L.+A.&author=Briens+C.+L.&author=Margaritis+A.&journal=392))

Tsang, V. L., Wang, A. X., Yusuf-Makagiansar, H. & Ryll, T. (2014), 'Development of a scale down cell culture model using multivariate analysis as a qualification tool', *Biotechnology Progress* 30(1), 152–160. CrossRef (<http://dx.doi.org/10.1002/btpr.1819>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Development+of+a+scale+down+cell+culture+model+using+multivariate+analysis+as+a+qualification+tool&publication+year=2014&author=Tsang+V.+L.&author=Wang+A.+X.&author=Makagiansar+H.&author=Ryll+T.&journal=Biotechnology+Progress&volume=30&doi=10.1002/btpr.1819&pages=152-160) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24124180>)

Tziampazis, E. & Sambanis, A. (1994), 'Modeling of cell culture processes', *Cytotechnology* 14(3), 191–204. CrossRef (<http://dx.doi.org/10.1007/BF00749616>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Modeling+of+cell+culture+processes&publication+year=1994&author=Tziampazis+E.&author=Sambanis+A.&journal=Cytotechnology&volume=14&doi=10.1007/BF00749616&pages=19204) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/7765590>)

Umaña, P. & Bailey, J. E. (1997), 'A mathematical model of N-linked glycoform biosynthesis', *Biotechnology and Bioengineering* 55(6), 890–908. CrossRef ([http://dx.doi.org/10.1002/\(SICI\)1097-0290\(19970920\)55:6<890::AID-BIT7>3.0.CO;2-B](http://dx.doi.org/10.1002/(SICI)1097-0290(19970920)55:6<890::AID-BIT7>3.0.CO;2-B)) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=A+mathematical+model+of+N-linked+glycoform+biosynthesis&publication+year=1997&author=Uma%C3%B1a+P.&author=Bailey+J.+E.&journal=Biotechnology+and+Bioengineering&volume=55&doi=10.1002/\(SICI\)1097-0290\(19970920\)55:6%3C890::AID-BIT7%3E3.0.CO;2-B&pages=890-908](https://scholar.google.com/scholar_lookup?title=A+mathematical+model+of+N-linked+glycoform+biosynthesis&publication+year=1997&author=Uma%C3%B1a+P.&author=Bailey+J.+E.&journal=Biotechnology+and+Bioengineering&volume=55&doi=10.1002/(SICI)1097-0290(19970920)55:6%3C890::AID-BIT7%3E3.0.CO;2-B&pages=890-908)) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/18636599>)

Undey, C., Low, D., Menezes, J. C. & Koch, M. (2011), *PAT Applied in Biopharmaceutical Process Development and Manufacturing: An Enabling Tool for Quality-by-Design*, Vol. 33, CRC Press. CrossRef (<http://dx.doi.org/10.1201/b11536>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=PAT+Applied+in+Biopharmaceutical+Process+Development+and+Manufacturing:+An+Enabling+Tool+for+Quality-by-Design&publication+year=2011&author=Undey+C.&author=Low+D.&author=Menezes+J.+C.&author=Koch+M.)

Van't Riet, K. (1979), 'Review of measuring methods and results in nonviscous gas-liquid mass transfer in stirred vessels', *Industrial and Engineering Chemistry Process Design and Development* 18(3), 357–364. CrossRef (<http://dx.doi.org/10.1021/i260071a001>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Review+of+measuring+methods+and+results+in+nonviscous+gas%E2%80%93liquid+mass+transfer+in+stirred+vessels&publication+year=1979&author=Van%E2%80%99t+Riet+K.&jour)

Velasco, A. (1993), 'Cell type-dependent variations in the subcellular distribution of alpha-mannosidase I and II', *The Journal of Cell Biology* 122(1), 39–51. CrossRef (<http://dx.doi.org/10.1083/jcb.122.1.39>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Cell+type+dependent+variations+in+the+subcellular+distribution+of+alpha-mannosidase+I+and+II&publication+year=1993&author=Velasco+A.&journal=The+Journal+of+Cell+Biology&volume=122&doi=10.1083/jcb.122.1.39&pages=39-51) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/8314846>)

Versteeg, H. K. & Malalasekera, W. (1995), *An Introduction to Computational Fluid Dynamics*, John Wiley and Sons. Google Scholar (https://scholar.google.com/scholar_lookup?title=An+Introduction+to+Computational+Fluid+Dynamics&publication+year=1995&author=Versteeg+H.+K.&author=Malalasekera+W.)

Villiger, T. K., Morbidelli, M. & Soos, M. (2015), 'Experimental determination of maximum effective hydrodynamic stress in multiphase flow using shear sensitive aggregates', *AIChE Journal* 61(5), 1735–1744. CrossRef (<http://dx.doi.org/10.1002/aic.14753>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Experimental+determination+of+maximum+effective+hydrodynamic+stress+in+multiphase+flow+using+shear+sensitive+aggregates&publication+year=2015&author=Villiger+T.+K.&a)

Villiger, T. K., Neunstoecklin, B., Karst, D. J., Lucas, E., Stettler, M., Broly, H., Morbidelli, M. & Soos, M. (2018), 'Experimental and CFD physical characterization of animal cell bioreactors: From micro-to production scale', *Biochemical Engineering Journal* 131, 84–94. CrossRef (<http://dx.doi.org/10.1016/j.bej.2017.12.004>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Experimental+and+CFD+physical+characterization+of+animal+cell+bioreactors:+From+micro-to+production+scale&publication+year=2018&author=Villiger+T.+K.&author=Neunstoecklin+B.&author=Karst+D.+J.&author=Lucas+E.&author=Stettler+M.&author=Broly+H.&author=Morb)

Villiger, T. K., Roulet, A., Périlleux, A., Stettler, M., Broly, H., Morbidelli, M., Soos, M., Scibona, E., Stettler, M., Broly, H., Morbidelli, M., Soos, M., Roulet, A., Périlleux, A., Stettler, M., Broly, H., Morbidelli, M. & Soos, M. (2016a), 'Controlling the time evolution of mAb N-linked glycosylation – Part I: Micro-bioreactor experiments', *Biotechnology Progress* 32(5), 1123–1134. CrossRef (<http://dx.doi.org/10.1002/btpr.2305>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Controlling+the+time+evolution+of+mAb+N-linked+glycosylation+%E2%80%93+Part+I:+Micro-bioreactor+experiments&publication+year=2016&author=Villiger+T.+K.&author=Roulet+A.&author=P%C3%A9rilleux+A.&author=Stettler+M.&author=Broly+H.&author=Morb)

Villiger, T. K., Scibona, E., Stettler, M., Broly, H., Morbidelli, M. & Soos, M. (2016b), 'Controlling the time evolution of mAb N-linked glycosylation – Part II: Model-based predictions', *Biotechnology Progress* 32(5), 1135–1148. CrossRef (<http://dx.doi.org/10.1002/btpr.2315>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Controlling+the+time+evolution+of+mAb+N-linked+glycosylation+%E2%80%93+Part+II:+Model-based+predictions&publication+year=2016&author=Villiger+T.+K.&author=Scibona+E.&author=Stettler+M.&author=Broly+H.&author=Morb)

Villiger, T. K., Steinhoff, R. F., Ivarsson, M., Solacroup, T., Stettler, M., Broly, H., Krismer, J., Pabst, M., Zenobi, R., Morbidelli, M. & Soos, M. (2016c), 'High-throughput profiling of nucleotides and nucleotide sugars to evaluate their impact on antibody N-glycosylation', *Journal of Biotechnology* 229, 3–12. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2016.04.039>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=High-throughput+profiling+of+nucleotides+and+nucleotide+sugars+to+evaluate+their+impact+on+antibody+N-glycosylation&publication+year=2016&author=Villiger+T.+K.&author=Steinhoff+R.+F.&author=Ivarsson+M.&author=Solacroup+T.&author=Stettler+M.&author=Broly+H.&author=Krismer+J.+12)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27131894>)

Villiger-Oberbek, A., Yang, Y., Zhou, W. & Yang, J. (2015), 'Development and application of a high-throughput platform for perfusion-based cell culture processes', *Journal of Biotechnology* 212, 21–29. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2015.06.428>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Development+and+application+of+a+high-throughput+platform+for+perfusion-based+cell+culture+processes&publication+year=2015&author=Villiger-Oberbek+A.&author=Yang+Y.&author=Zhou+W.&author=Yang+J.&journal=Journal+of+Biotechnology&volume=212&doi=10.1016/j.jbiotec.2015.06.428&pages=21-29)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26197419>)

Vogg, S., Müller-Späth, T. & Morbidelli, M. (2018), 'Current status and future challenges in continuous biochromatography', *Current Opinion in Chemical Engineering* 22, 138–144. CrossRef (<http://dx.doi.org/10.1016/j.coche.2018.09.001>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Current+status+and+future+challenges+in+continuous+biochromatography&publication+year=2018&author=Vogg+S.&author=M%C3%BCller-Sp%C3%A4th+T.&author=Morbidelli+M.&journal=Current+Opinion+in+Chemical+Engineering&volume=22&doi=10.1016/j.coche.2018.09.001&pages=138-144)

Voisard, D., Meuwly, F., Ruffieux, P. A., Baer, G. & Kadouri, A. (2003), 'Potential of cell retention techniques for large-scale high-density perfusion culture of suspended mammalian cells', *Biotechnology and Bioengineering* 82(7), 751–765. CrossRef (<http://dx.doi.org/10.1002/bit.10629>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Potential+of+cell+retention+techniques+for+large-scale+high-density+perfusion+culture+of+suspended+mammalian+cells&publication+year=2003&author=Voisard+D.&author=Meuwly+F.&author=Ruffieux+P.+A.&author=Baer+G.&author=Kadouri+A.765)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/12701141>)

Vojinović, V., Cabral, J. M. S. & Fonseca, L. P. (2006), 'Real-time bioprocess monitoring: Part I: In situ sensors', *Sensors and Actuators, B: Chemical* 114, 1083–1091. CrossRef (<http://dx.doi.org/10.1016/j.snb.2005.07.059>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Real-time+bioprocess+monitoring:+Part+I:+In+situ+sensors&publication+year=2006&author=Vojinovi%C4%87+V.&author=Cabral+J.+M.+S.&author=Fonseca+L.+P.&journal=Sensors+and+Actuators+1091)

Von Stosch, M., Hamelink, J.-M. & Oliveira, R. (2016), 'Hybrid modeling as a QbD/PAT tool in process development: An industrial E. coli case study', *Bioprocess and Biosystems Engineering* 39(5), 773–784. CrossRef (<http://dx.doi.org/10.1007/s00449-016-1557-1>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Hybrid+modeling+as+a+QbD/PAT+tool+in+process+development:+An+industrial+E.+coli+case+study&publication+year=2016&author=Von+Stosch+M.&author=Hamelink+J.-M.&author=Oliveira+R.&journal=Bioprocess+and+Biosystems+Engineering&volume=39&doi=10.1007/s00449-016-1557-1&pages=773-784)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26879643>)

Von Stosch, M. & Willis, M. J. (2017), 'Intensified design of experiments for upstream bioreactors', *Engineering in Life Sciences* 17(11), 1173–1184. CrossRef (<http://dx.doi.org/10.1002/elsc.201600037>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Intensified+design+of+experiments+for+upstream+bioreactors&publication+year=2017&author=Von+Stosch+M.&author=Willis+M.+J.&journal=Engineering+in+Life+Sciences&volume=1184)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/32624745>)

Vulto, A. G. & Jaquez, O. A. (2017), 'The process defines the product: What really matters in biosimilar design and production?', *Rheumatology* 56(suppl 4), 14–29. CrossRef (<http://dx.doi.org/10.1093/rheumatology/kex278>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=The+process+defines+the+product:+What+really+matters+in+biosimilar+design+and+production?&publication+year=2017&author=Vulto+A.+G.&author=Jaquez+O.+A.&journal=Rheumatology&volume=56&doi=10.1093/rheumatology/kex278&pages=14-29)

Walsh, G. (2010), 'Post-translational modifications of protein biopharmaceuticals', *Drug Discovery Today* 15(17–18), 773–780. CrossRef (<http://dx.doi.org/10.1016/j.drudis.2010.06.009>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Post-translational+modifications+of+protein+biopharmaceuticals&publication+year=2010&author=Walsh+G.&journal=Drug+Discovery+Today&volume=15&doi=10.1016/j.drudis.2010.06.009&pa780)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/20599624>)

Walsh, G. (2014), 'Biopharmaceutical benchmarks 2014', *Nature Biotechnology* 32(7), 992–1000. CrossRef (<http://dx.doi.org/10.1038/nbt.3040>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Biopharmaceutical+benchmarks+2014&publication+year=2014&author=Walsh+G.&journal=Nature+Biotechnology&volume=32&doi=10.1038/nbt.3040&pages=992-1000)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25299917>)

Walther, J., Godawat, R., Hwang, C., Abe, Y., Sinclair, A. & Konstantinov, K. (2015), 'The business impact of an integrated continuous biomanufacturing platform for recombinant protein production', *Journal of Biotechnology* 213, 3–12. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2015.05.010>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=The+business+impact+of+an+integrated+continuous+biomanufacturing+platform+for+recombinant+protein+production&publication+year=2015&author=Walther+J.&author=Godawa12)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26014522>)

Walther, J., Lu, J., Hollenbach, M., Yu, M., Hwang, C., McLarty, J. & Brower, K. (2018), 'Perfusion cell culture decreases process and product heterogeneity in a head-to-head comparison with fed-batch', *Biotechnology Journal* 14(2), 1700733.

Google Scholar (https://scholar.google.com/scholar_lookup?title=Perfusion+cell+culture+decreases+process+and+product+heterogeneity+in+a+head-to-head+comparison+with+fed-batch&publication+year=2018&author=Walther+J.&author=Lu+J.&author=Hollenbach+M.&author=Yu+M.&author=Hwang+C.&author=McLarty+J.&author=Brower+K.&journal=Biotechnolog)

Walther, J., McLarty, J. & Johnson, T. (2019), 'The effects of alternating tangential flow (ATF) residence time, hydrodynamic stress, and filtration flux on high-density perfusion cell culture', *Biotechnology and Bioengineering* 116(2), 320–332. CrossRef (<http://dx.doi.org/10.1002/bit.26811>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?title=The+effects+of+alternating+tangential+flow+\(ATF\)+residence+time+hydrodynamic+stress+and+filtration+flux+on+high-density+perfusion+cell+culture&publication+year=2019&author=Walther+J.&author=McLarty+J.&author=Johnson+T.&journal=Biotechnology+and+Bioengineering&volume=116&doi=10.100332](https://scholar.google.com/scholar_lookup?title=The+effects+of+alternating+tangential+flow+(ATF)+residence+time+hydrodynamic+stress+and+filtration+flux+on+high-density+perfusion+cell+culture&publication+year=2019&author=Walther+J.&author=McLarty+J.&author=Johnson+T.&journal=Biotechnology+and+Bioengineering&volume=116&doi=10.100332))
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/30080929>)

-
- Walther, J., Shah, N., Hollenbach, M., Wang, J., Yu, M., Lu, J., Yang, Y., Konstantinov, K. B. & Hwang, C. (2016), 'Overcoming process intensification challenges to deliver a manufacturable and competitive integrated continuous biomanufacturing platform', in *Cell Culture Engineering XV*, Robert Kiss, Genentech Sarah Harcum, Clemson University Jeff Chalmers, Ohio State University Eds, ECT Symposium Series.
Google Scholar ([https://scholar.google.com/scholar?q=Walther+J.+Shah+N.+Hollenbach+M.+Wang+J.+Yu+M.+Lu+J.+Yang+Y.+Konstantinov+K.+B.+&+Hwang+C.+ \(2016\)+%E2%80%98Overcoming+process+intensification+challenges+to+deliver+a+manufacturable+and+competitive+integrated+continuous+biomanufacturing+platform%E2%80%99+in+](https://scholar.google.com/scholar?q=Walther+J.+Shah+N.+Hollenbach+M.+Wang+J.+Yu+M.+Lu+J.+Yang+Y.+Konstantinov+K.+B.+&+Hwang+C.+ (2016)+%E2%80%98Overcoming+process+intensification+challenges+to+deliver+a+manufacturable+and+competitive+integrated+continuous+biomanufacturing+platform%E2%80%99+in+))
-
- Wang, J., Liu, L., Ball, T., Yu, L., Li, Y. & Xing, F. (2016), 'Revealing a 5,000-y-old beer recipe in China', *Proceedings of the National Academy of Sciences* 113(23), 6444–6448.
CrossRef (<http://dx.doi.org/10.1073/pnas.1601465113>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Revealing+a+5+000-y-old+beer+recipe+in+China&publication+year=2016&author=Wang+J.&author=Liu+L.&author=Ball+T.&author=Yu+L.&author=Li+Y.&author=Xing+F.&journal=Proceedings+of+the+National+A6448)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27217567>)
-
- Wang, S. B., Lee-Goldman, A., Ravikrishnan, J., Zheng, L. & Lin, H. (2018), 'Manipulation of the sodium-potassium ratio as a lever for controlling cell growth and improving cell specific productivity in perfusion CHO cell cultures', *Biotechnology and Bioengineering* 115(4), 921–931. CrossRef (<http://dx.doi.org/10.1002/bit.26527>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Manipulation+of+the+sodium-potassium+ratio+as+a+lever+for+controlling+cell+growth+and+improving+cell+specific+productivity+in+perfusion+CHO+cell+cultures&publication+year=2018&author=Wang+S.+B.+&author+Goldman+A.&author=Ravikrishnan+J.&author=Zheng+L.&author=Lin+H.&journal=Biotechnology+and+Bioengineering&volume=115&doi=10.1002/bit.26527&pages=921-931)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/29278412>)
-
- Wang, S., Godfrey, S., Ravikrishnan, J., Lin, H., Vogel, J. & Coffman, J. (2017), 'Shear contributions to cell culture performance and product recovery in ATF and TFF perfusion systems', *Journal of Biotechnology* 246, 52–60. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2017.01.020>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Shear+contributions+to+cell+culture+performance+and+product+recovery+in+ATF+and+TFF+perfusion+systems&publication+year=2017&author=Wang+S.&author=Godfrey+S.&authc60)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28159614>)
-
- Wang, Z., Zhuge, J., Fang, H. & Prior, B. A. (2001), 'Glycerol production by microbial fermentation: A review', *Biotechnology Advances* 19(3), 201–223.
CrossRef ([http://dx.doi.org/10.1016/S0734-9750\(01\)00060-X](http://dx.doi.org/10.1016/S0734-9750(01)00060-X)) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Glycerol+production+by+microbial+fermentation:+A+review&publication+year=2001&author=Wang+Z.&author=Zhuge+J.&author=Fang+H.&author=Prior+B.+A.&journal=Biotechnolog9750(01)00060-X&pages=201-223)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/14538083>)
-
- Warikoo, V., Godawat, R., Brower, K., Jain, S., Cummings, D., Simons, E., Johnson, T., Walther, J., Yu, M., Wright, B., McLarty, J., Karey, K. P., Hwang, C., Zhou, W., Riske, F. & Konstantinov, K. (2012), 'Integrated continuous production of recombinant therapeutic proteins', *Biotechnology and Bioengineering* 109(12), 3018–3029. CrossRef (<http://dx.doi.org/10.1002/bit.24584>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Integrated+continuous+production+of+recombinant+therapeutic+proteins&publication+year=2012&author=Warikoo+V.&author=Godawat+R.&author=Brower+K.&author=Jain+S.&aut3029)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/22729761>)
-
- Watanabe, S., Shuttleworth, J. & Al-Rubeai, M. (2002), 'Regulation of cell cycle and productivity in NS0 cells by the over-expression of p21CIP1', *Biotechnology and Bioengineering* 77(1), 1–7.
CrossRef (<http://dx.doi.org/10.1002/bit.10112>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Regulation+of+cell+cycle+and+productivity+in+NS0+cells+by+the+over-expression+of+p21CIP1&publication+year=2002&author=Watanabe+S.&author=Shuttleworth+J.&author=Al-Rubeai+M.&journal=Biotechnology+and+Bioengineering&volume=77&doi=10.1002/bit.10112&pages=1-7)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/11745168>)
-
- Webster, T. A., Hadley, B. C., Hilliard, W., Jaques, C. & Mason, C. (2018), 'Development of generic raman models for a GS-KO™ CHO platform process', *Biotechnology Progress* 34(3), 730–737.
CrossRef (<http://dx.doi.org/10.1002/btpr.2633>) |
Google Scholar ([https://scholar.google.com/scholar?q=Webster+T.+A.+Hadley+B.+C.+Hilliard+W.+Jaques+C.+&+Mason+C.+ \(2018\)+%E2%80%98Development+of+generic+raman+models+for+a+GS-KOTM+CHO+platform+process%E2%80%99+Biotechnology+Progress+34\(3\)+730%E2%80%93737.](https://scholar.google.com/scholar?q=Webster+T.+A.+Hadley+B.+C.+Hilliard+W.+Jaques+C.+&+Mason+C.+ (2018)+%E2%80%98Development+of+generic+raman+models+for+a+GS-KOTM+CHO+platform+process%E2%80%99+Biotechnology+Progress+34(3)+730%E2%80%93737.))
-
- Whelan, J., Craven, S. & Glennon, B. (2012), 'In situ Raman spectroscopy for simultaneous monitoring of multiple process parameters in mammalian cell culture bioreactors', *Biotechnology Progress* (5), 1355–1362. CrossRef (<http://dx.doi.org/10.1002/btpr.1590>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=In+situ+Raman+spectroscopy+for+simultaneous+monitoring+of+multiple+process+parameters+in+mammalian+cell+culture+bioreactors&publication+year=2012&author=Whelan+J.&1362)
-
- Whitford, W. G. (2014), 'Single-use systems support continuous bioprocessing by perfusion culture', in *Continuous Processing in Pharmaceutical Manufacturing*, Wiley-VCH Verlag GmbH and Co. KGaA, Weinheim, chapter 9, pp. 183–226.
Google Scholar ([https://scholar.google.com/scholar?q=Whitford+W.+G.+ \(2014\)+%E2%80%98Single-use+systems+support+continuous+bioprocessing+by+perfusion+culture%E2%80%99+in+Continuous+Processing+in+Pharmaceutical+Manufacturing+Wiley-VCH+Verlag+GmbH+and+Co.+KGaA+Weinheim+chapter+9+pp.+183%E2%80%93226.](https://scholar.google.com/scholar?q=Whitford+W.+G.+ (2014)+%E2%80%98Single-use+systems+support+continuous+bioprocessing+by+perfusion+culture%E2%80%99+in+Continuous+Processing+in+Pharmaceutical+Manufacturing+Wiley-VCH+Verlag+GmbH+and+Co.+KGaA+Weinheim+chapter+9+pp.+183%E2%80%93226.))
-
- Wold, S., Sjöström, M. & Eriksson, L. (2001), 'PLS-regression: A basic tool of chemometrics', *Chemometrics and Intelligent Laboratory Systems* 58(2), 109–130.
CrossRef ([http://dx.doi.org/10.1016/S0169-7439\(01\)00155-1](http://dx.doi.org/10.1016/S0169-7439(01)00155-1)) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=PLS-regression:+A+basic+tool+of+chemometrics&publication+year=2001&author=Wold+S.&author=Sj%C3%B6str%C3%B6m+M.&author=Eriksson+L.&journal=Chemometrics+and+Intelligent+L7439(01)00155-1&pages=109-130)
-
- Wolf, M. K. F., Closet, A., Bzowska, M., Bielser, J.-M., Souquet, J., Broly, H. & Morbidelli, M. (2019a), 'Improved performance in mammalian cell perfusion cultures by growth inhibition', *Biotechnology Journal* 14(2), 1700722. CrossRef (<http://dx.doi.org/10.1002/biot.201700722>) |
Google Scholar (https://scholar.google.com/scholar_lookup?title=Improved+performance+in+mammalian+cell+perfusion+cultures+by+growth+inhibition&publication+year=2019&author=Wolf+M.+K.+F.+&author=Closet+A.&author=Bzowska+M.&authM.&author=Souquet+J.&author=Broly+H.&author=Morbidelli+M.&journal=Biotechnology+Journal&volume=14&doi=10.1002/biot.201700722)
| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/29781256>)
-
- Wolf, M. K. F., Lorenz, V., Karst, D. J., Souquet, J., Broly, H. & Morbidelli, M. (2018), 'Devel-opment of a shake tube-based scale-down model for perfusion cultures', *Biotechnology and Bioengineering* 115(11), 2703–2713. CrossRef (<http://dx.doi.org/10.1002/bit.26804>) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Development+of+a+shake+tube-based+scale-down+model+for+perfusion+cultures&publication+year=2018&author=Wolf+M.+K.+F.&author=Lorenz+V.&author=Karst+D.+J.&author=Souquet+J.&author=Broly+H.&author=Morbidelli+M.&doi=10.1016/j.jbiotec.2019.107295)

Wolf, M. K. F., Müller, A., Souquet, J., Broly, H. & Morbidelli, M. (2019b), 'Process design and development of a mammalian cell perfusion culture in shake-tube and benchtop bioreactors', *Biotechnology and Bioengineering* 116(8), 1973–1985. [CrossRef \(http://dx.doi.org/10.1002/bit.26999\)](https://doi.org/10.1002/bit.26999) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=Process+design+and+development+of+a+mammalian+cell+perfusion+culture+in+shake-tube+and+benchtop+bioreactors&publication+year=2019&author=Wolf+M.+K.+F.&author=M+K+C3+BCiler+A.&author=Souquet+J.&author=Broly+H.&author=Morbidelli+M.&journal=Biotechnol+1985)
 | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/31038199>)

Wolf, M. K. F., Pechlaner, A., Lorenz, V., Karst, D. J., Souquet, J., Broly, H., & Morbidelli, M. (2019c). A two-step procedure for the design of perfusion bioreactors. *Biochemical Engineering Journal*, 151, 107295. [CrossRef \(http://dx.doi.org/10.1016/j.bej.2019.107295\)](https://doi.org/10.1016/j.bej.2019.107295) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=A+two-step+procedure+for+the+design+of+perfusion+bioreactors&publication+year=2019&author=Wolf+M.+K.+F.&author=Pechlaner+A.&author=Lorenz+V.&author=Karst+D.+J.&author=Souquet+J.&doi=10.1016/j.bej.2019.107295)

Wolton, A. D. & Rayner, A. (2014), 'Lessons learned in the ballroom', *Pharmaceutical Engineering* 34(4), 1–5.
 Google Scholar (https://scholar.google.com/scholar_lookup?title=Lessons+learned+in+the+ballroom&publication+year=2014&author=Wolton+A.+D.&author=Rayner+A.&journal=Pharmaceutical+Engineering&volume=34&pages=1-5)

Wong, Y. H., Krishnaswamy, P. R. & Teo, W. K. (1992), 'Advanced control of pH in mammalian cell culture', in Furusaki, S., Endo, I. & Matsuno, R., eds, *Biochemical Engineering for 2001*, Springer Japan, pp. 689–691. [CrossRef \(http://dx.doi.org/10.1007/978-4-431-68180-9_184\)](https://doi.org/10.1007/978-4-431-68180-9_184) |
 Google Scholar (https://scholar.google.com/scholar?q=Wong+Y.+H.+Krishnaswamy+P.+R.+&Teo+W.+K.+&doi=10.1007/978-4-431-68180-9_184)

Woodcock, J. (2014), 'Modernizing pharmaceutical manufacturing: Continuous manufacturing as a key enabler', in *International Symposium on Continuous Manufacturing of Pharmaceuticals*, Cambridge, MA.
 Google Scholar (<https://scholar.google.com/scholar?q=Woodcock+J.+&doi=10.1016/j.jbiotec.2016.06.019>)

Wright, B., Bruninghaus, M., Vrabell, M., Walther, J. & Shah, N. (2015), 'A novel seed-train process', *BioProcess International* 13(3), 16–25.
 Google Scholar (https://scholar.google.com/scholar_lookup?title=A+novel+seed-train+process&publication+year=2015&author=Wright+B.&author=Bruninghaus+M.&author=Vrabell+M.&author=Walther+J.&author=Shah+N.&journal=BioProcess+International&volume=13&pages=16-25)

Wurm, F. M. (2004), 'Production of recombinant protein therapeutics in cultivated mammalian cells', *Nature Biotechnology* (11), 1393–1398. [CrossRef \(http://dx.doi.org/10.1038/nbt1026\)](https://doi.org/10.1038/nbt1026) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=Production+of+recombinant+protein+therapeutics+in+cultivated+mammalian+cells&publication+year=2004&author=Wurm+F.+M.&journal=Nature+Biotechnology&volume=11&doi=10.1038/nbt1026)

Xu, P., Clark, C., Ryder, T., Sparks, C., Zhou, J., Wang, M., Russell, R. & Scott, C. (2017a), 'Characterization of TAP Ambr 250 disposable bioreactors, as a reliable scale-down model for biologics process development', *Biotechnology Progress* 33(2), 478–489. [CrossRef \(http://dx.doi.org/10.1002/btpr.2417\)](https://doi.org/10.1002/btpr.2417) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=Characterization+of+TAP+Ambr+250+disposable+bioreactors+as+a+reliable+scale-down+model+for+biologics+process+development&publication+year=2017&author=Xu+P.&author=Clark+C.&author=Ryder+T.&author=Sparks+C.&author=Zhou+J.&author=Wang+M.&author=Scott+C.&doi=10.1002/btpr.2417)
 | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27977912>)

Xu, S. & Chen, H. (2016), 'High-density mammalian cell cultures in stirred-tank bioreactor without external pH control', *Journal of Biotechnology* 231, 149–159. [CrossRef \(http://dx.doi.org/10.1016/j.jbiotec.2016.06.019\)](https://doi.org/10.1016/j.jbiotec.2016.06.019) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=High-density+mammalian+cell+cultures+in+stirred-tank+bioreactor+without+external+pH+control&publication+year=2016&author=Xu+S.&author=Chen+H.&journal=Journal+of+Biotechnology&volume=231&doi=10.1016/j.jbiotec.2016.06.019)
 | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27320019>)

Xu, S., Gavin, J., Jiang, R. & Chen, H. (2017b), 'Bioreactor productivity and media cost comparison for different intensified cell culture processes', *Biotechnology Progress* 33(4), 867–878. [CrossRef \(http://dx.doi.org/10.1002/btpr.2415\)](https://doi.org/10.1002/btpr.2415) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=Bioreactor+productivity+and+media+cost+comparison+for+different+intensified+cell+culture+processes&publication+year=2017&author=Xu+S.&author=Gavin+J.&author=Jiang+R.&author=Chen+H.&doi=10.1002/btpr.2415)
 | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27977910>)

Xu, S., Jiang, R., Chen, Y., Wang, F. & Chen, H. (2017c), 'Impact of Pluronic® F68 on hollow fiber filter-based perfusion culture performance', *Bioprocess and Biosystems Engineering* 40(9), 1317–1326. [CrossRef \(http://dx.doi.org/10.1007/s00449-017-1790-2\)](https://doi.org/10.1007/s00449-017-1790-2) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=Impact+of+Pluronic+C2+AE+F68+on+hollow+fiber+filter-based+perfusion+culture+performance&publication+year=2017&author=Xu+S.&author=Jiang+R.&author=Chen+Y.&author=Wang+F.&author=Chen+H.&journal=Bioprocess+and+Biosystem+1790-2&pages=1317-1326)
 | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/28577048>)

Yang, W. C., Lu, J., Kwiatkowski, C., Yuan, H., Kshirsagar, R., Ryll, T. & Huang, Y. M. (2014), 'Perfusion seed cultures improve biopharmaceutical fed-batch production capacity and product quality', *Biotechnology Progress* 30(3), 616–625. [CrossRef \(http://dx.doi.org/10.1002/btpr.1884\)](https://doi.org/10.1002/btpr.1884) |
 Google Scholar (https://scholar.google.com/scholar_lookup?title=Perfusion+seed+cultures+improve+biopharmaceutical+fed-batch+production+capacity+and+product+quality&publication+year=2014&author=Yang+W.+C.&author=Lu+J.&author=Kwiatkowski+C.&author=Yuan+H.&author=Kshirsagar+R.&author=Ry+625)
 | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/24574326>)

Yang, W. C., Minkler, D. F., Kshirsagar, R., Ryll, T. & Huang, Y. M. (2016), 'Concentrated fed-batch cell culture increases manufacturing capacity without additional volumetric capacity', *Journal of Biotechnology* 217, 1–11. [CrossRef \(http://dx.doi.org/10.1016/j.jbiotec.2015.10.009\)](https://doi.org/10.1016/j.jbiotec.2015.10.009) |

Google Scholar (https://scholar.google.com/scholar_lookup?title=Concentrated+fed-batch+cell+culture+increases+manufacturing+capacity+without+additional+volumetric+capacity&publication+year=2016&author=Yang+W.+C.&author=Minkler+D.+F.&author=Kshirsagar+R+11) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26521697>)

Yao, T. & Asayama, Y. (2017), 'Animal-cell culture media: History, characteristics, and current issues', *Reproductive Medicine and Biology* 16(2), 99–117. CrossRef (<http://dx.doi.org/10.1002/rmb2.12024>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Animal-cell+culture+media:+History+characteristics+and+current+issues&publication+year=2017&author=Yao+T.&author=Asayama+Y.&journal=Reproductive+Medicine+and+Biology&volume=16&117) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/29259457>)

Yeung, K. S. Y., Hoare, M., Thornhill, N. F., Williams, T. & Vaghjiani, J. D. (2002), 'Near-infrared spectroscopy for bioprocess monitoring and control', *Biotechnology and Bioengineering* 63(6), 684–693. CrossRef ([http://dx.doi.org/10.1002/\(SICI\)1097-0290\(19990620\)63:6<684::AID-BIT6>3.0.CO;2-Q](http://dx.doi.org/10.1002/(SICI)1097-0290(19990620)63:6<684::AID-BIT6>3.0.CO;2-Q)) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Near%3%A2%C5%81%C5%82infrared+spectroscopy+for+bioprocess+monitoring+and+control&publication+year=2002&author=Yeung+K.+S.+Y.&author=Hoare+M.&author=Thornhill+0290\(19990620\)63:6%3C684::AID-BIT6%3E3.0.CO;2-Q&pages=684-693](https://scholar.google.com/scholar_lookup?title=Near%3%A2%C5%81%C5%82infrared+spectroscopy+for+bioprocess+monitoring+and+control&publication+year=2002&author=Yeung+K.+S.+Y.&author=Hoare+M.&author=Thornhill+0290(19990620)63:6%3C684::AID-BIT6%3E3.0.CO;2-Q&pages=684-693))

Yoon, S. K., Choi, S. L., Song, J. Y. & Lee, G. M. (2005), 'Effect of culture pH on erythropoietin production by Chinese hamster ovary cells grown in suspension at 32.5 and 37.0C', *Biotechnology and Bioengineering* 89(3), 345–356. CrossRef (<http://dx.doi.org/10.1002/bit.20353>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Effect+of+culture+pH+on+erythropoietin+production+by+Chinese+hamster+ovary+cells+grown+in+suspension+at+32.5+and+37.0C&publication+year=2005&author=Yoon+S.+K.&author=Choi+S.+L.&author=Song+J.+Y.&author=Lee+G.+M)

Yoon, S. & Konstantinov, K. B. (1994), 'Continuous, real-time monitoring of the oxygen uptake rate (OUR) in animal cell bioreactors', *Biotechnology and Bioengineering* 44, 983–990. CrossRef (<http://dx.doi.org/10.1002/bit.260440815>) | Google Scholar ([https://scholar.google.com/scholar_lookup?title=Continuous+real-time+monitoring+of+the+oxygen+uptake+rate+\(OUR\)+in+animal+cell+bioreactors&publication+year=1994&author=Yoon+S.&author=Konstantinov+K.+B.&journal=Biotechnology+and+Bioengineering&volume=44&doi=10.1002/bit.260440990](https://scholar.google.com/scholar_lookup?title=Continuous+real-time+monitoring+of+the+oxygen+uptake+rate+(OUR)+in+animal+cell+bioreactors&publication+year=1994&author=Yoon+S.&author=Konstantinov+K.+B.&journal=Biotechnology+and+Bioengineering&volume=44&doi=10.1002/bit.260440990))

Yu, L. X., Baker, J., Berlam, S. C., Boam, A., Brandreth, E. J., Buhse, L., Cosgrove, T., Doleski, D., Ensor, L., Famulare, J., Ganapathy, M., Grampp, G., Hussong, D., Iser, R., Johnston, G., Kesisisoglou, F., Khan, M., Kozlowski, S., Lacana, E., Lee, S. L., Miller, S., Miksinski, S. P., Moore, C. M. V., Mullin, T., Raju, G. K., Raw, A., Rosencrance, S., Rosolowsky, M., Stinavage, P., Thomas, H., Wesdyk, R., Windisch, J. & Vaithiyalingam, S. (2015), 'Advancing product quality: A summary of the inaugural FDA/PQRI Conference', *The AAPS Journal* 17(4), 1011–1018. CrossRef (<http://dx.doi.org/10.1208/s12248-015-9754-4>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Advancing+product+quality:+A+summary+of+the+inaugural+FDA/PQRI+Conference&publication+year=2015&author=Yu+L.+X.&author=Baker+J.&author=Berlam+S.+C.&author=Boam+015-9754-4&pages=1011-1018) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/25840884>)

Zagari, F., Jordan, M., Stettler, M., Broly, H. & Wurm, F. M. (2013), 'Lactate metabolism shift in CHO cell culture: The role of mitochondrial oxidative activity', *New Biotechnology* 30(2), 238–245. CrossRef (<http://dx.doi.org/10.1016/j.nbt.2012.05.021>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Lactate+metabolism+shift+in+CHO+cell+culture:+The+role+of+mitochondrial+oxidative+activity&publication+year=2013&author=Zagari+F.&author=Jordan+M.&author=Stettler+M.&author=Wurm+F.+M)

Zalai, D., Tobak, T. & Putics, Á. (2015), 'Impact of apoptosis on the on-line measured dielectric properties of CHO cells', *Bioprocess and Biosystems Engineering* 38(12), 2427–2437. CrossRef (<http://dx.doi.org/10.1007/s00449-015-1479-3>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Impact+of+apoptosis+on+the+on-line+measured+dielectric+properties+of+CHO+cells&publication+year=2015&author=Zalai+D.&author=Tobak+T.&author=Putics+A.&journal=Bioprocess+and+Biosystems+Engineering&volume=38&issue=12&pages=2427-2437) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26440966>)

Zhang, A., Tsang, V. L., Moore, B., Shen, V., Huang, Y. M., Kshirsagar, R. & Ryll, T. (2015a), 'Advanced process monitoring and feedback control to enhance cell culture process production and robustness', *Biotechnology and Bioengineering* 112(12), 2495–2504. CrossRef (<http://dx.doi.org/10.1002/bit.25684>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Advanced+process+monitoring+and+feedback+control+to+enhance+cell+culture+process+production+and+robustness&publication+year=2015&author=Zhang+A.&author=Tsang+V.+L.&author=Moore+B.&author=Shen+V.&author=Huang+Y.+M.&author=Kshirsagar+R.&author=Ryll+T) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/26108810>)

Zhang, Y. H. P., Sun, J. & Ma, Y. (2016), 'Biomanufacturing: History and perspective', *Journal of Industrial Microbiology and Biotechnology* 44(4–5), 773–784. CrossRef (<http://dx.doi.org/10.1007/s12095-016-1863-2>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Biomanufacturing:+History+and+perspective&publication+year=2016&author=Zhang+Y.+H.+P.&author=Sun+J.&author=Ma+Y.&journal=Journal+of+Industrial+Microbiology+and+Biotechnology&volume=44&issue=4-5&pages=773-784) | PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27837351>)

Zhang, Y., Stobbe, P., Silvander, C. O. & Chotteau, V. (2015b), 'Very high cell density perfusion of CHO cells anchored in a non-woven matrix-based bioreactor', *Journal of Biotechnology* 213, 28–41. CrossRef (<http://dx.doi.org/10.1016/j.jbiotec.2015.07.006>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Very+high+cell+density+perfusion+of+CHO+cells+anchored+in+a+non-woven+matrix-based+bioreactor&publication+year=2015&author=Zhang+Y.&author=Stobbe+P.&author=Silvander+C.+O.&author=Chotteau+V.&journal=Journal+of+Biotechnology&volume=213&doi=10.1016/j.jbiotec.2015.07.006)

Zhou, W. & Kantardjiev, A. (2014), *Mammalian Cell Cultures for Biologics Manufacturing*, Springer. CrossRef (<http://dx.doi.org/10.1007/978-3-642-54050-9>) | Google Scholar (https://scholar.google.com/scholar_lookup?title=Mammalian+Cell+Cultures+for+Biologics+Manufacturing&publication+year=2014&author=Zhou+W.&author=Kantardjiev+A)

Zhu, L. K., Song, B. Y., Wang, Z. L., Montell, D. T., Shen, X., Hacker, D. L., De Jesus, M. & Wurm, F. M. (2017), 'Studies on fluid dynamics of the flow field and gas transfer in orbitally shaken tubes', *Biotechnology Progress* 33(1), 192–200. CrossRef (<http://dx.doi.org/10.1002/btpr.2375>) |

Google Scholar ([https://scholar.google.com/scholar_lookup?](https://scholar.google.com/scholar_lookup?title=Studies+on+fluid+dynamics+of+the+flow+field+and+gas+transfer+in+orbitally+shaken+tubes&publication+year=2017&author=Zhu+L.+K.&author=Song+B.+Y.&author=Wang+Z.+L.&au200)

[title=Studies+on+fluid+dynamics+of+the+flow+field+and+gas+transfer+in+orbitally+shaken+tubes&publication+year=2017&author=Zhu+L.+K.&author=Song+B.+Y.&author=Wang+Z.+L.&au200](https://scholar.google.com/scholar_lookup?title=Studies+on+fluid+dynamics+of+the+flow+field+and+gas+transfer+in+orbitally+shaken+tubes&publication+year=2017&author=Zhu+L.+K.&author=Song+B.+Y.&author=Wang+Z.+L.&au200))

| PubMed (<https://www.ncbi.nlm.nih.gov/pubmed/27690196>)

Zhu, M. M., Goyal, A., Rank, D. L., Gupta, S. K., Vanden Boom, T. & Lee, S. S. (2005), 'Effects of elevated pCO₂ and osmolality on growth of CHO cells and production of antibody-fusion protein B1: A case study', *Biotechnology Progress* 21(1), 70–77. CrossRef (<http://dx.doi.org/10.1021/bp049815s>) |

Google Scholar (<https://scholar.google.com/scholar?q=Zhu+M.+M.+Goyal+A.+Rank+D.+L.+Gupta+S.+K.+Vanden+Boom+T.+&+Lee+S.+S.+>

[2005\)+%E2%80%98Effects+of+elevated+pCO2+and+osmolality+on+growth+of+CHO+cells+and+production+of+antibody-](https://scholar.google.com/scholar?q=Zhu+M.+M.+Goyal+A.+Rank+D.+L.+Gupta+S.+K.+Vanden+Boom+T.+&+Lee+S.+S.+)

[fusion+protein+B1:+A+case+study%E2%80%99+Biotechnology+Progress+21\(1\)+70%E2%80%9377.](https://scholar.google.com/scholar?q=Zhu+M.+M.+Goyal+A.+Rank+D.+L.+Gupta+S.+K.+Vanden+Boom+T.+&+Lee+S.+S.+))

Zoro, B. & Tait, A. (2017), 'Development of a novel automated perfusion mini bioreactor ambr[®] 250 perfusion', in *Integrated Continuous Biomanufacturing III*, Suzanne Farid, University College London, United Kingdom Chetan Goudar, Amgen, USA Paula Alves, IBET, Portugal Veena Warikoo, Axcella Health, Inc., USA Eds, ECI Symposium Series 2017', p. 250.

Google Scholar (<https://scholar.google.com/scholar?q=Zoro+B.+&+Tait+A.+>

[2017\)+%E2%80%98Development+of+a+novel+automated+perfusion+mini+bioreactor+ambr%E2%93%87+250+perfusion%E2%80%99+in+Integrated+Continuous+Biomanufacturing+III+Su](https://scholar.google.com/scholar?q=Zoro+B.+&+Tait+A.+)